δ - J/t phase diagram of t-J model

ED and DMRG results

Rongyang Sun IASTU 20200421

Outline

- t-J model
- Phase separation
- Phase diagram
  - 1D chain
  - Two-leg ladder
  - Three-leg ladder
  - Four-leg ladder
  - 2D
- Outlook

t-J model

t-J model

$$H = H_t + H_J$$

$$H_t = -t \sum_{\langle ij \rangle \sigma} c_{i\sigma} c_{j\sigma}^{\dagger} + h.c.$$

$$H_J = J \sum_{\langle ij \rangle} \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} n_i n_j \right)$$

$$n_i \leq 1$$

Phase separation

Phase separation

- Rommer, S., White, S. R., & Scalapino, D. (2000). Phase separation in ladders. *Physical Review B*, *61*(20), 13424–13430.
- Emery, V. J., Kivelson, S. A., & Lin, H. Q. (1990). Phase separation in the t - J model. *Physical Review Letters*, *64*(4), 475–478.

Phase separation

How to measure

- Charge density distribution
- Inverse compressibility $1/\kappa \sim \partial^2 E(N)/\partial N^2$

Phase separation

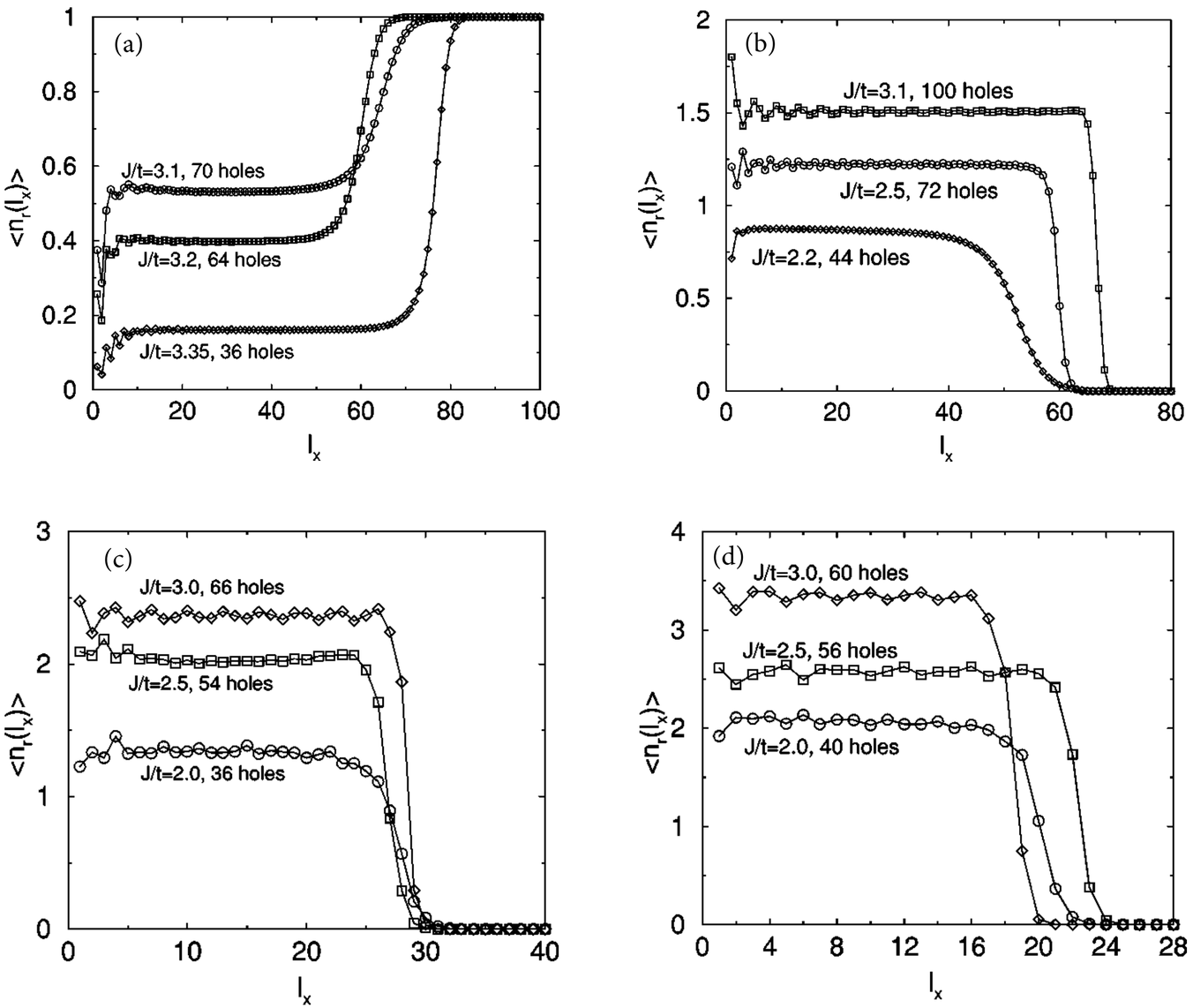
Charge density distribution

(a) 1D chain

(b) Two-leg ladder

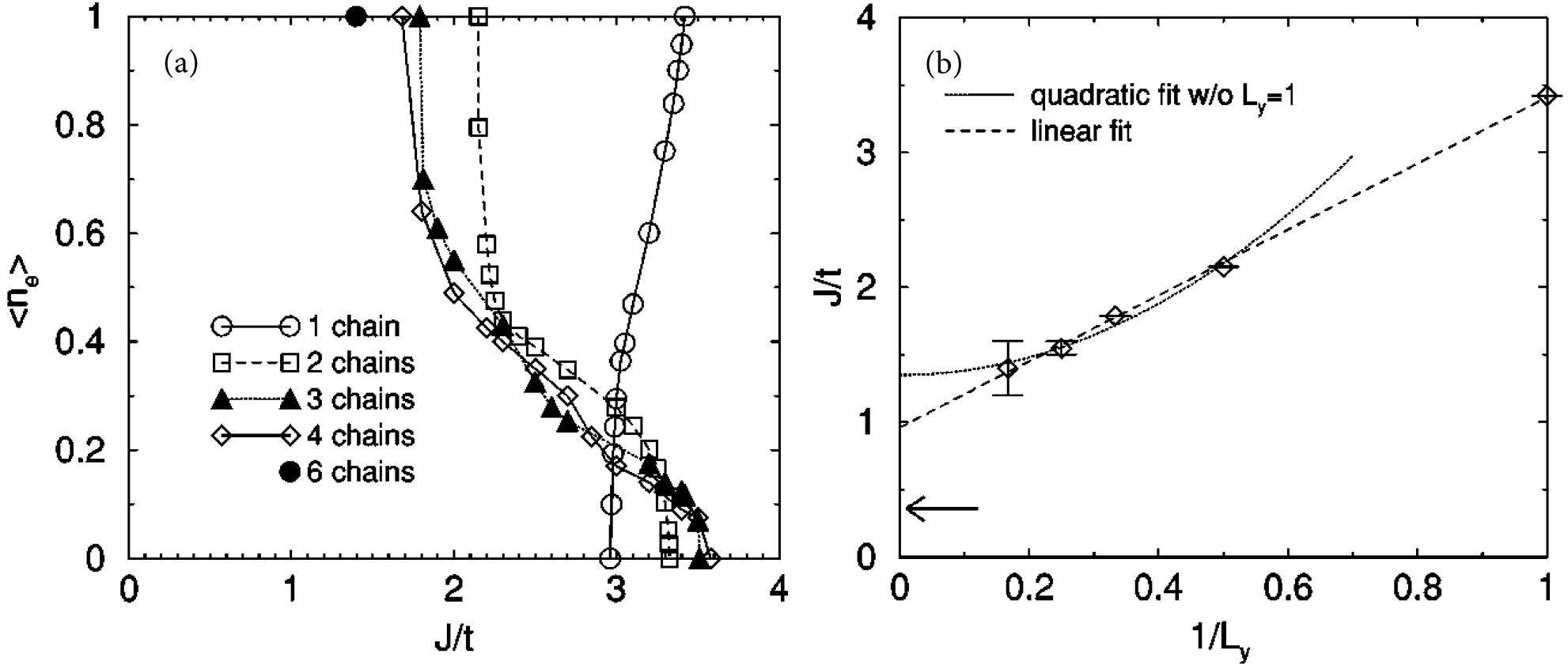
(c) Three-leg ladder

(d) Four-leg ladder



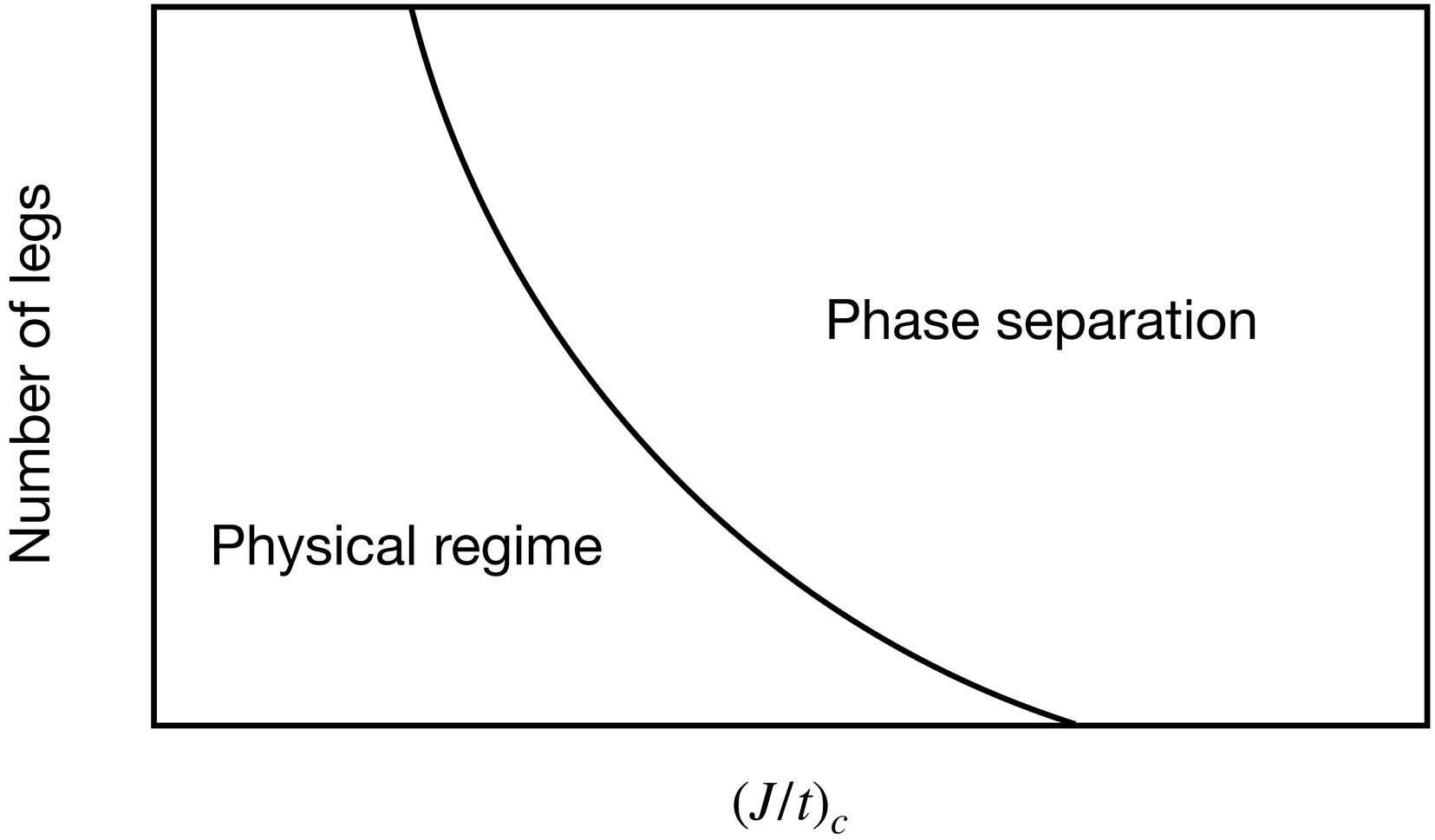
Phase separation

Phase boundaries



Phase separation

Physical regime



Phase diagram: 1D

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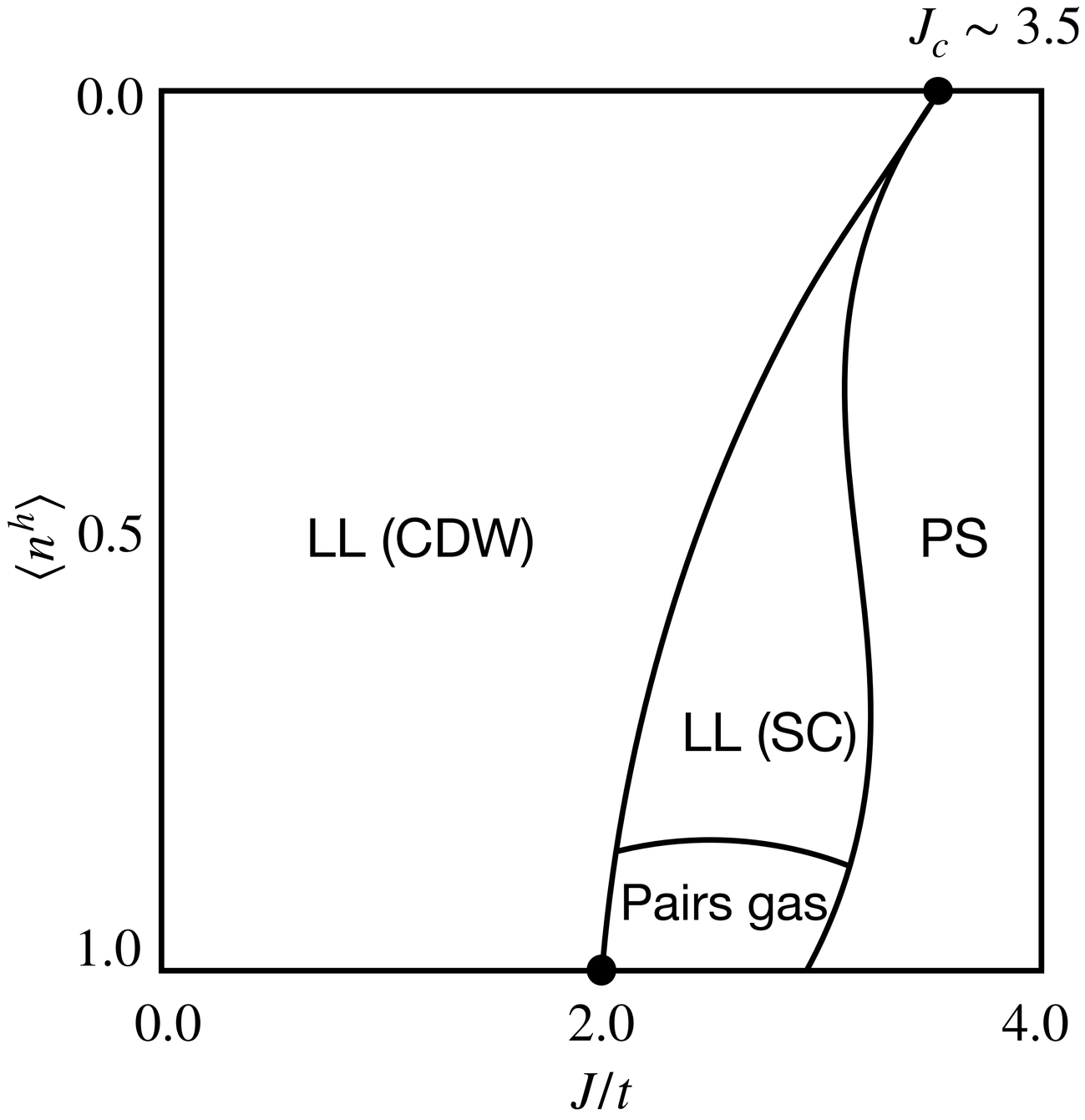
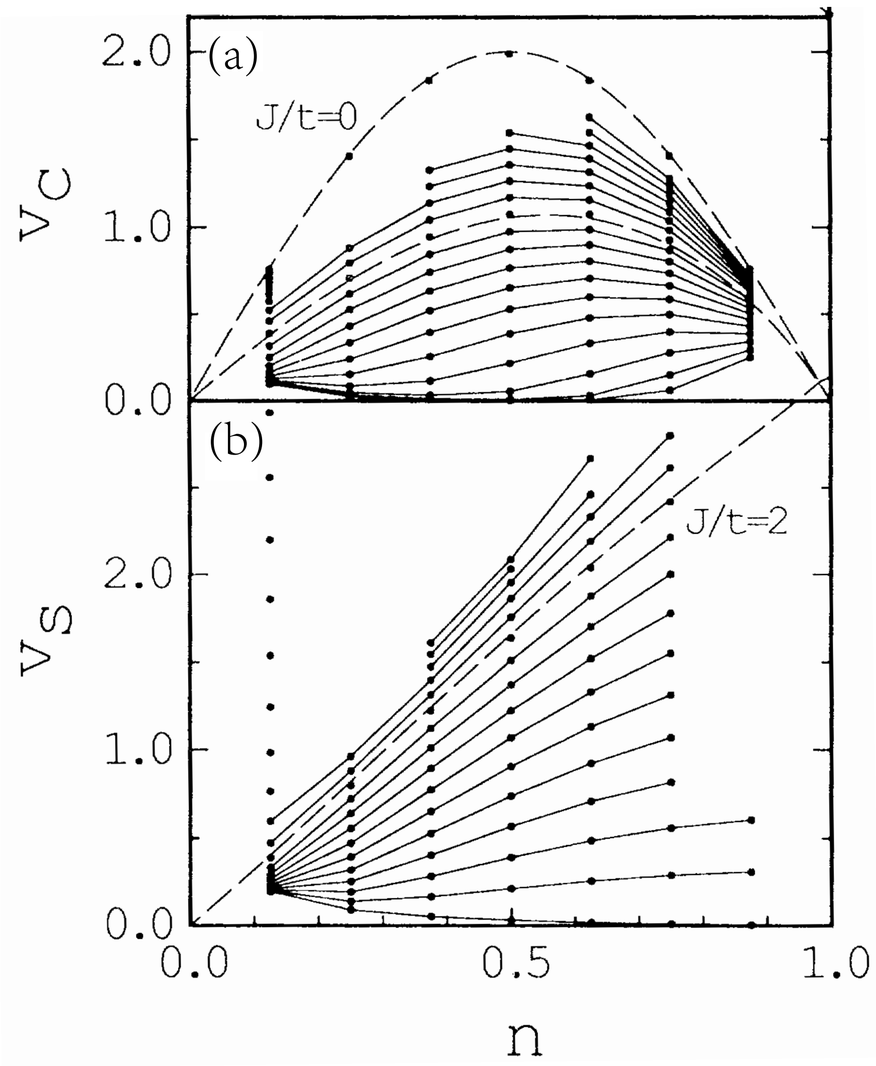
- Ogata, M., Luchini, M. U., Sorella, S., & Assaad, F. F. (1991). Phase diagram of the one-dimensional t - J model. *Physical Review Letters*, *66*(18), 2388–2391.
- Kawakami, N., & Yang, S. K. (1990). Correlation functions in the one-dimensional t-J model. *Physical Review Letters*, *65*(18), 2309–2311.

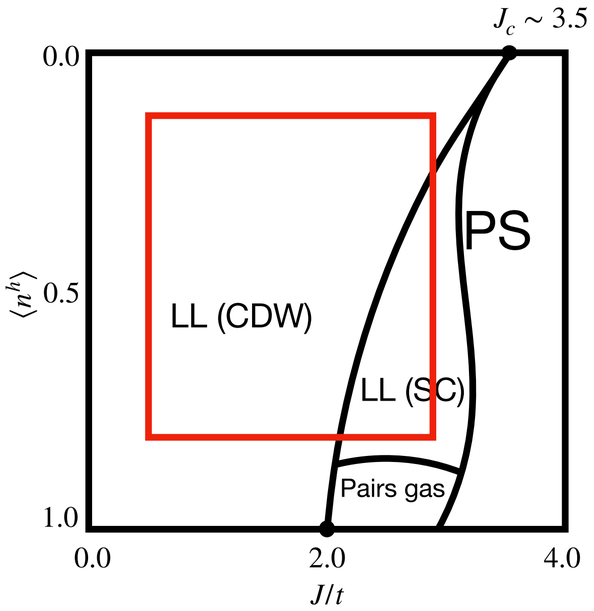
Phase diagram: 1D



Phase diagram: 1D

Charge velocity v_c and spin velocity v_s



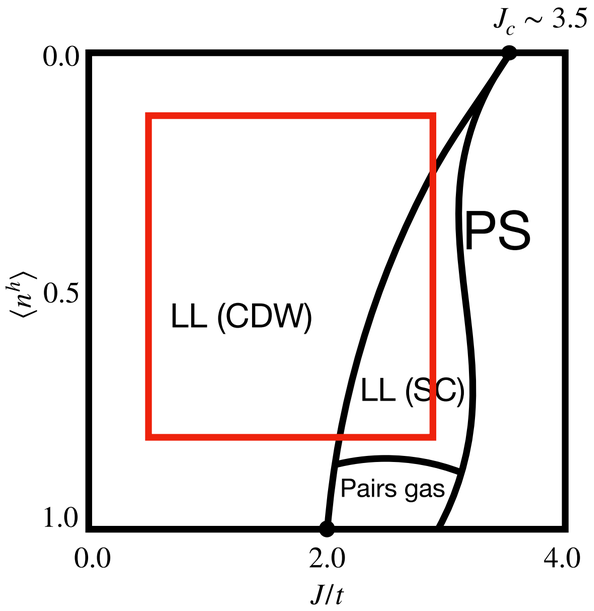


Phase diagram: 1D

Central charge c

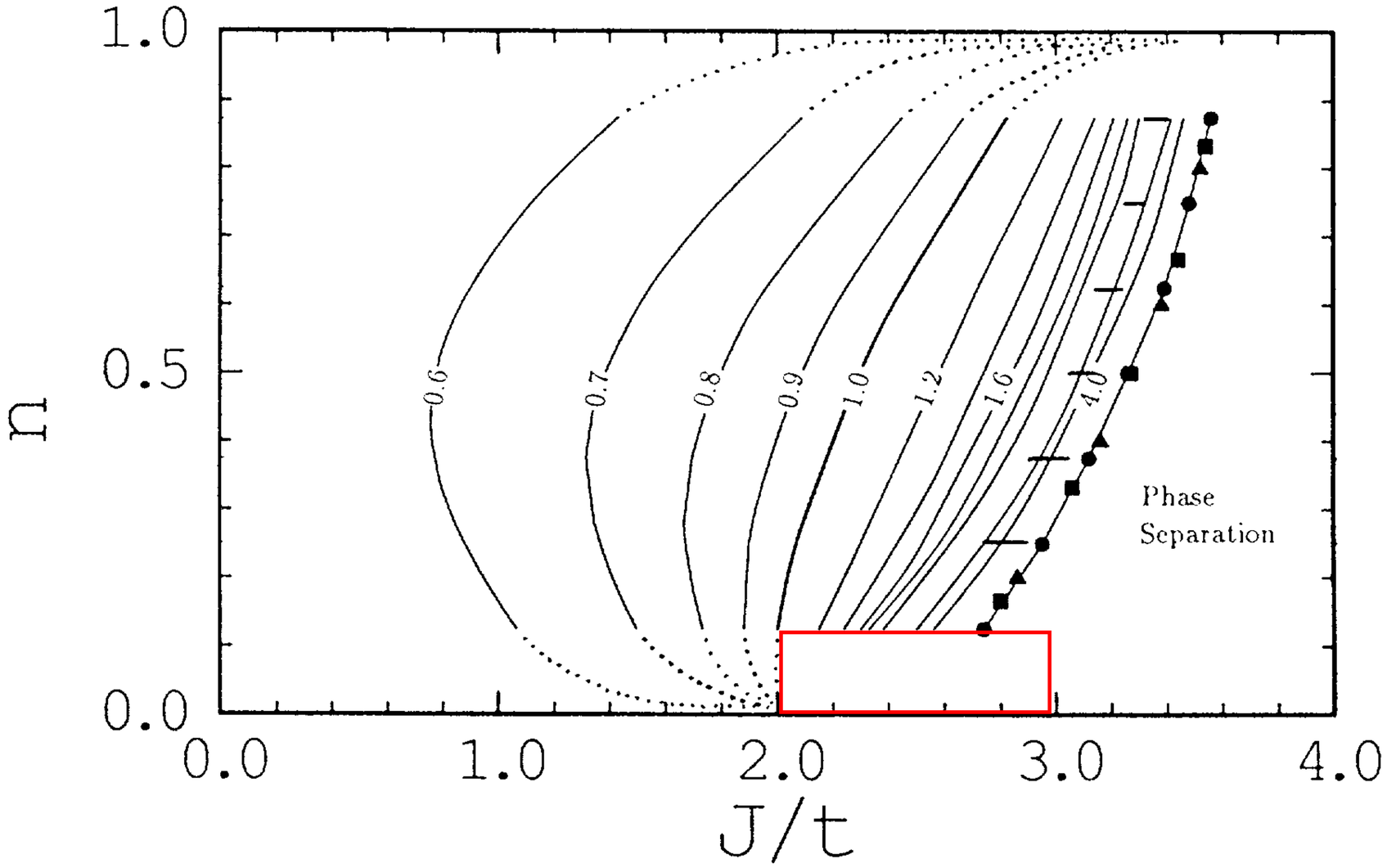
$$E/N_a = \epsilon_\infty - \pi\left(v_c + v_s\right) c/6N_a^2$$

$$c \sim 1.01$$

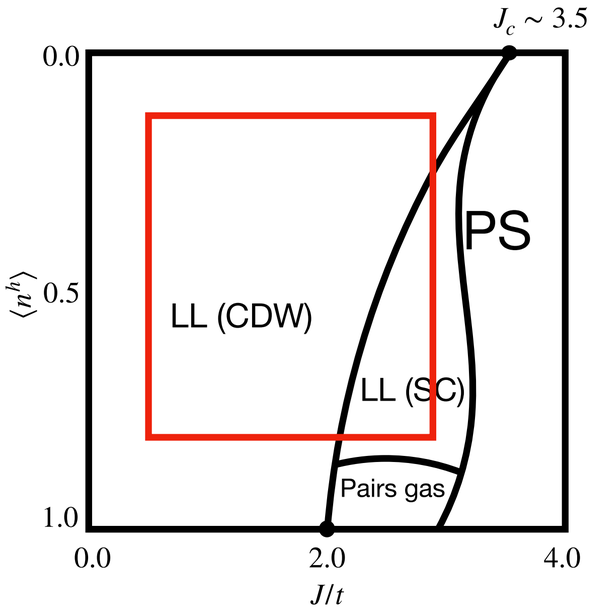


Phase diagram: 1D

Luttinger exponent K_ρ



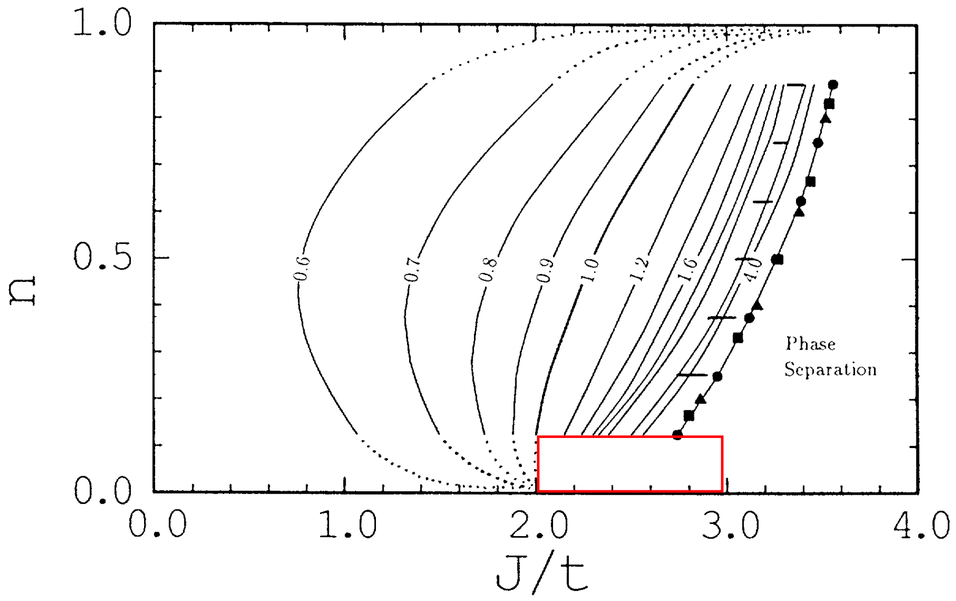
$$\frac{1}{n^2\kappa} = \frac{\pi}{2}\frac{v_c}{K_\rho}$$

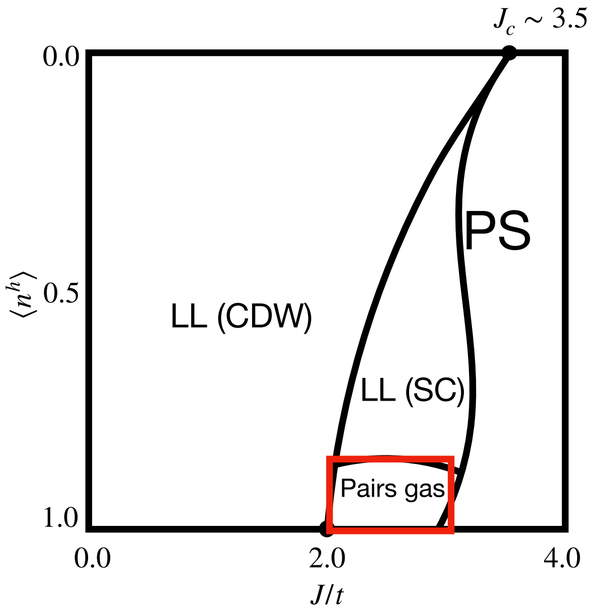


Phase diagram: 1D

Pairs gas

- High doping concentration regime
- Energy of a four electron system equals energy of two singlet pairs.
- Unclear





Phase diagram: two-leg ladder

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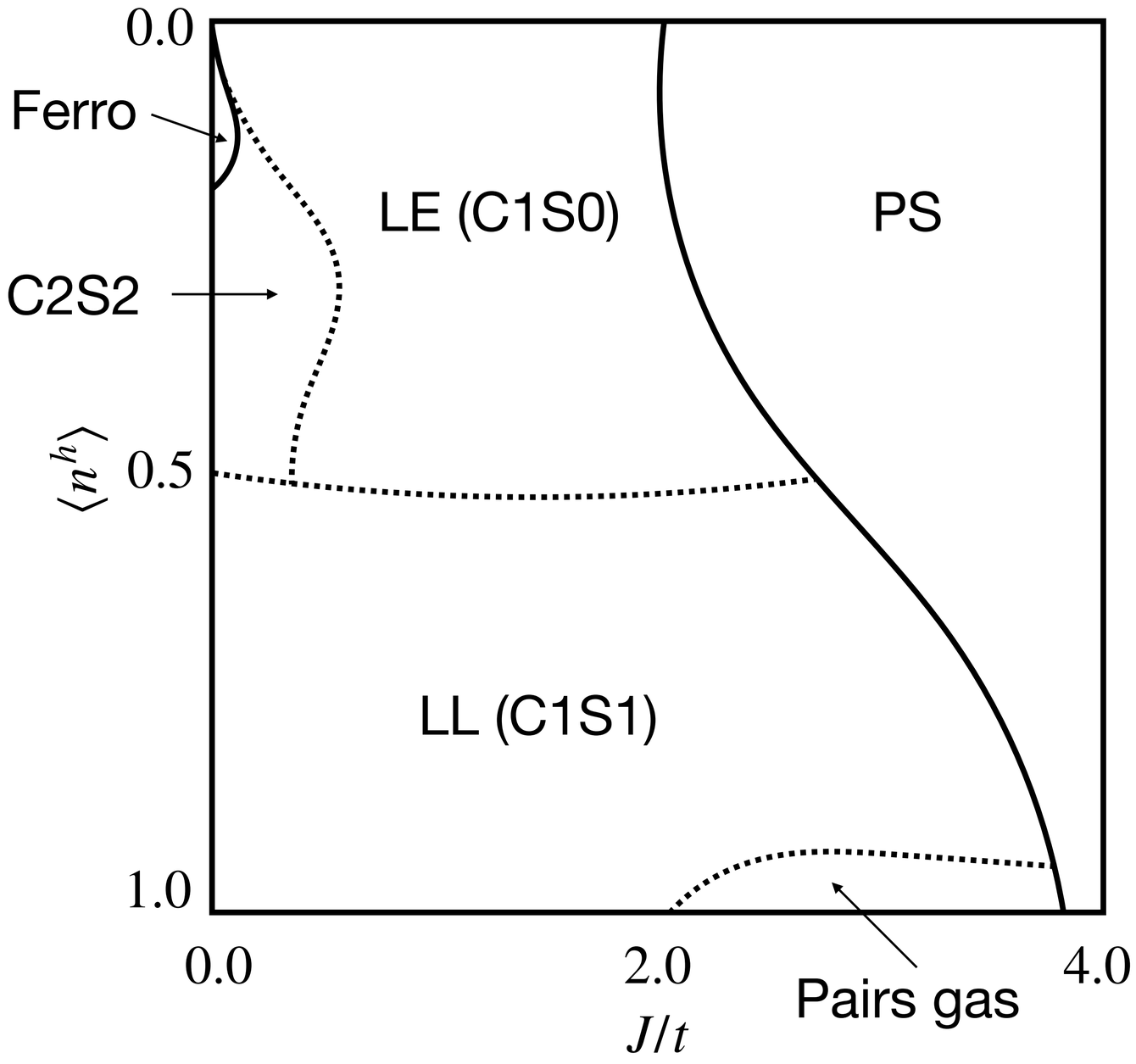
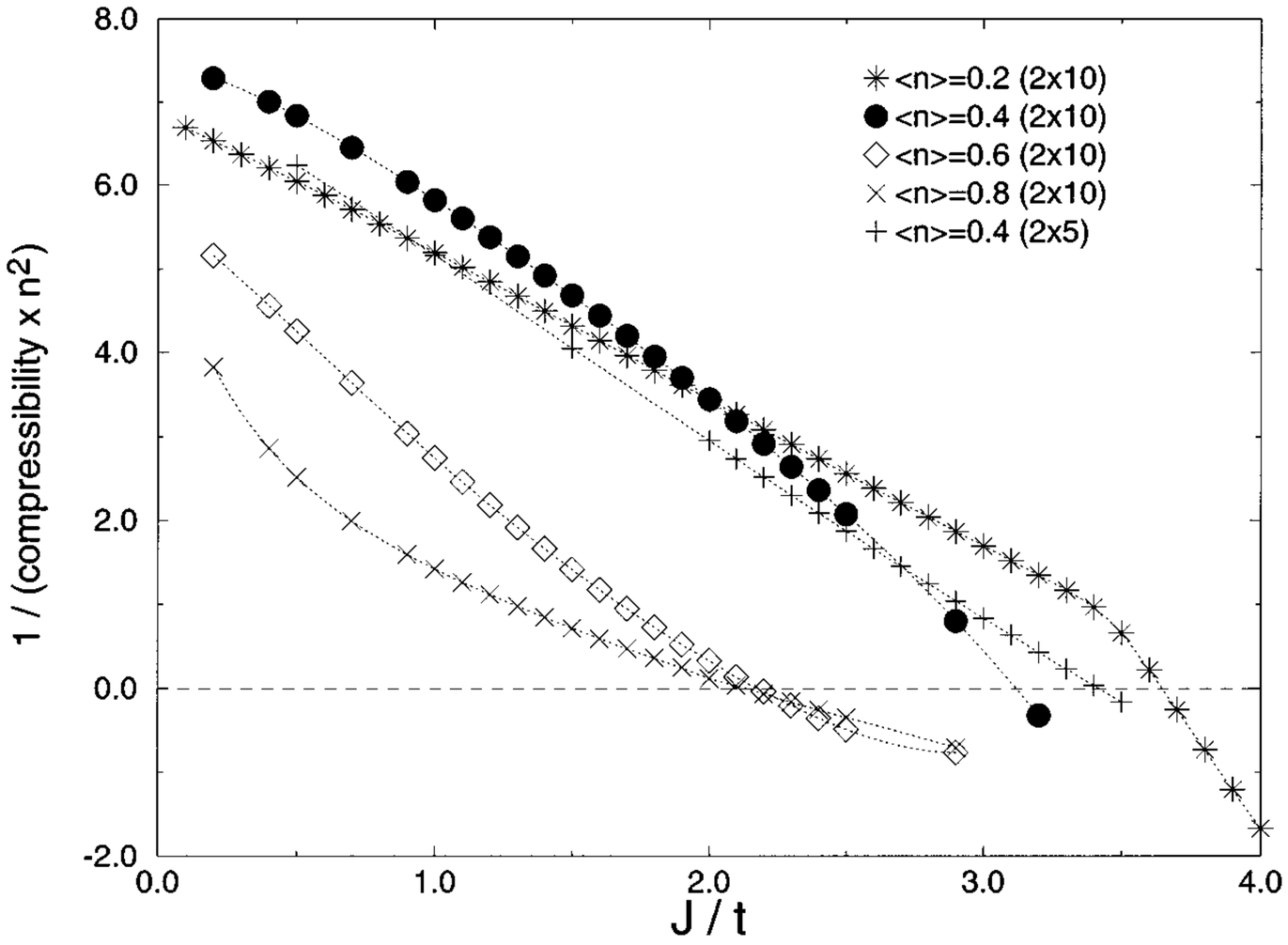
- Dagotto, E., Riera, J., & Scalapino, D. (1992). Superconductivity in ladders and coupled planes. *Physical Review B*, *45*(10), 5744–5747.
- Tsunetsugu, H., Troyer, M., & Rice, T. M. (1994). Pairing and excitation spectrum in doped t-J ladders. *Physical Review B*, *49*(22), 16078–16081.
- Poilblanc, D., Scalapino, D. J., & Hanke, W. (1995). Spin and charge modes of the t - J ladder. *Physical Review B*, *52*(9), 6796–6800.
- Troyer, M., Tsunetsugu, H., Troyer, M., Tsunetsugu, H., & Rice, T. (1996). Properties of lightly doped t-J two-leg ladders. *Physical Review B*, *53*(1), 251–267.
- Hayward, C., & Poilblanc, D. (1996). Luttinger-liquid behavior and superconducting correlations in t-J ladders. *Physical Review B*, *53*(17), 11721–11728.
- Müller, T., & Rice, T. (1998). Phase diagram of coupled ladders. *Physical Review B*, *58*(6), 3425–3437.
- Jiang, H., Chen, S., & Weng, Z. (2019). Critical role of the sign structure in the doped Mott insulator: Superconductivity vs. Fermi liquid, 1909.01475

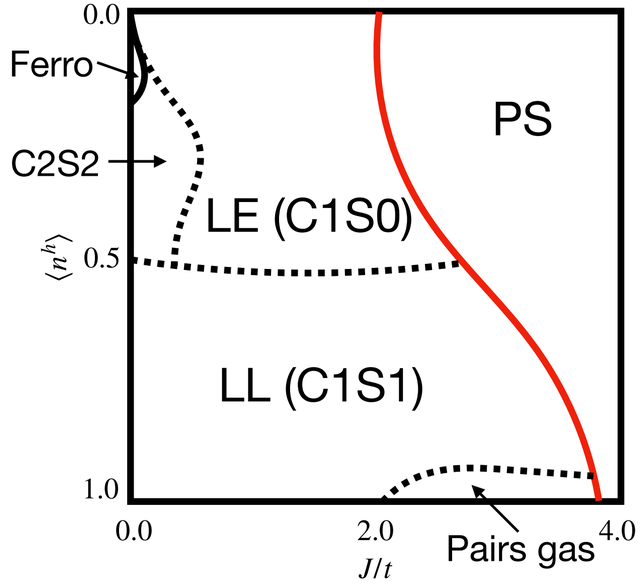
Phase diagram: two-leg ladder



Phase diagram: two-leg ladder

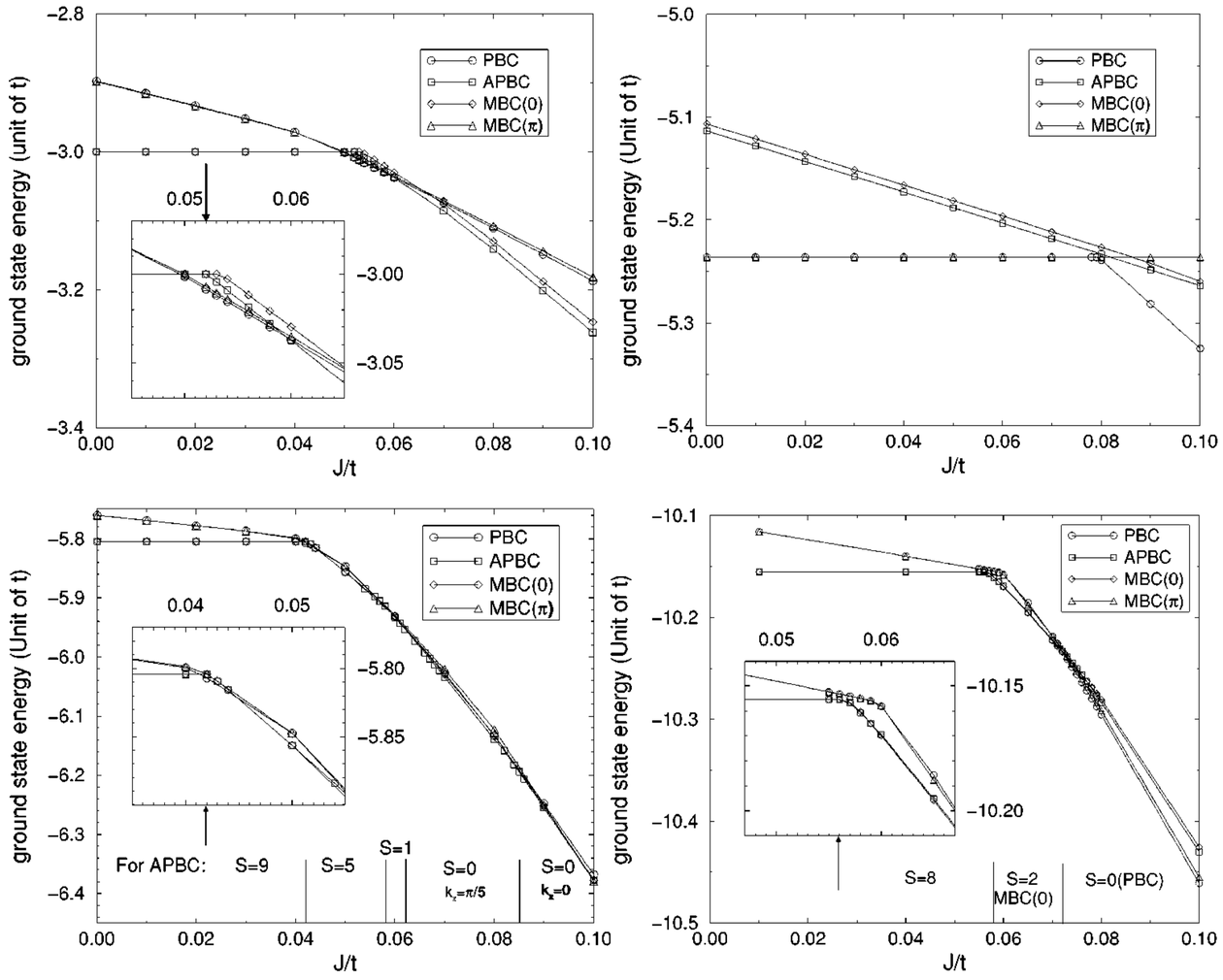
Inverse compressibility

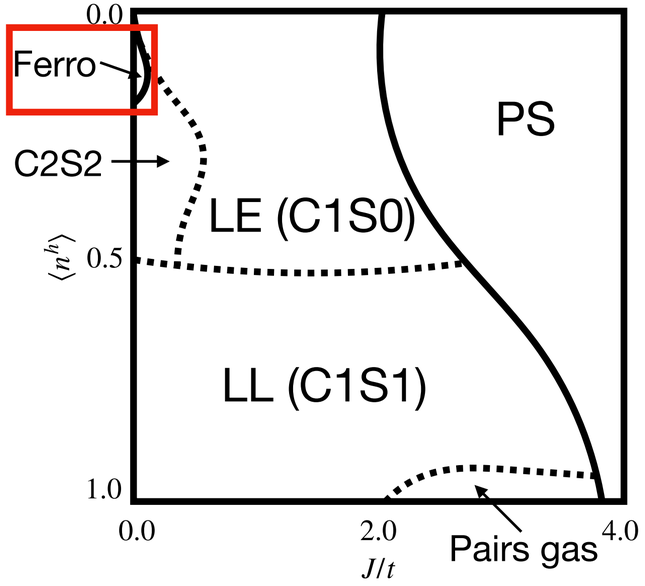




Phase diagram: two-leg ladder

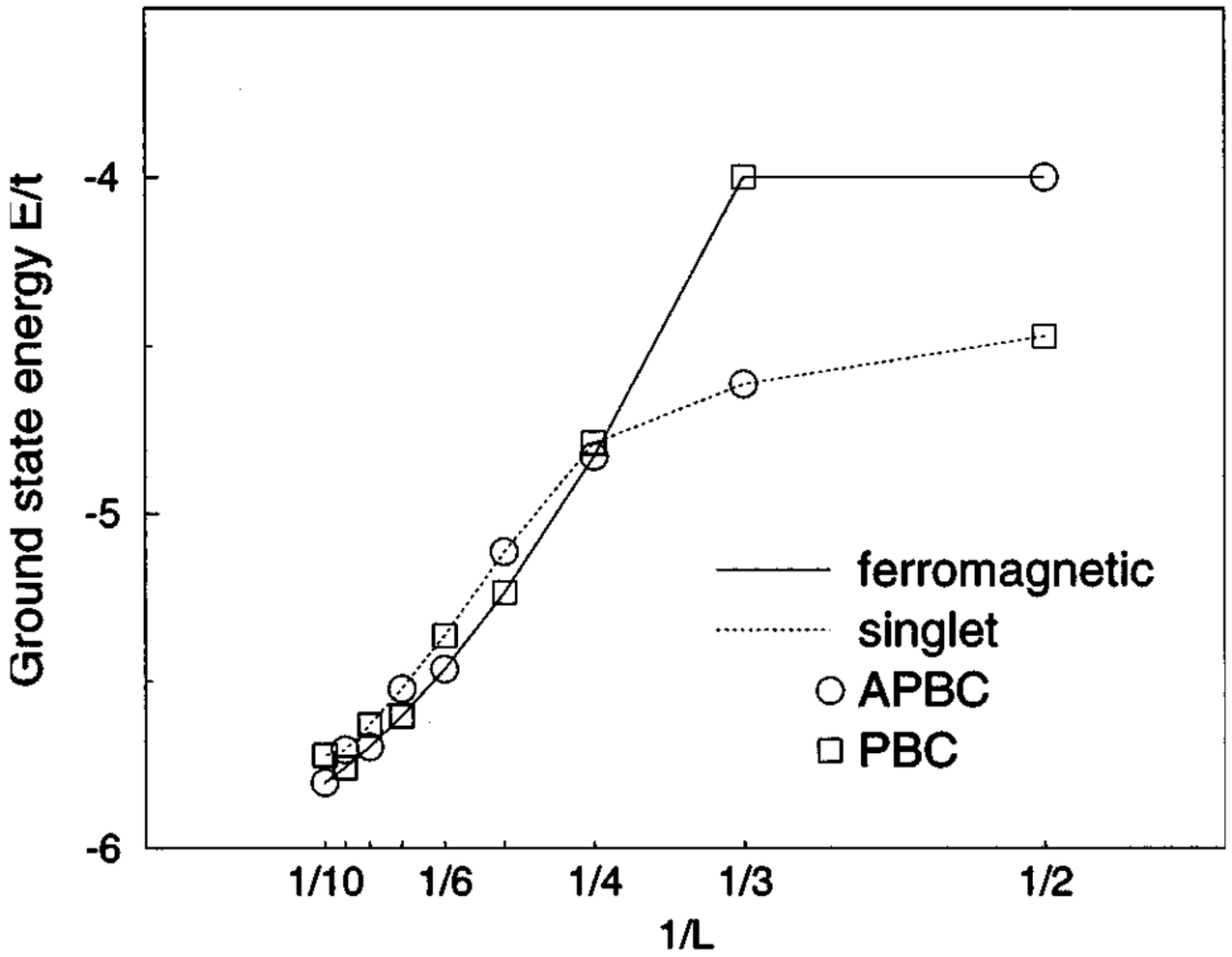
$J/t \to 0$ Nagaoka ferromagnetic

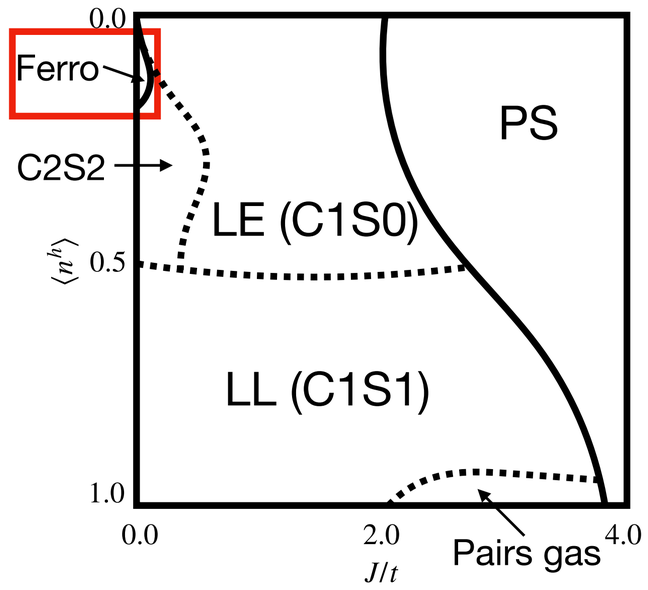




Phase diagram: two-leg ladder

$J/t \to 0$ **Nagaoka ferromagnetic**

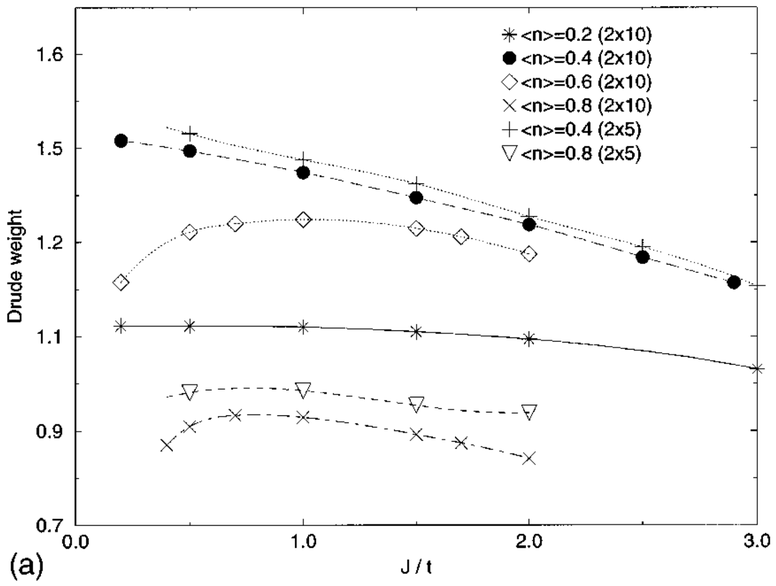


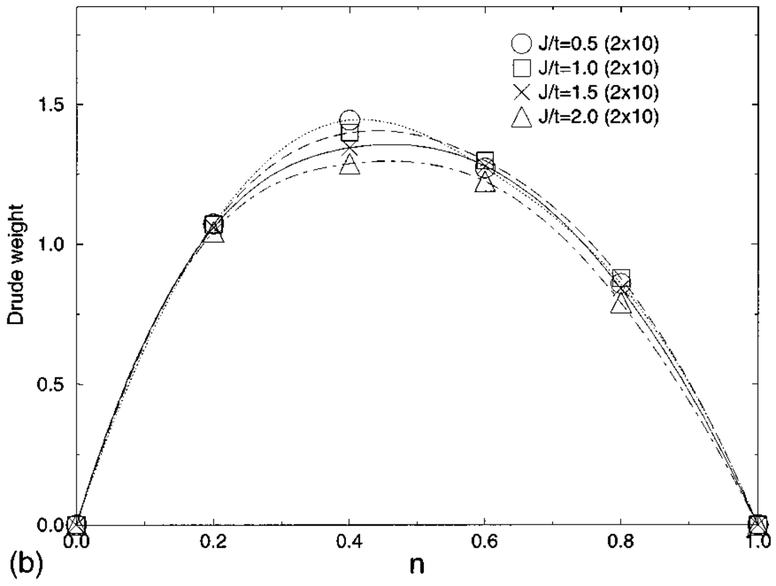


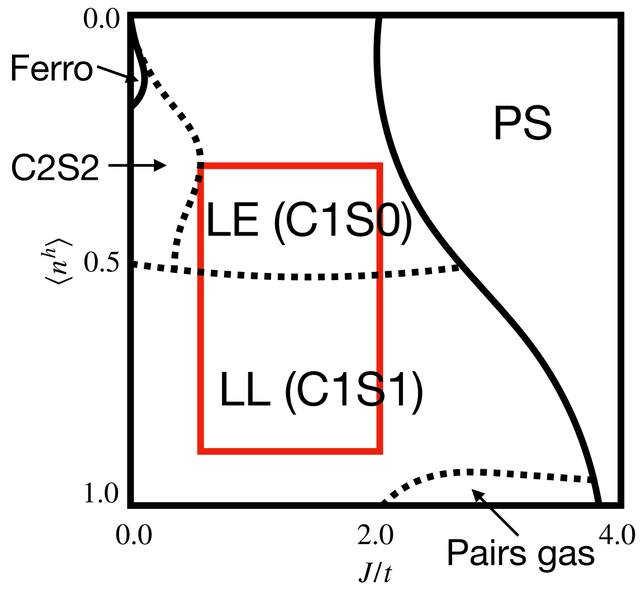
Phase diagram: two-leg ladder

Drude weight σ_0

$$\sigma_0 = \frac{L^2}{4\pi} \frac{\partial^2 \left(E_0/N\right)}{\partial \Phi^2}$$

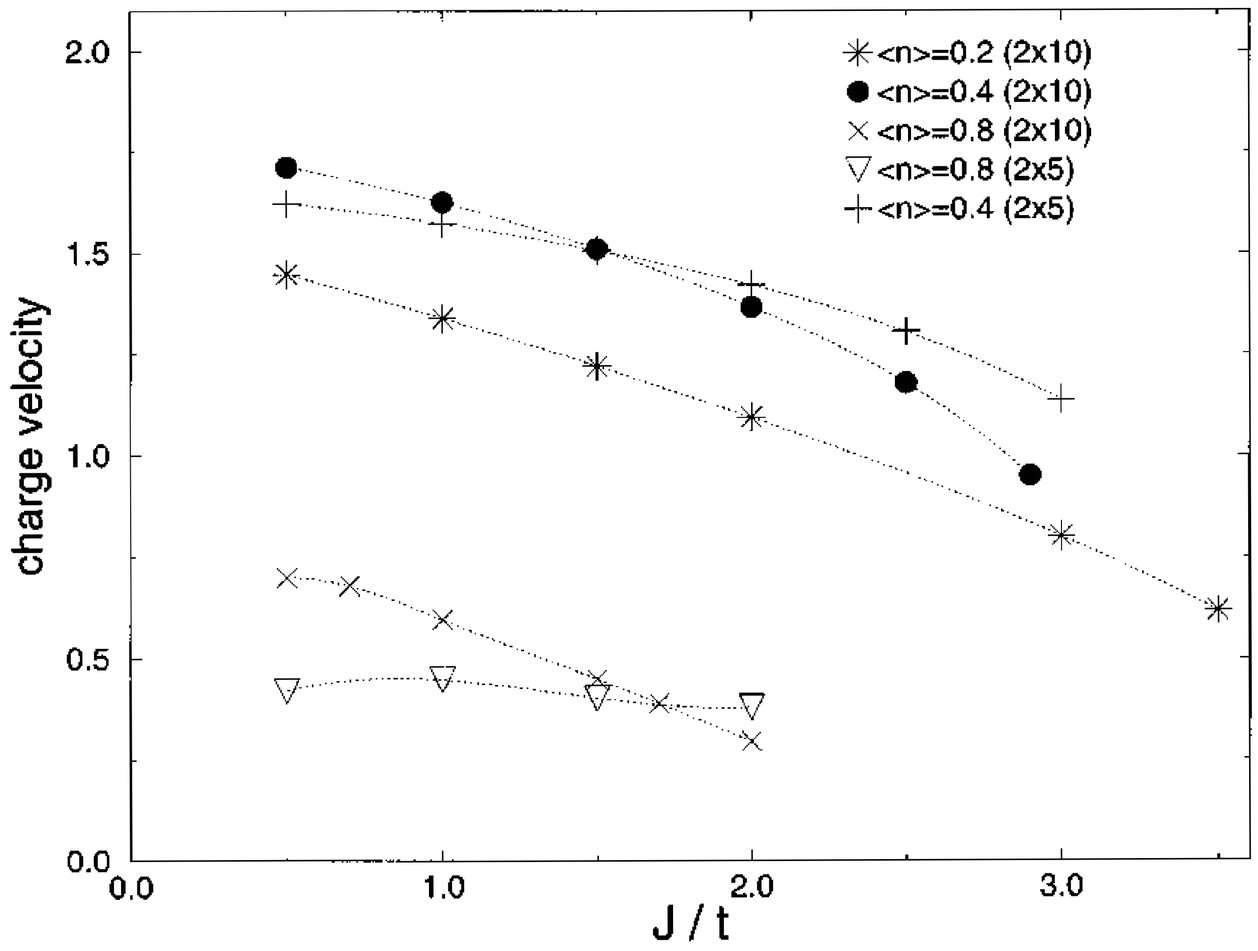


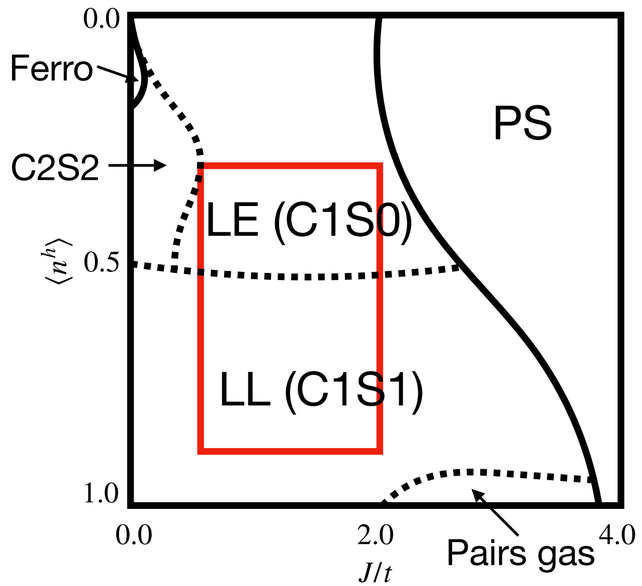




Phase diagram: two-leg ladder

Charge velocity v_c

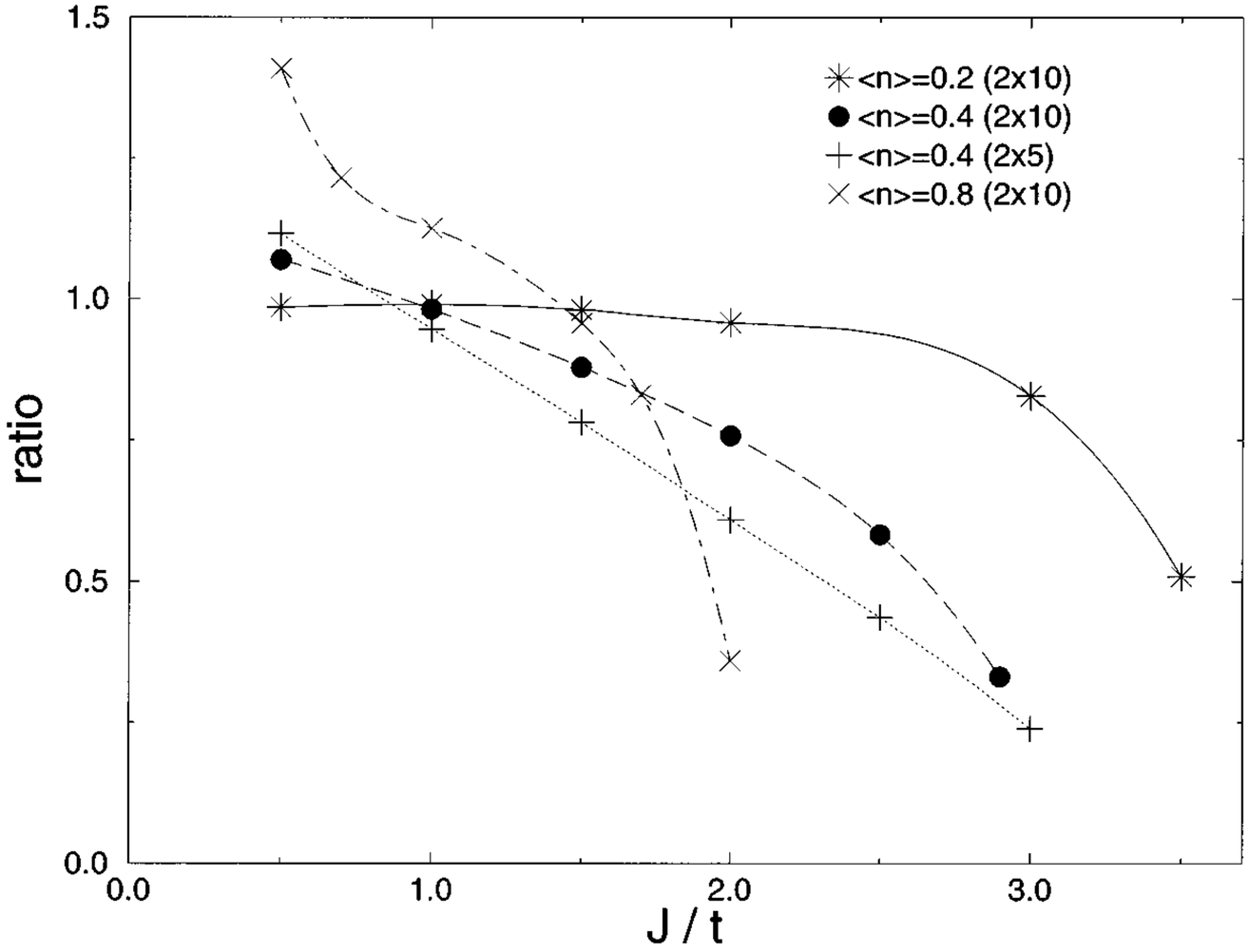


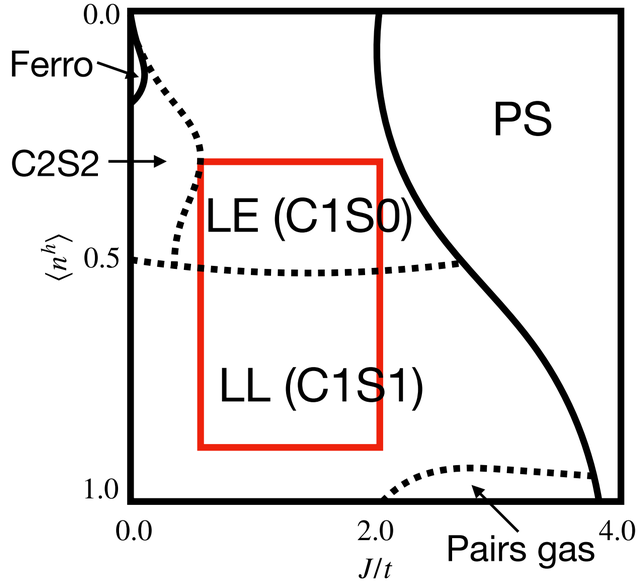


Phase diagram: two-leg ladder

Luttinger liquid ?

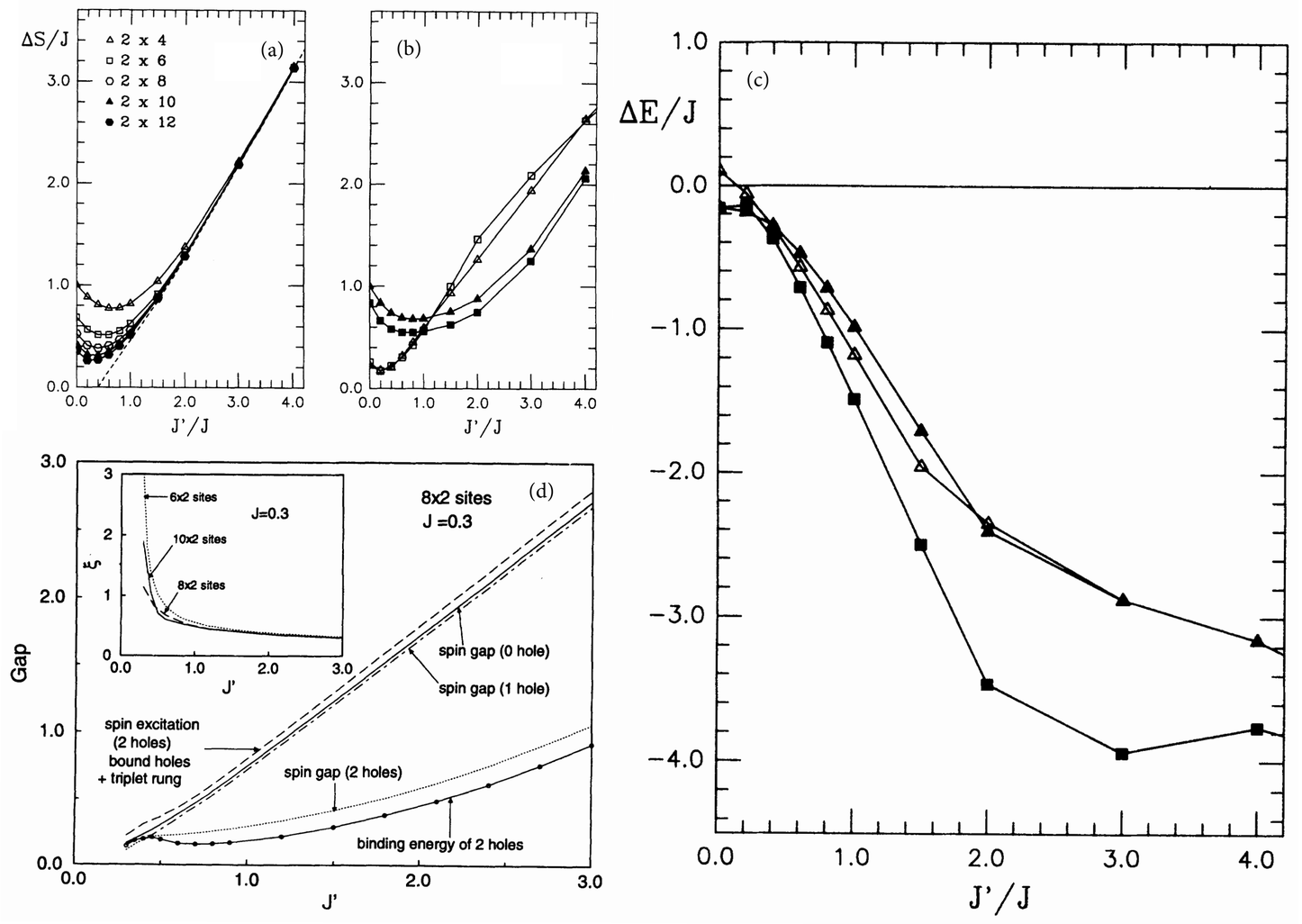
$$\pi n^2 \kappa v_c^2 / \sigma_0 = 1$$

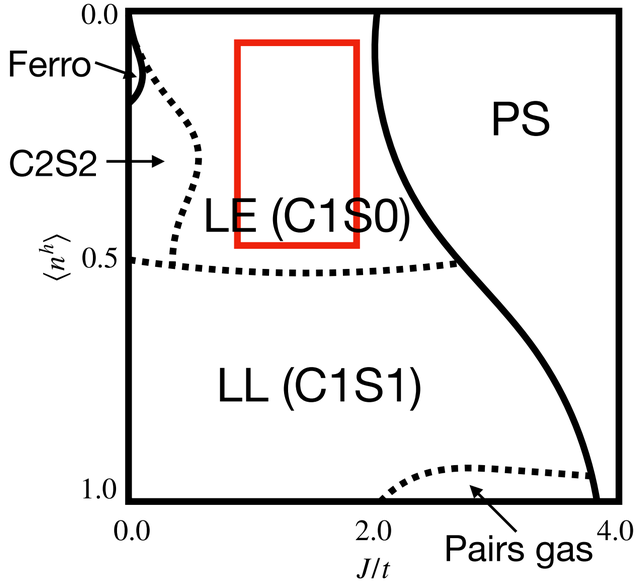




Phase diagram: two-leg ladder

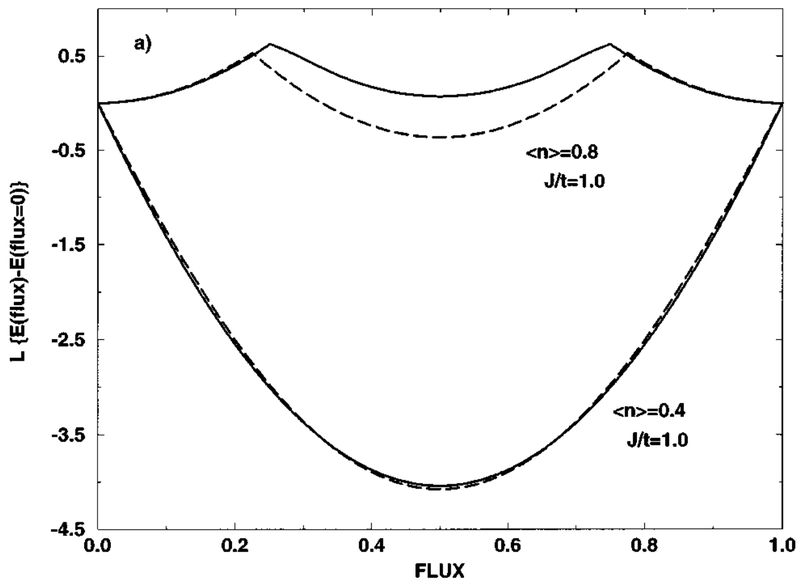
Luther-Emery liquid: spin gap and pairing

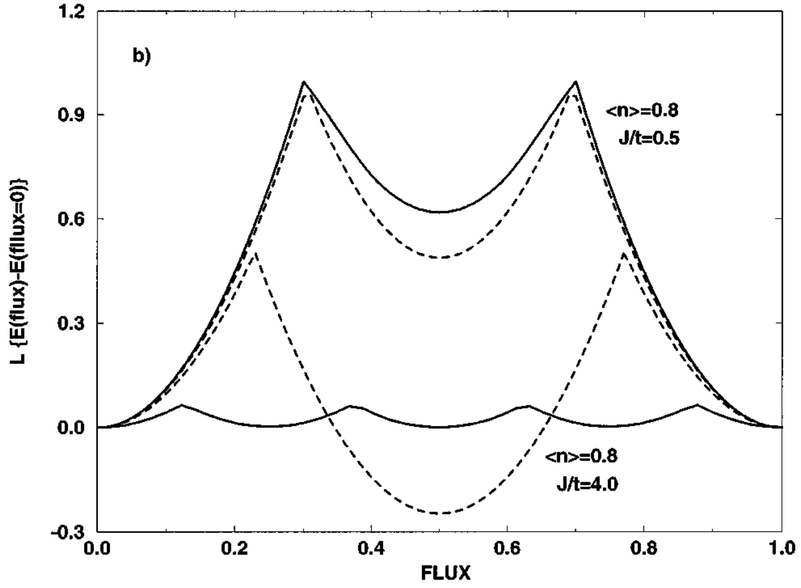


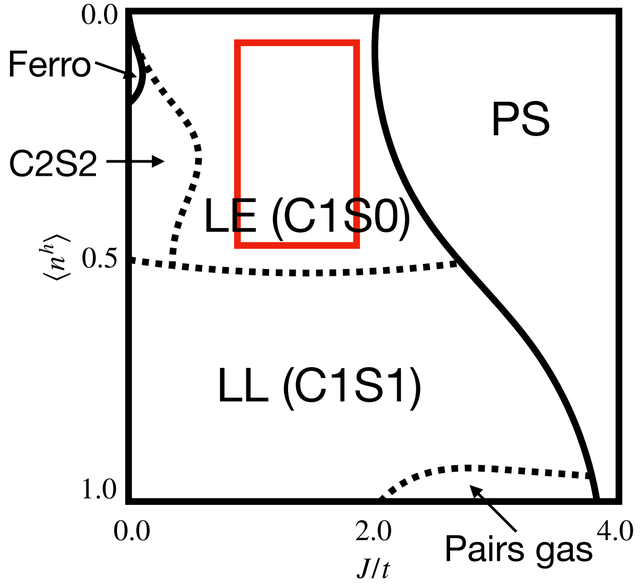


Phase diagram: two-leg ladder

Luther-Emery liquid: Flux response

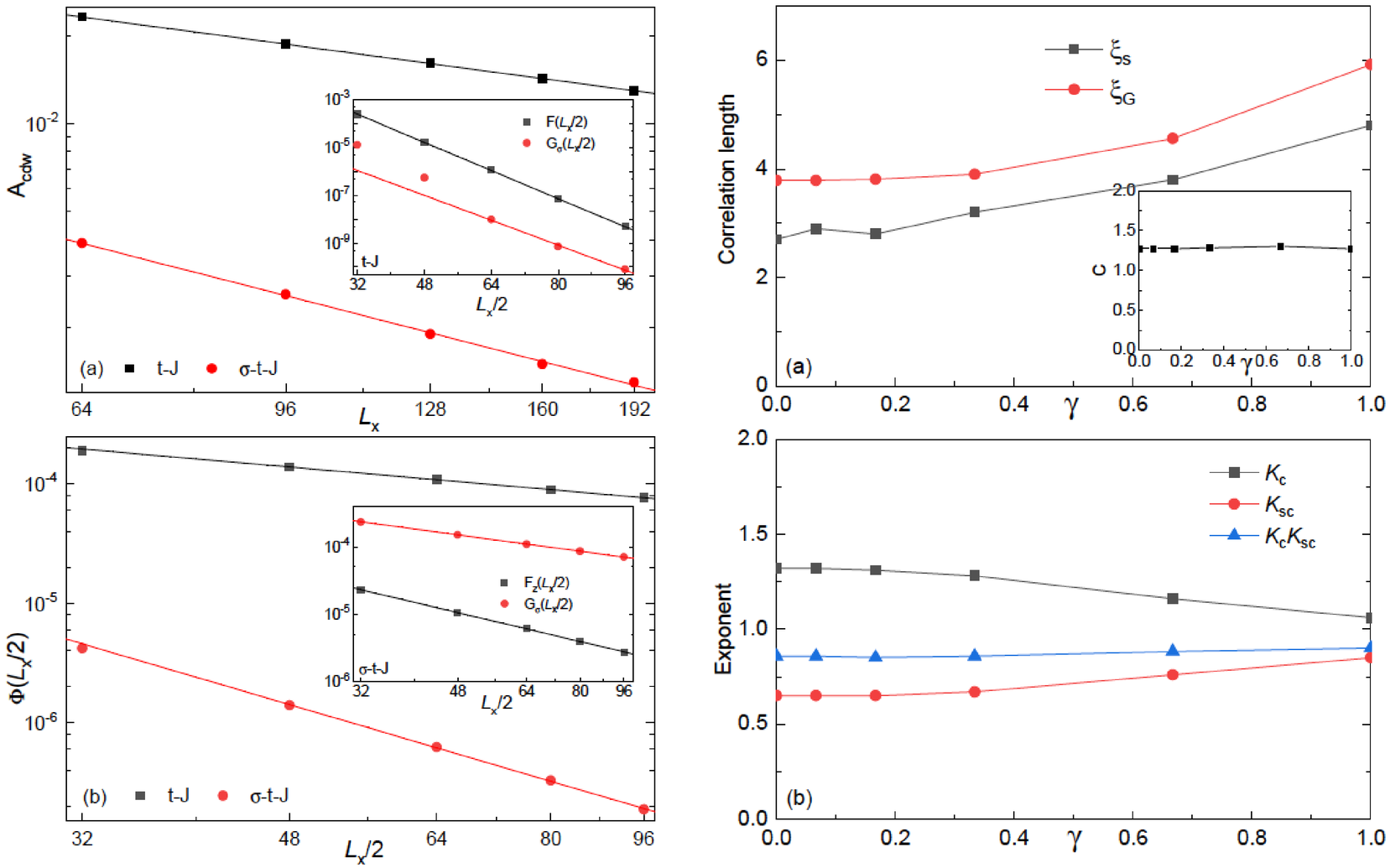


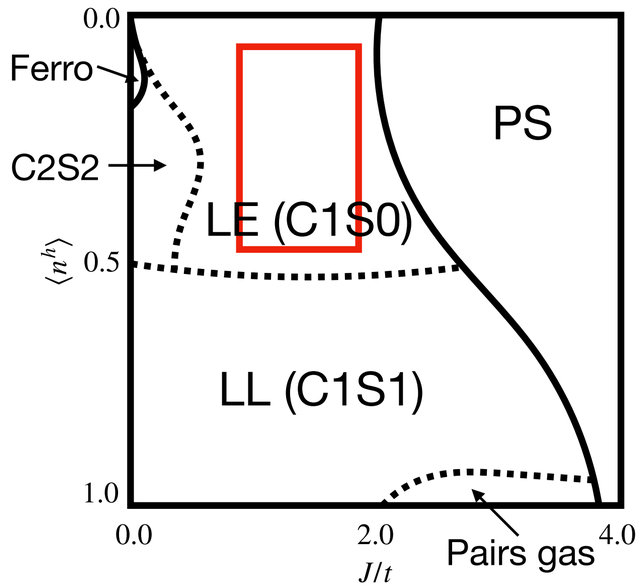




Phase diagram: two-leg ladder

Luther-Emery liquid: direct exploration

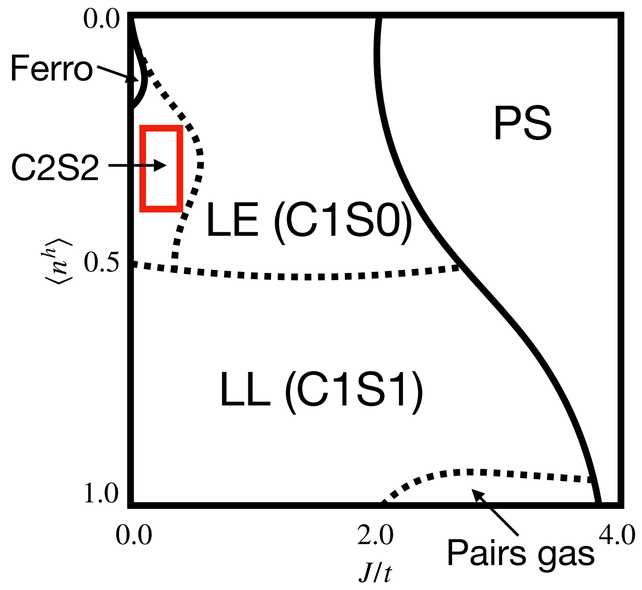




Phase diagram: two-leg ladder

C2S2 phase ?

- No charge pair
- Spin gap closed
- Müller, T., & Rice, T. (1998). Phase diagram of coupled ladders. *Physical Review B*, *58*(6), 3425–3437.



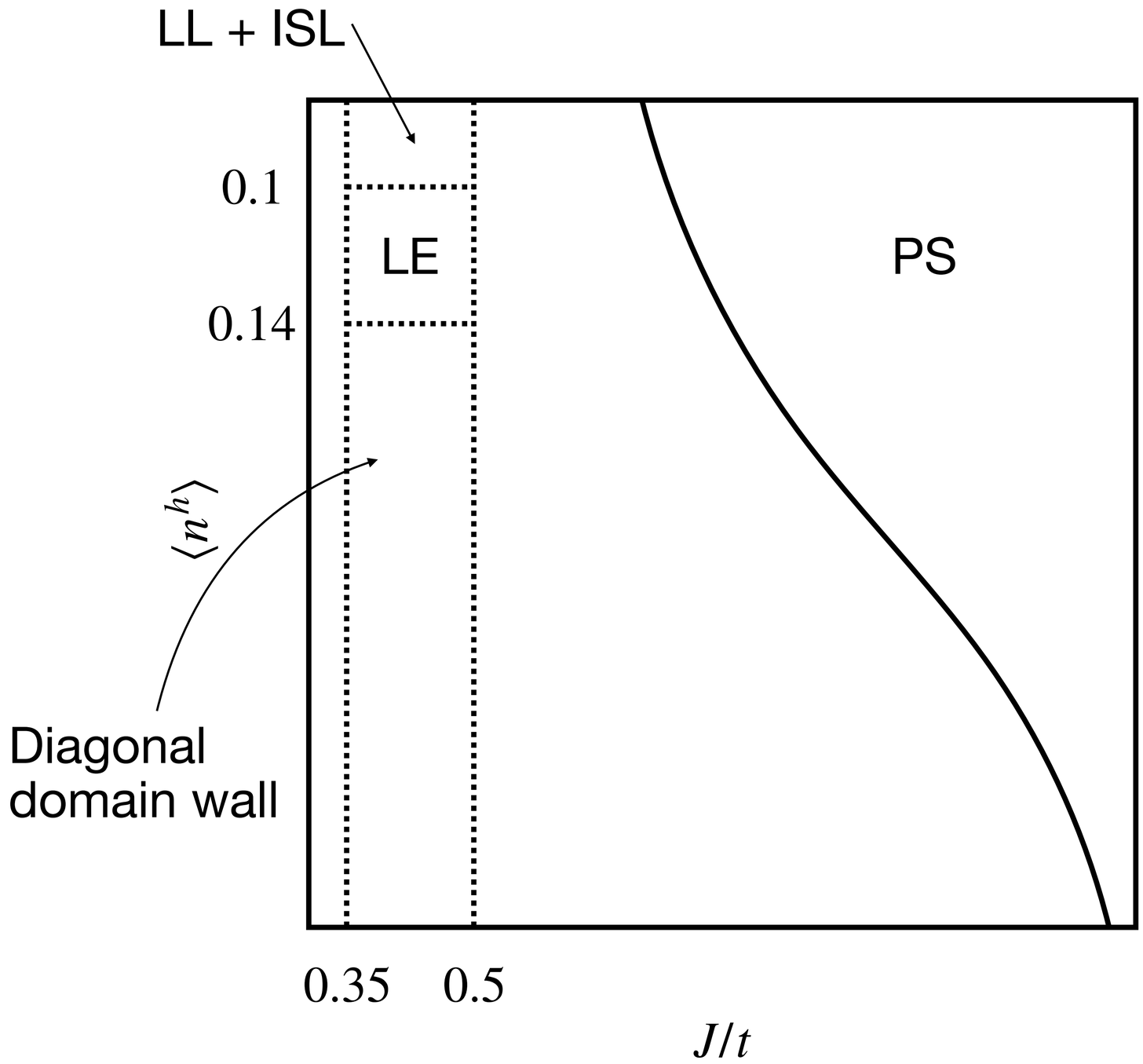
Phase diagram: three-leg ladder

Phase diagram: three-leg ladder

- Rice, T., Haas, S., Sigrist, M., & Zhang, F. C. (1997). Lightly doped three-leg ladders: An analog for the underdoped cuprates. *Physical Review B*, *56*(22), 14655–14667.

- White, S. R., & Scalapino, D. (1998). Ground-state properties of the doped three-leg ladder. *Physical Review B*, *57*(5), 3031–3037.

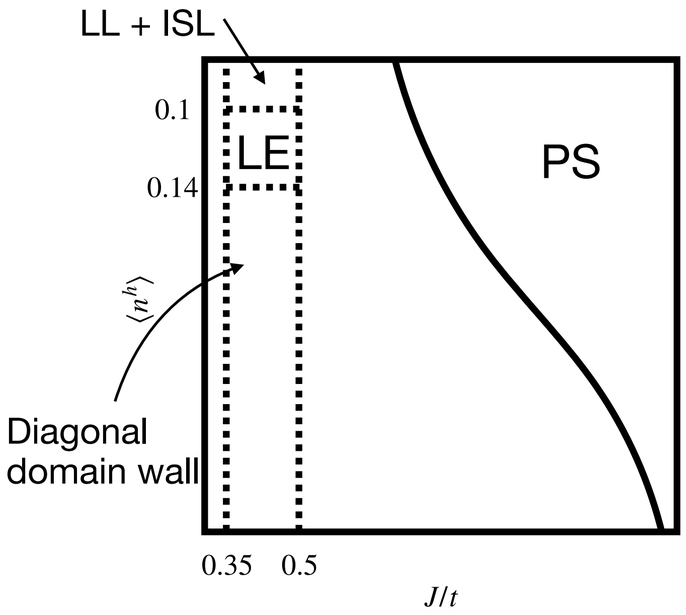
Phase diagram: three-leg ladder



Phase diagram: three-leg ladder

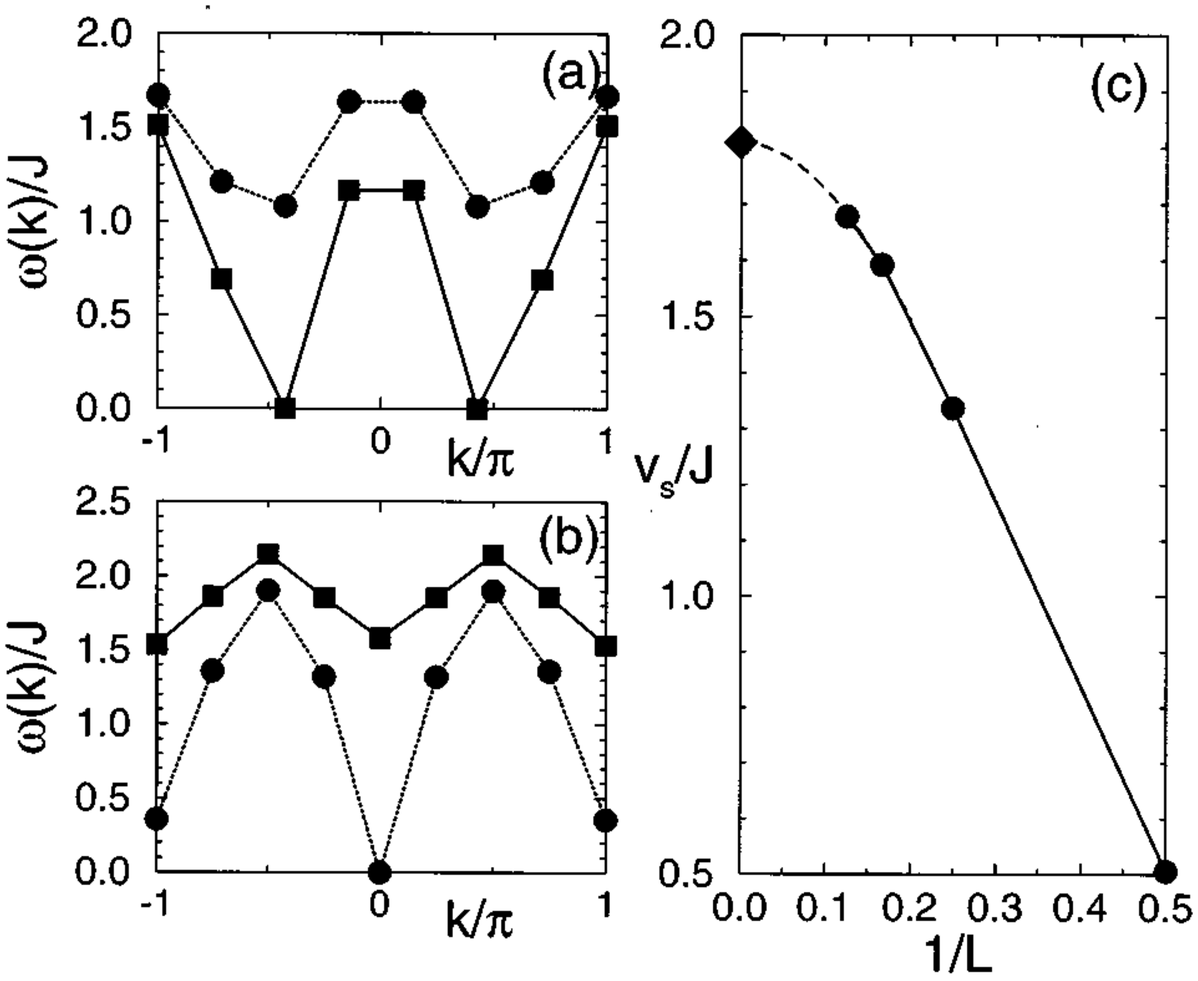
Y direction parity

- OBC at y direction
- Two even parity modes
- One odd parity mode

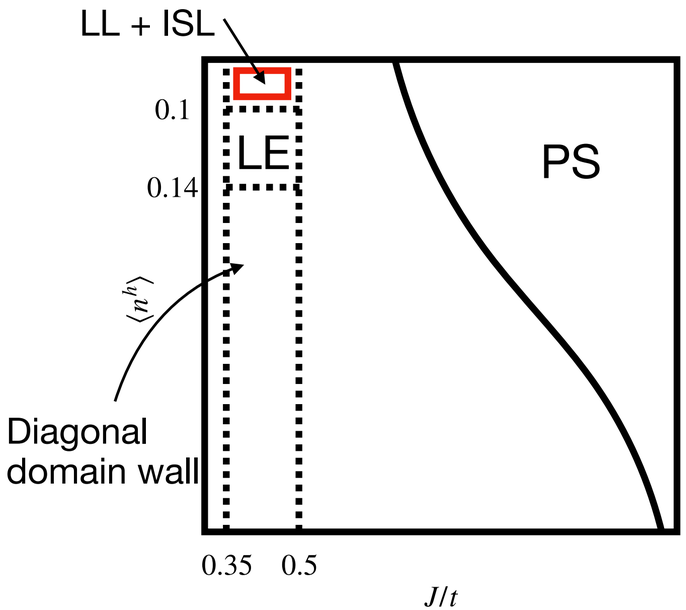


Phase diagram: three-leg ladder

LL + ISL: spinon excitation spectrum

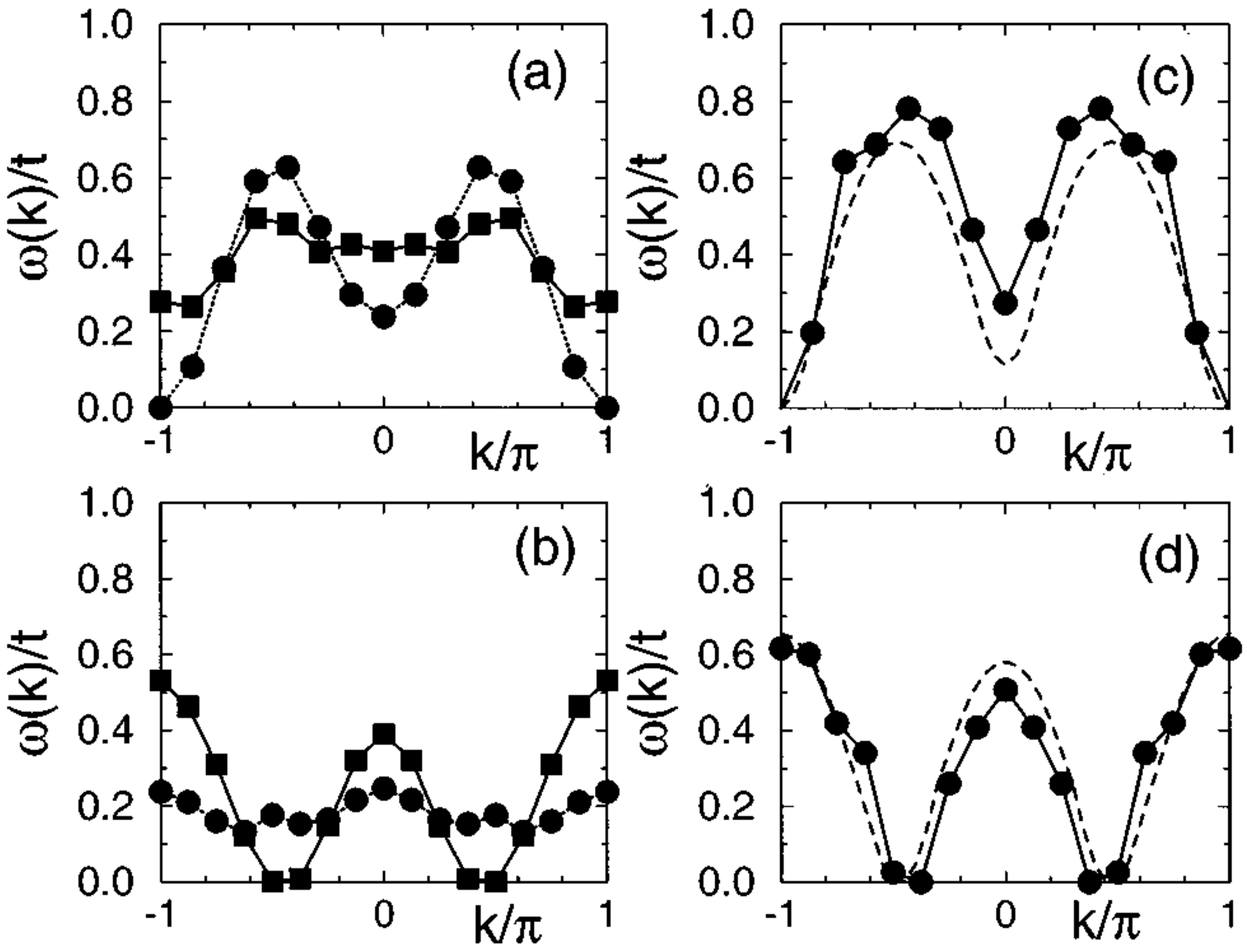


- Odd parity

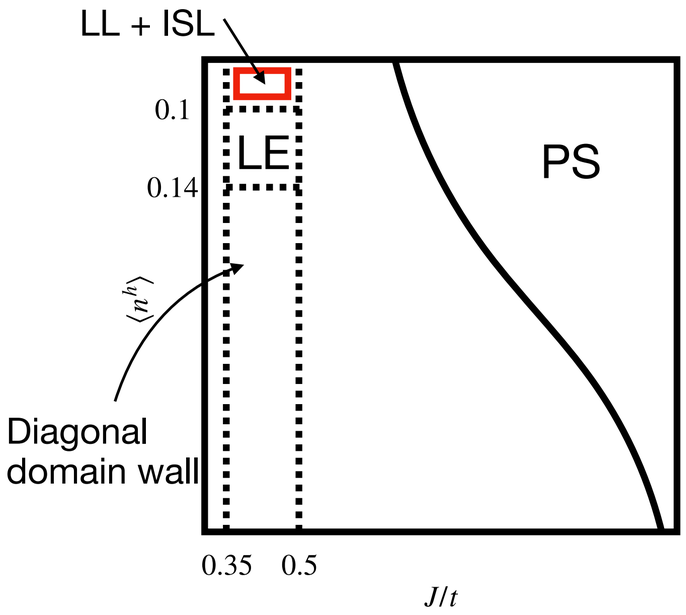


Phase diagram: three-leg ladder

LL + ISL: holon excitation spectrum

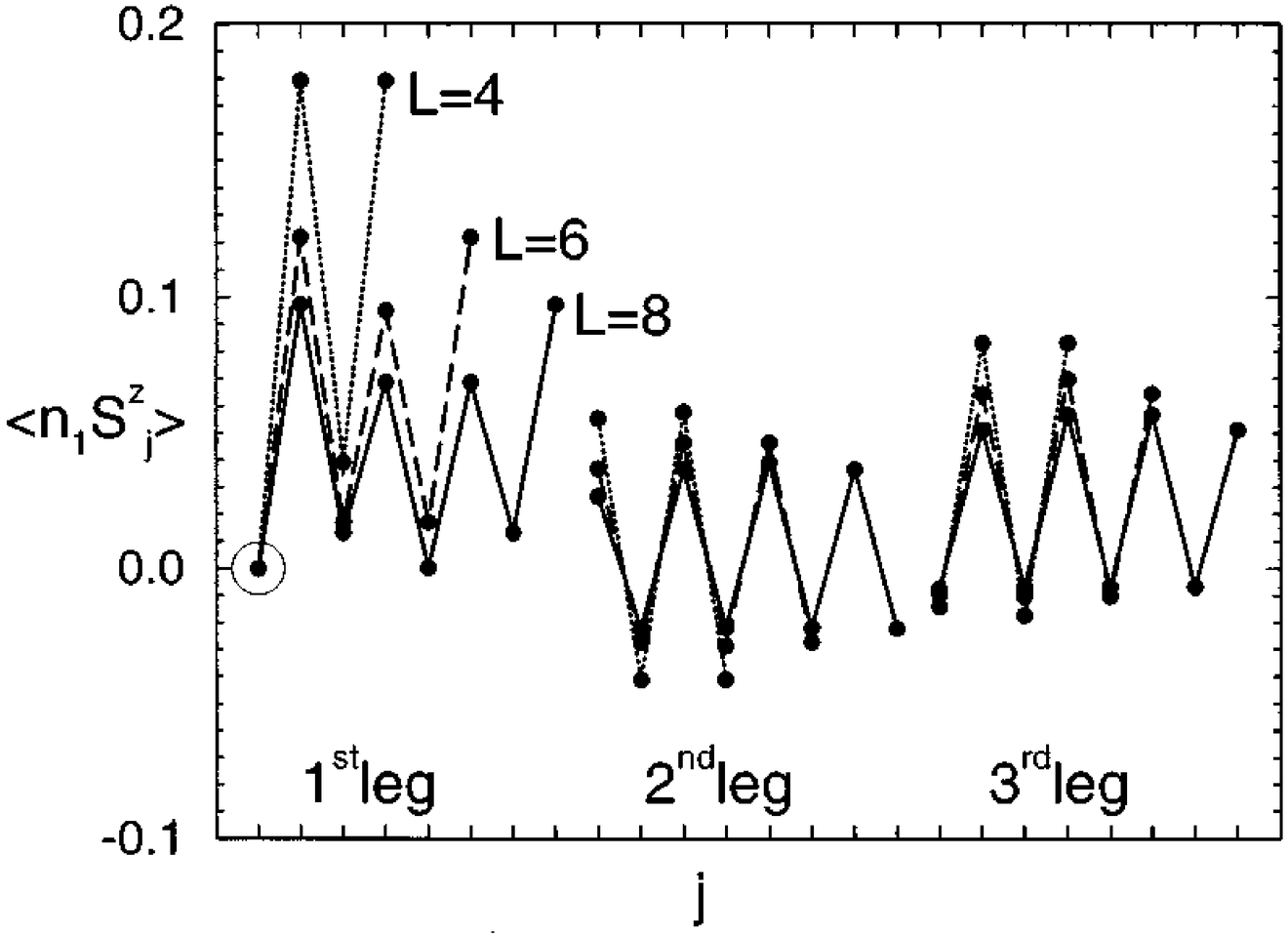


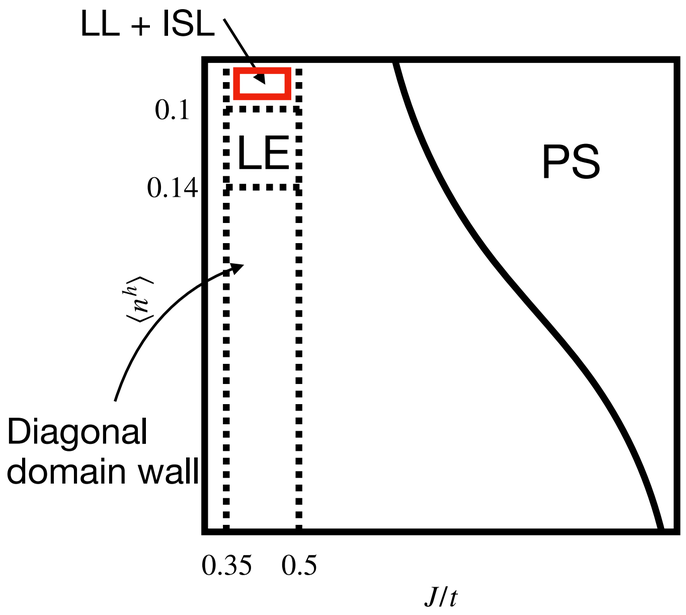
- Even parity



Phase diagram: three-leg ladder

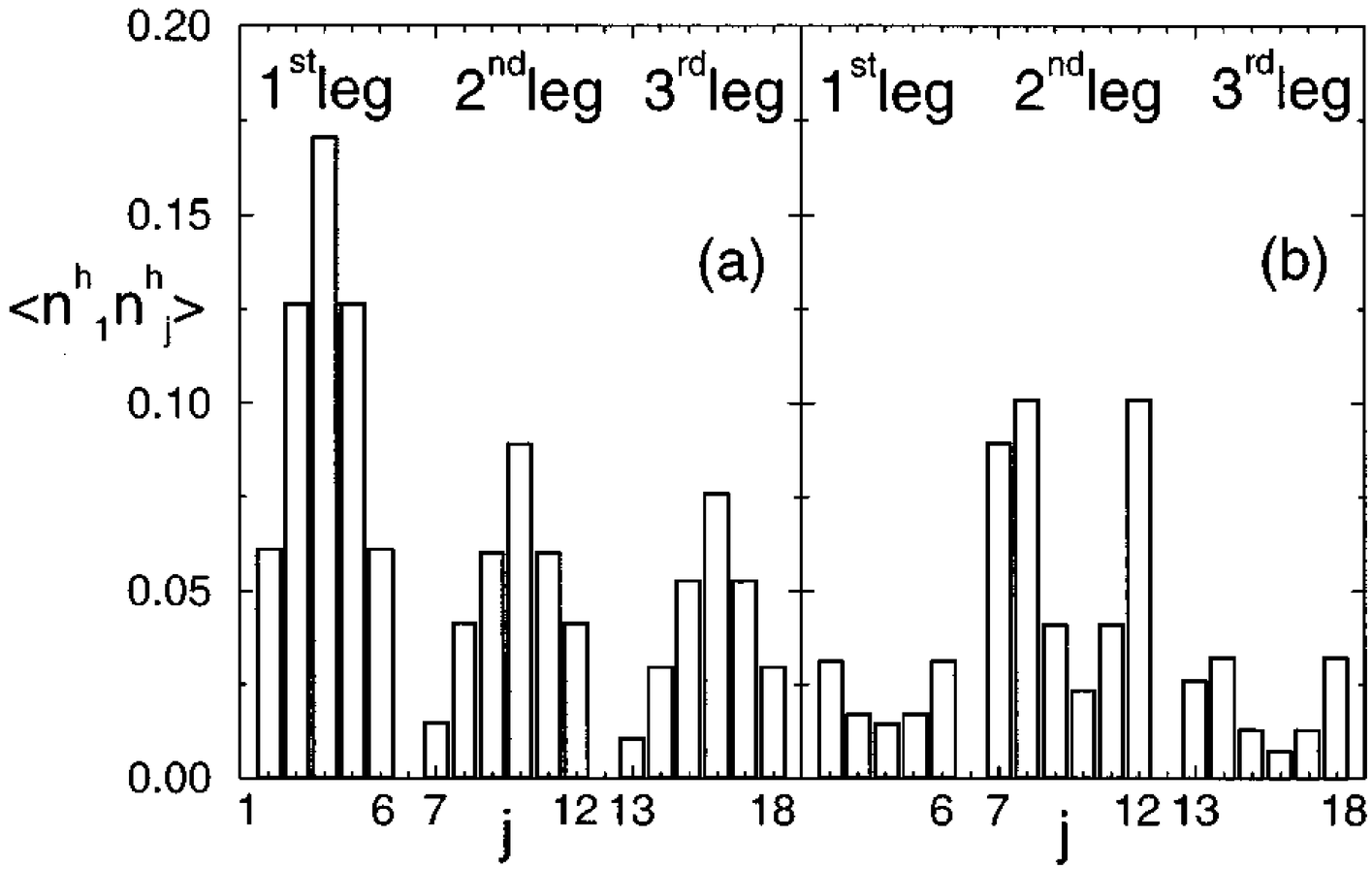
LL + ISL: spin-charge separation

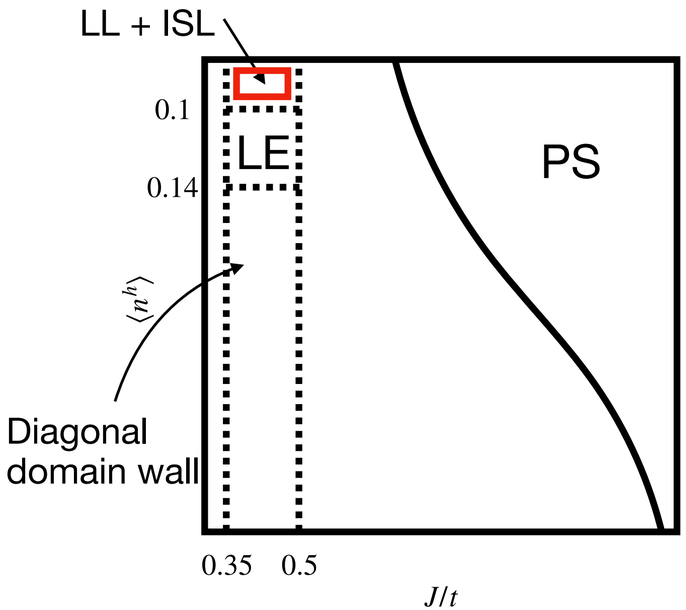




Phase diagram: three-leg ladder

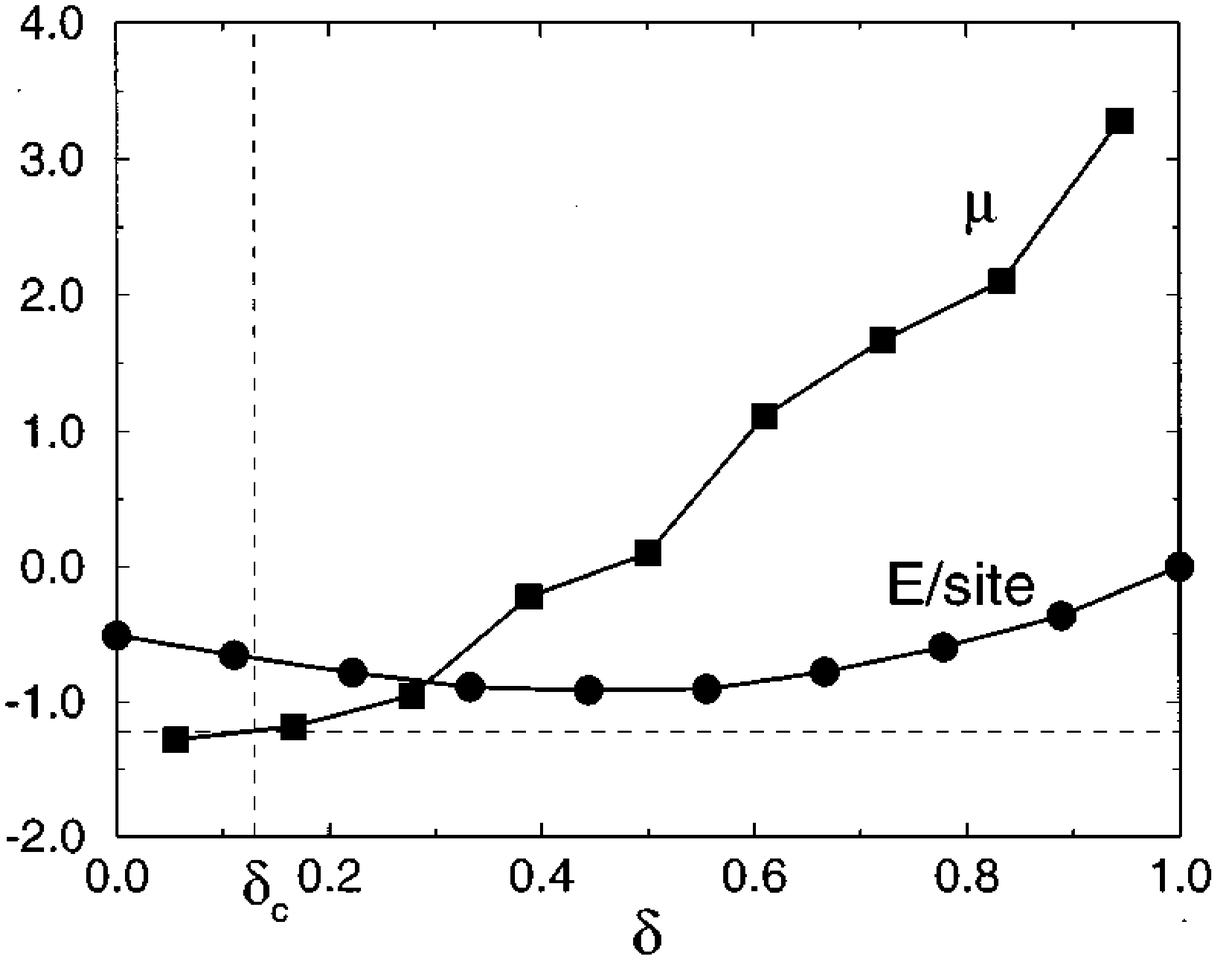
LL + ISL: no pairing



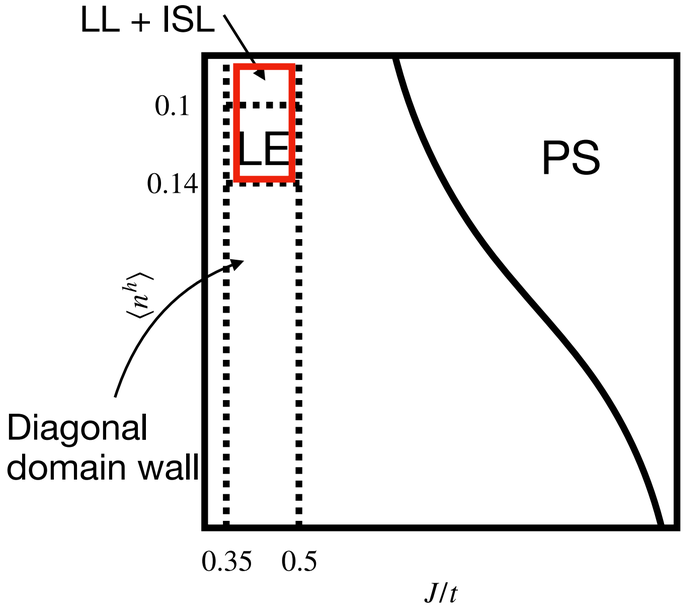


Phase diagram: three-leg ladder

LL + ISL to LE: chemical potential vs excited state (LE) energy

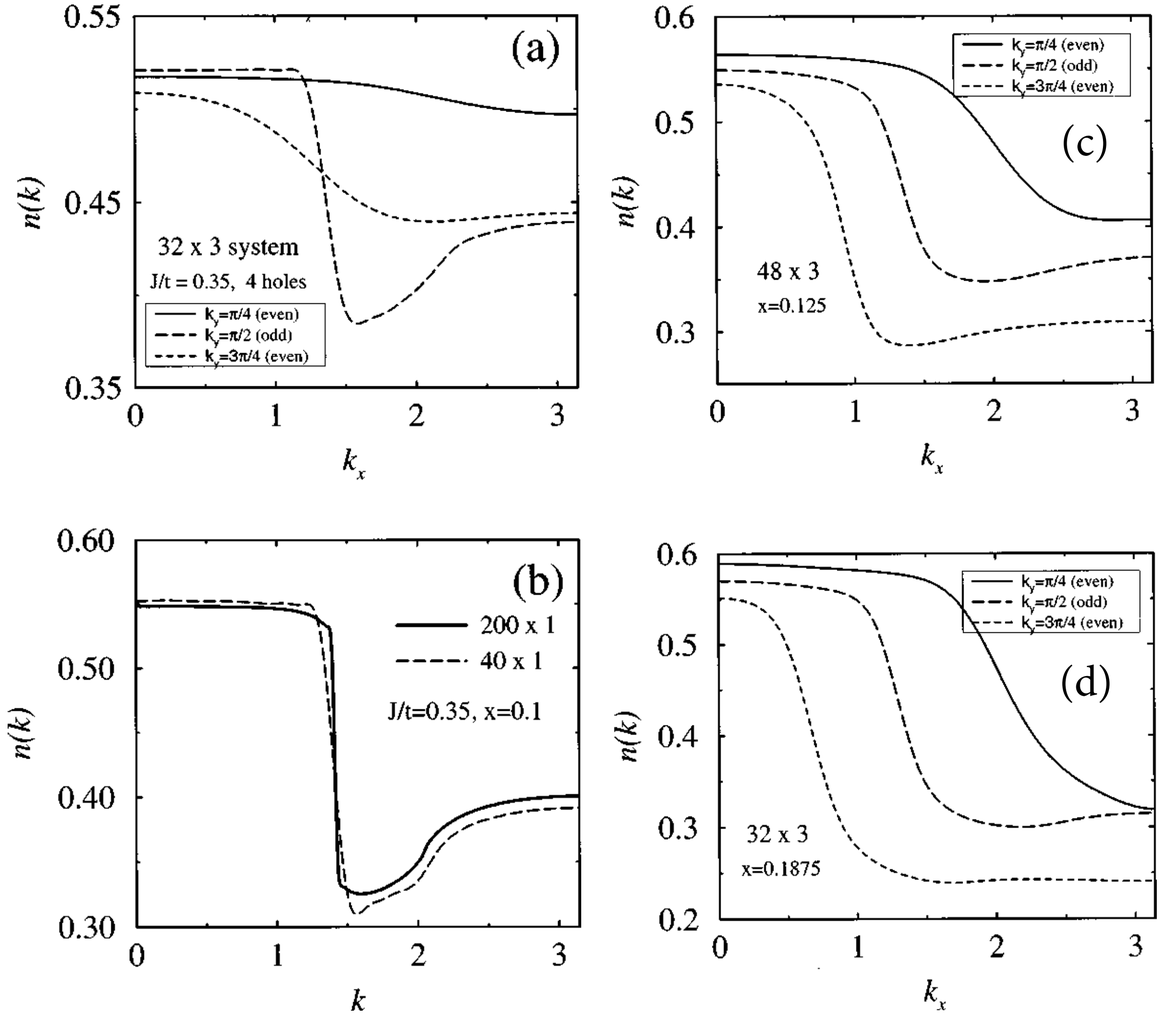


$$\delta_c \sim 0.13$$

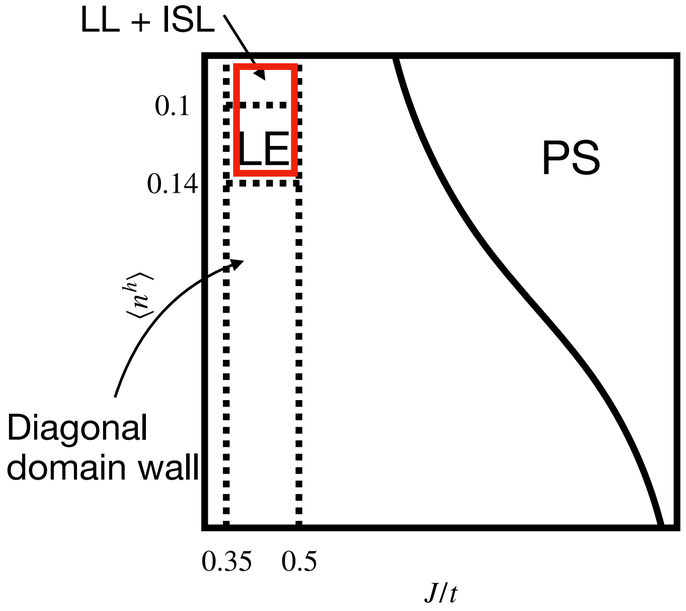


Phase diagram: three-leg ladder

LL + ISL to LE: $n(k)$ distribution

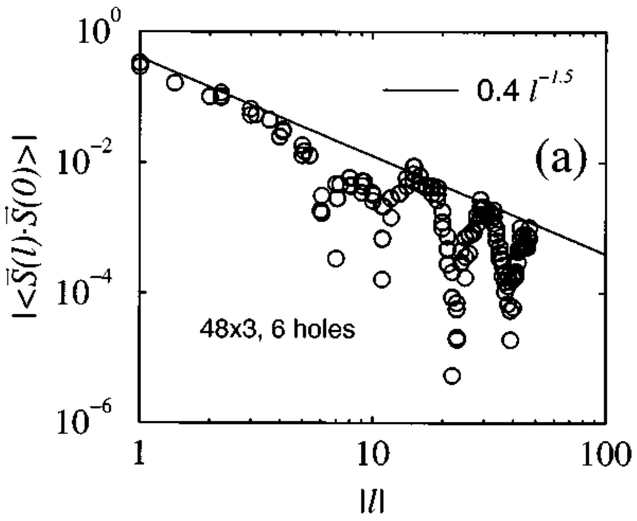


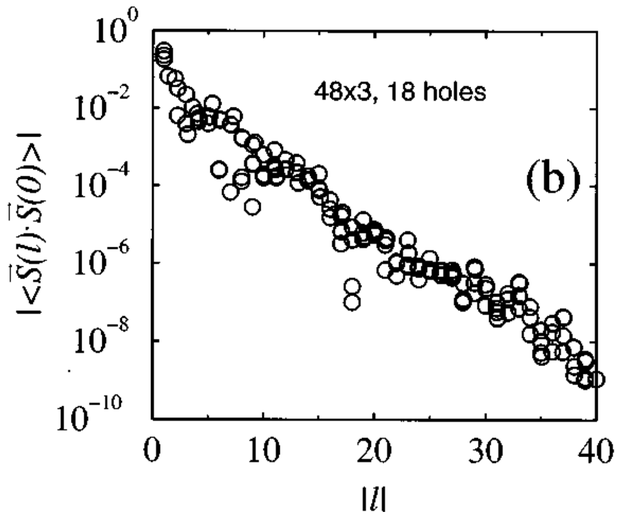
- LL + ISL: Luttinger theorem violation



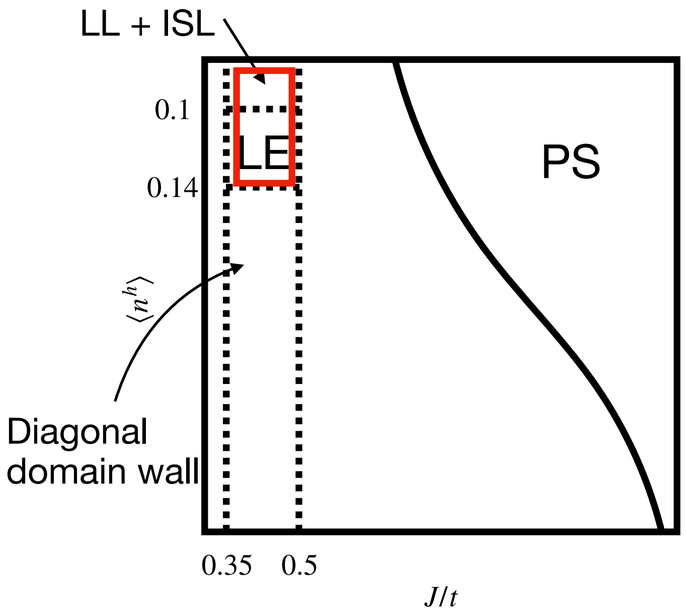
Phase diagram: three-leg ladder

LL + ISL to LE: spin-spin correlation



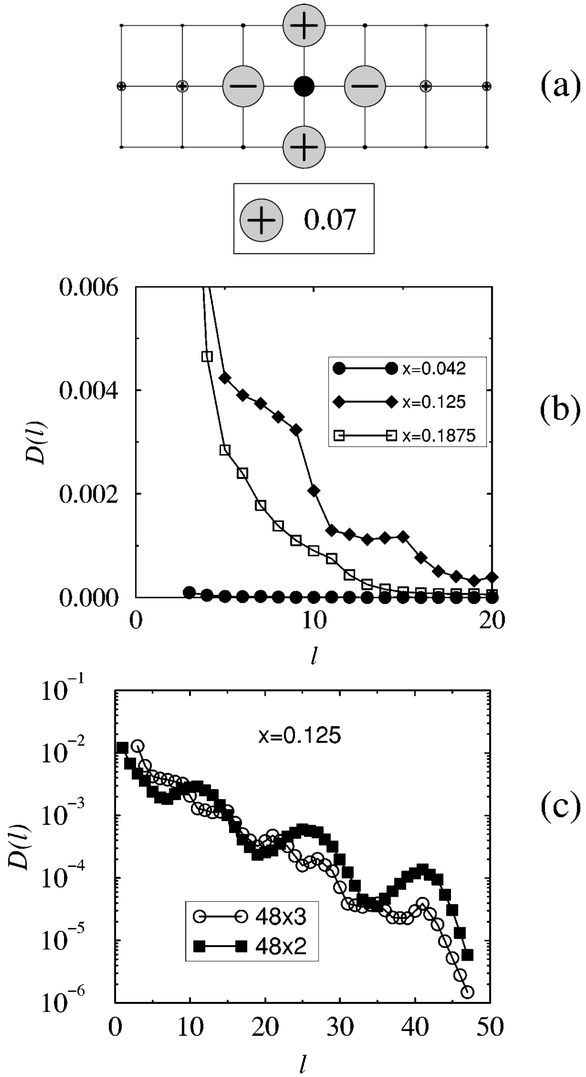


- Spin gap open

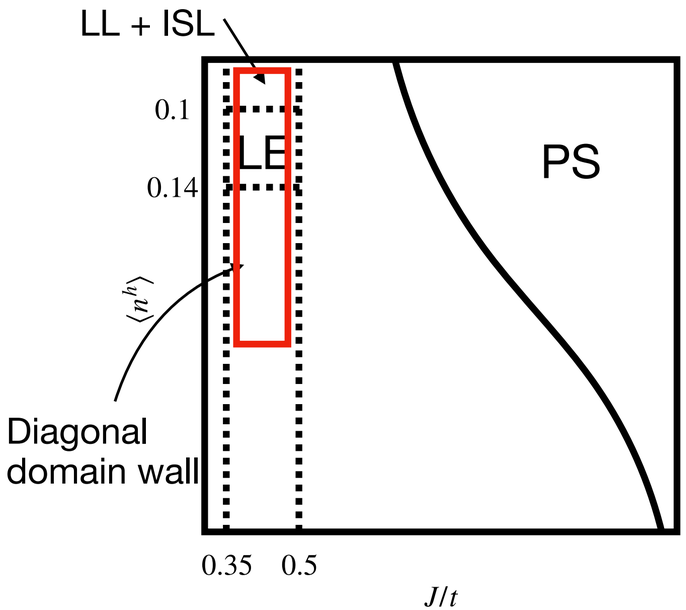


Phase diagram: three-leg ladder

D-wave pairing

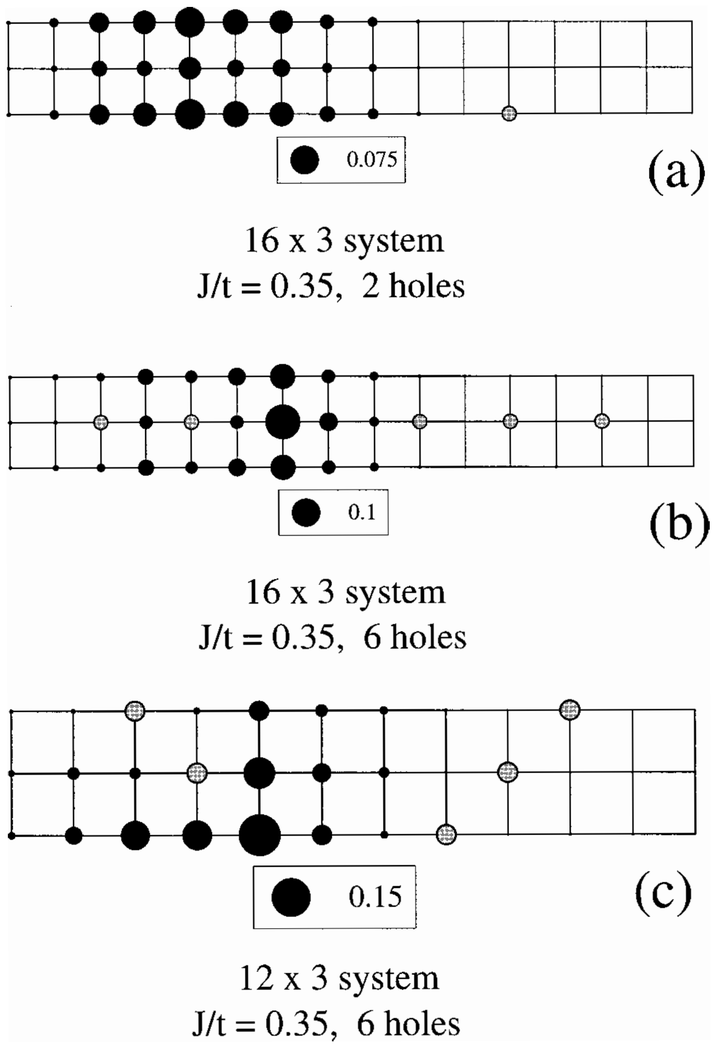


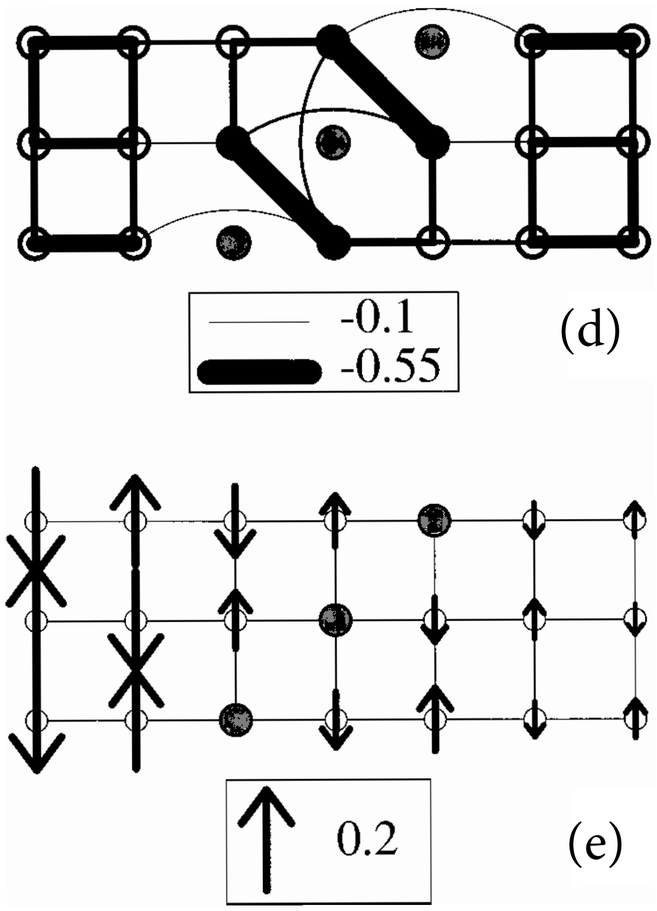
- SC dome?

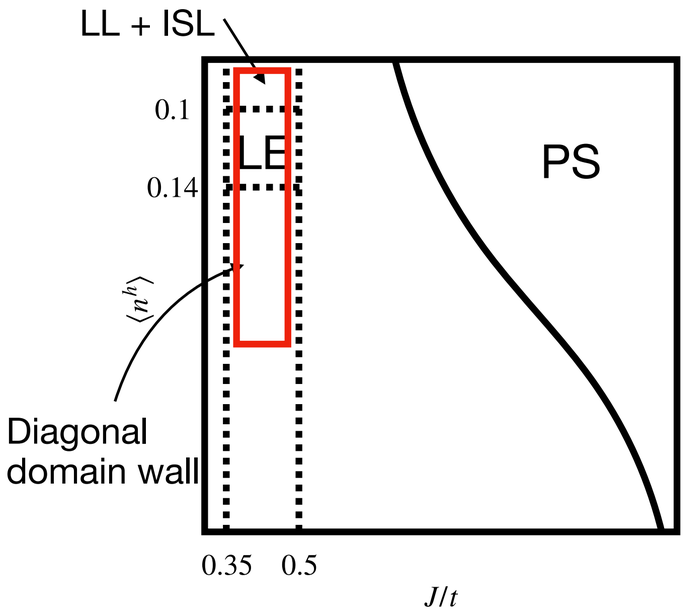


Phase diagram: three-leg ladder

Domain walls





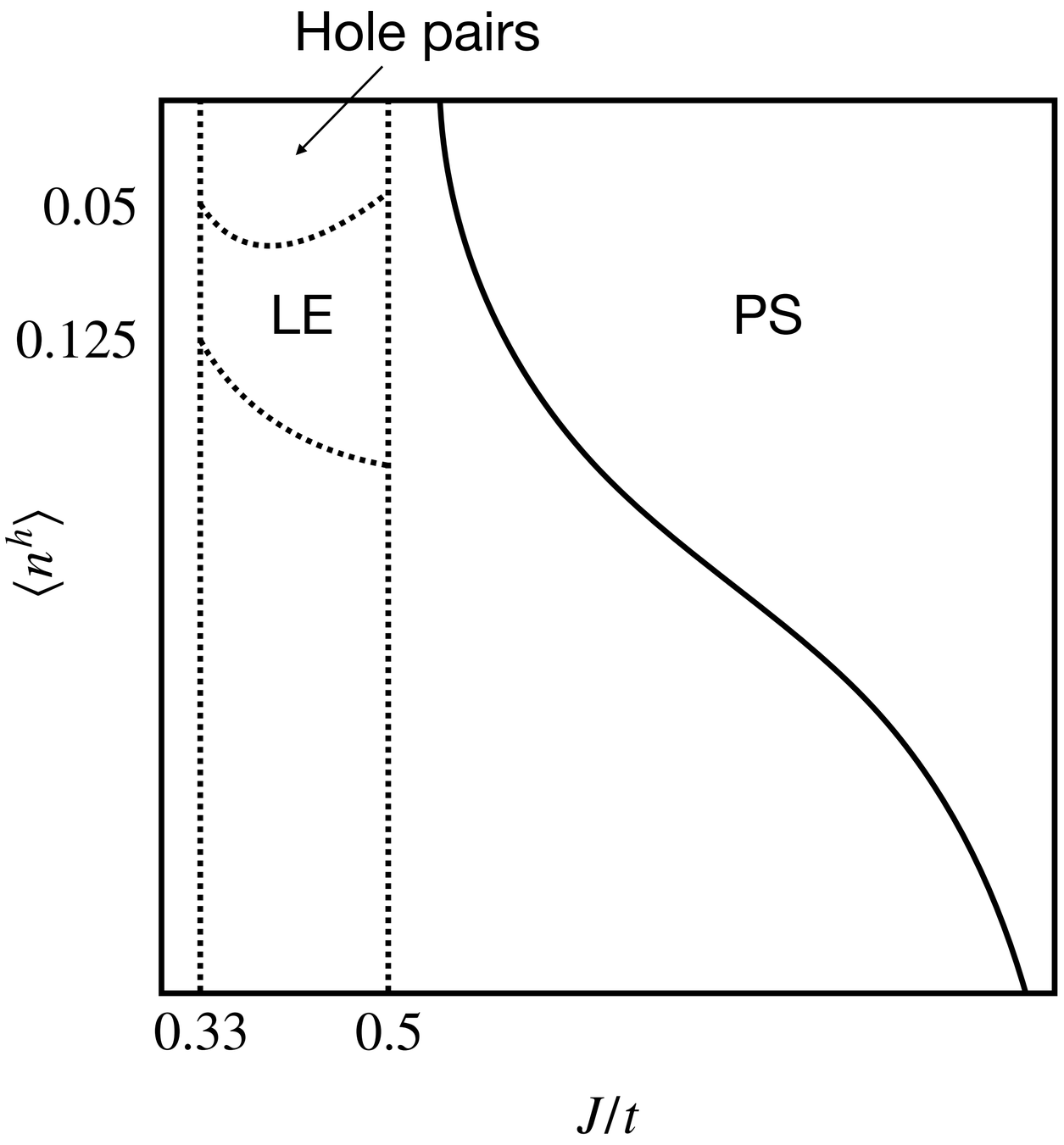


Phase diagram: four-leg ladder

Phase diagram: four-leg ladder

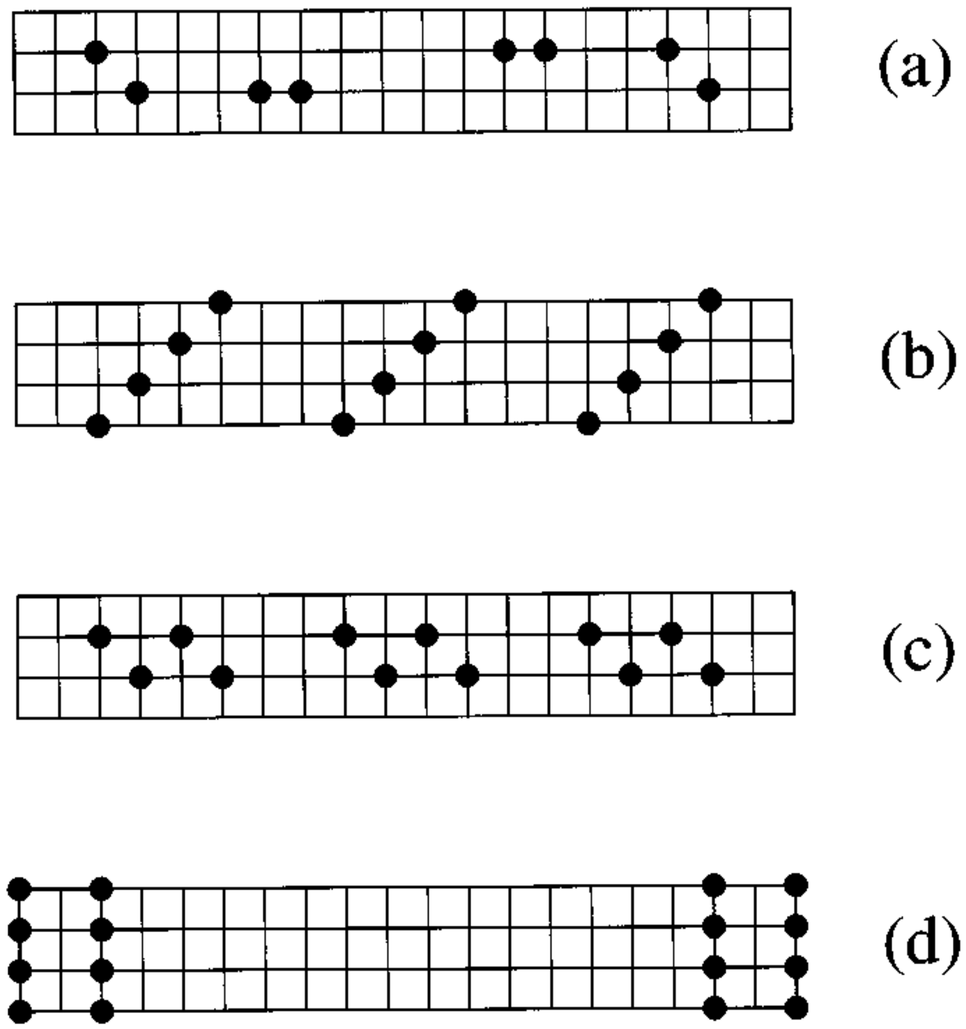
- White, S. R., & Scalapino, D. (1997). Ground states of the doped four-leg t-J ladder. *Physical Review B*, *55*(22), R14701–R14704.
- White, S. R., & Scalapino, D. J. (1999). Competition between stripes and pairing in a t-t'-J model. *Physical Review B*, *60*(2), R753–R756.
- Dodaro, J. F., Jiang, H.-C., & Kivelson, S. A. (2017). Intertwined order in a frustrated four-leg t-J cylinder. *Physical Review B*, *95*(15), 155116.
- Jiang, H.-C., Weng, Z.-Y., & Kivelson, S. A. (2018). Superconductivity in the doped t-J model: Results for four-leg cylinders. *Physical Review B*, *98*(14), 140505.
- Jiang, Y.-F., Zaanen, J., Devereaux, T. P., & Jiang, H.-C. (2019). Ground state phase diagram of the doped Hubbard model on the 4-leg cylinder, 1907.11728.

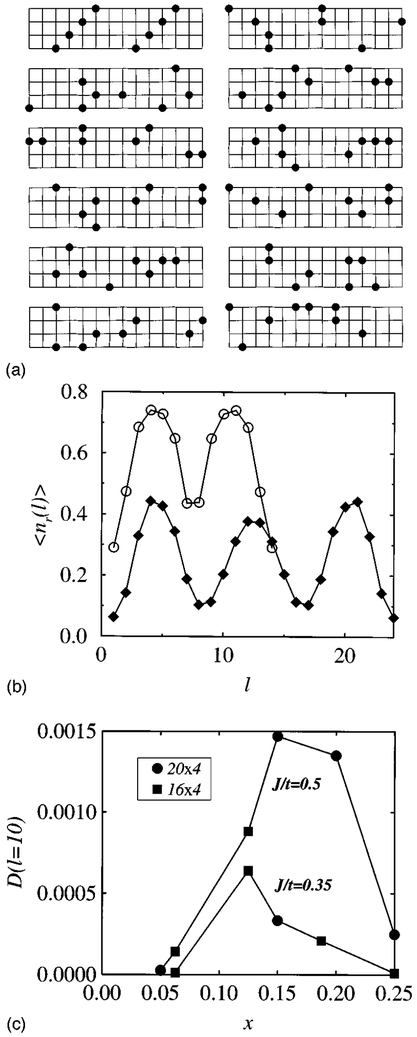
Phase diagram: four-leg ladder

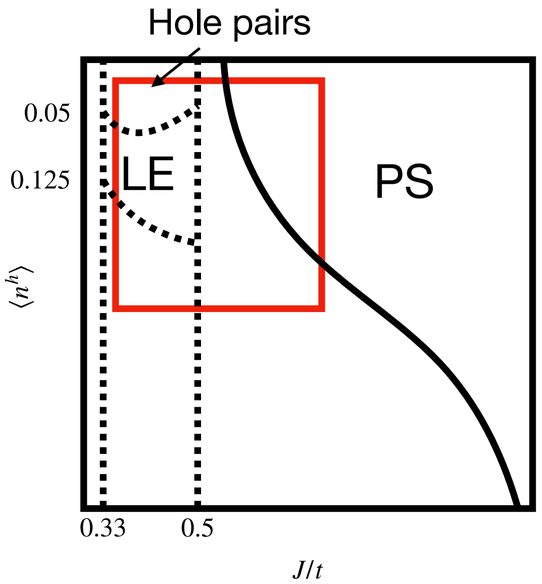


Phase diagram: four-leg ladder

Hole profile and pairing

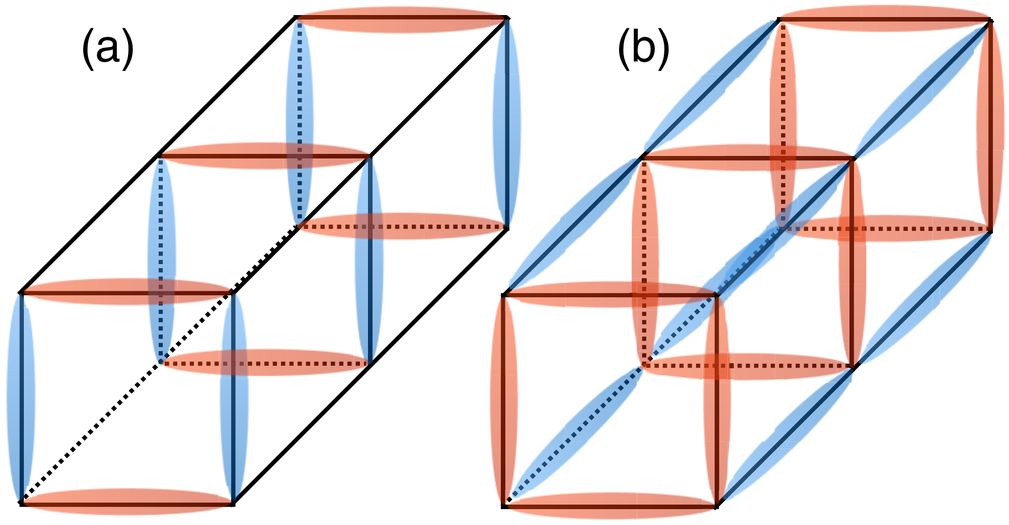




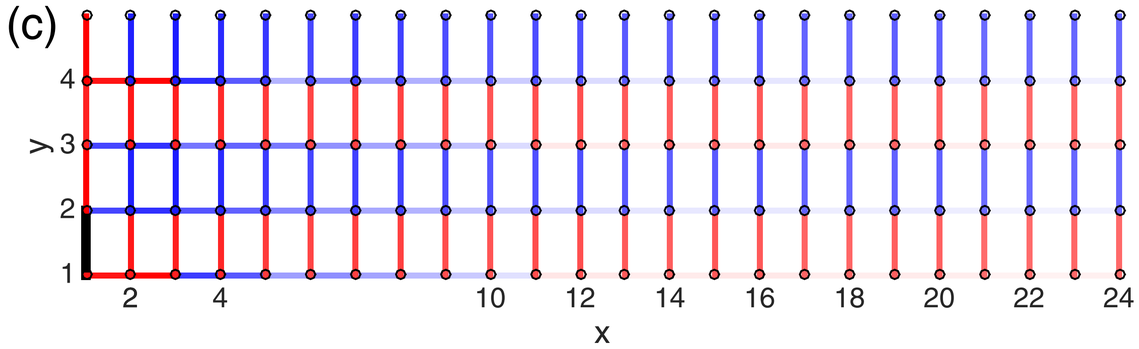


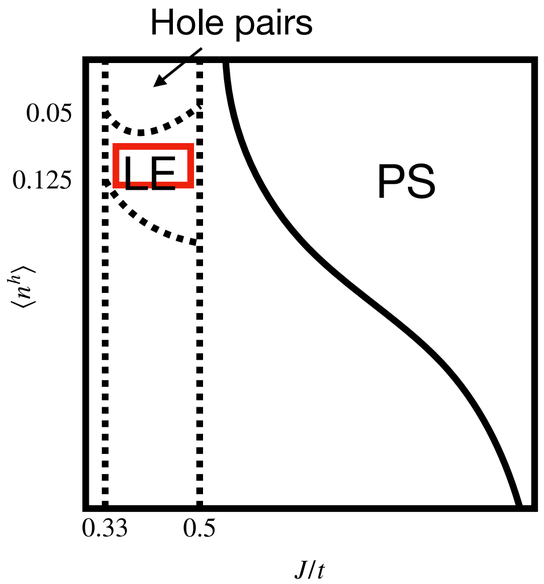
Phase diagram: four-leg ladder

Pairing symmetry



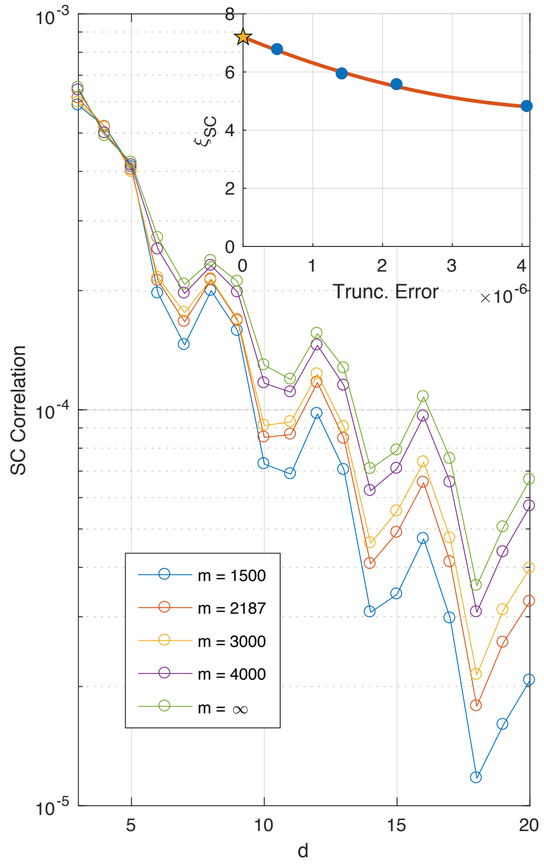
d-wave **d-wave-like**

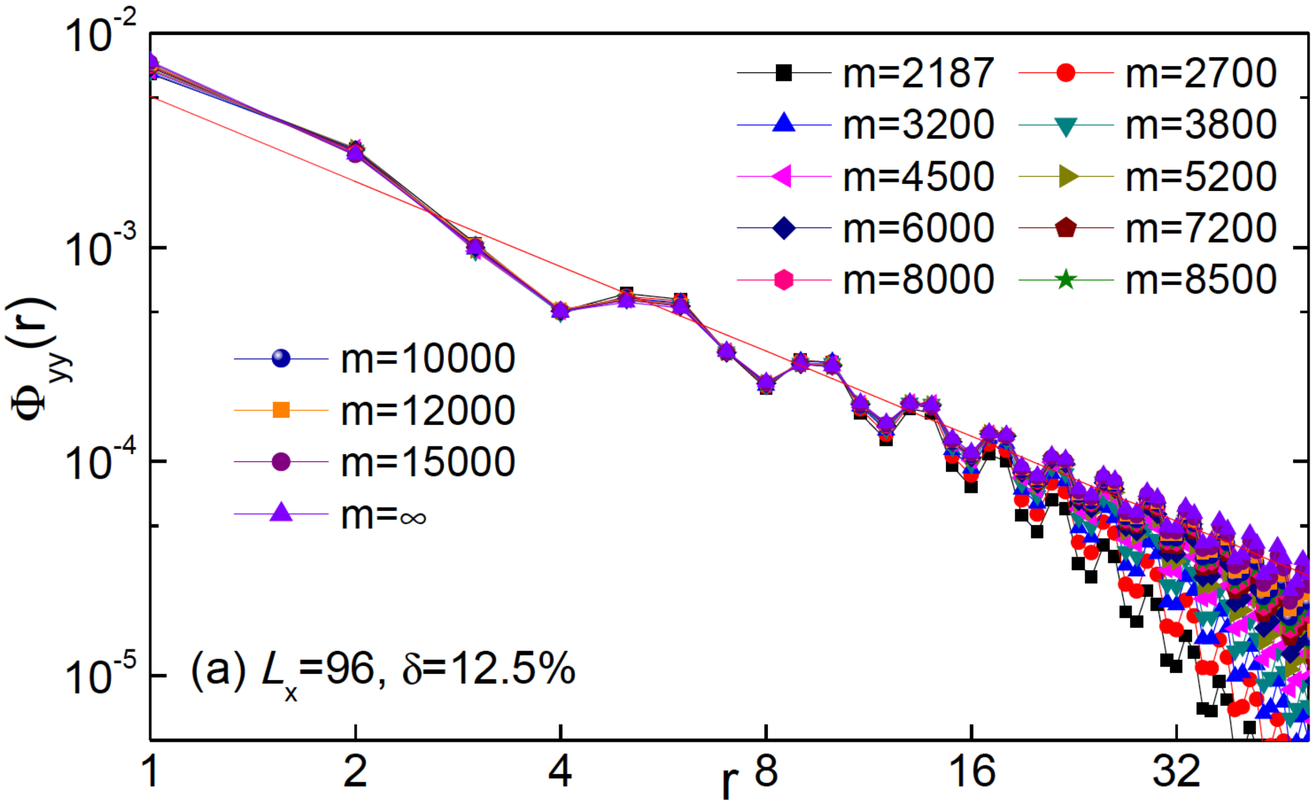


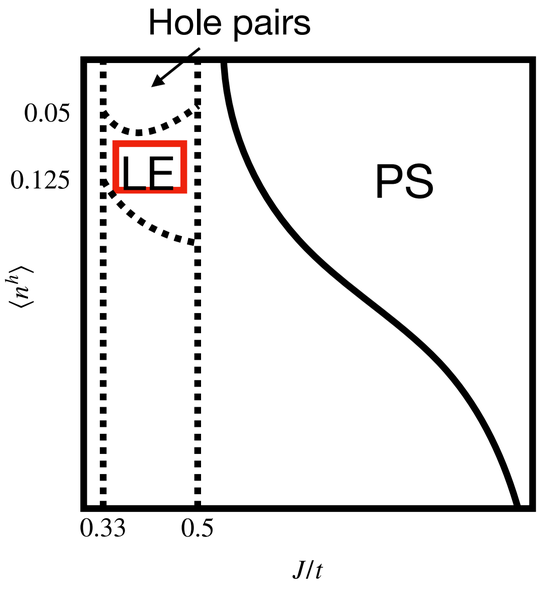


Phase diagram: four-leg ladder

Miss the LE phase

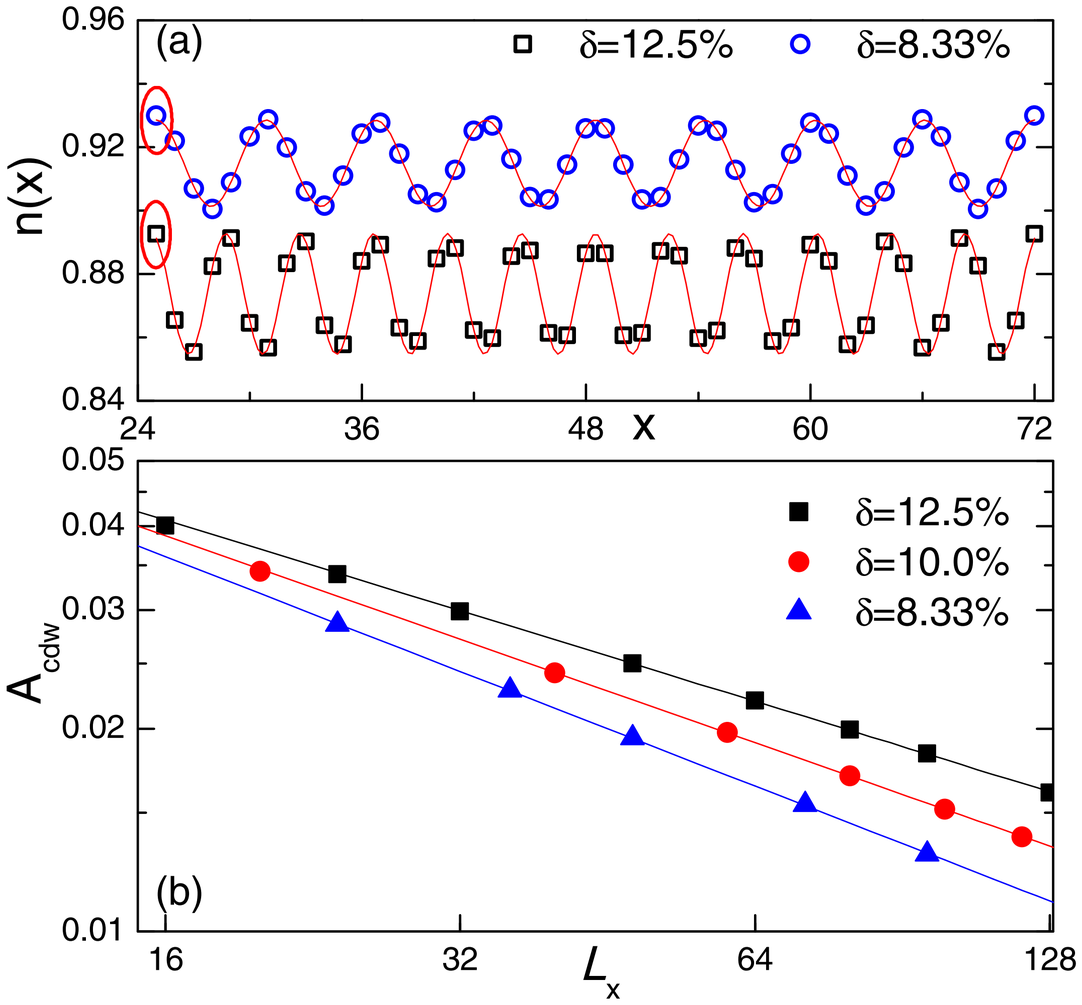


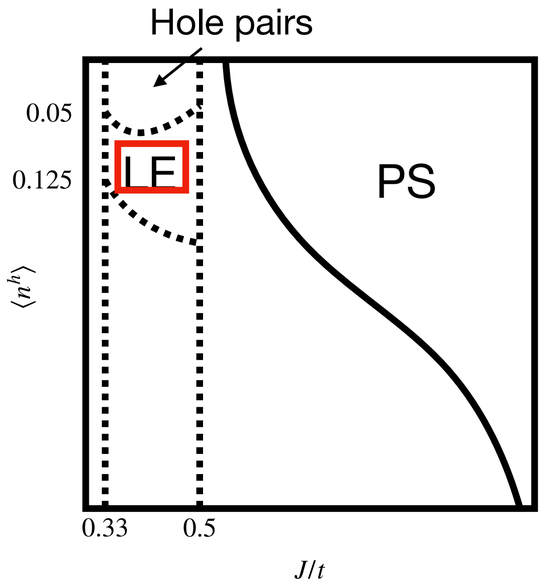




Phase diagram: four-leg ladder

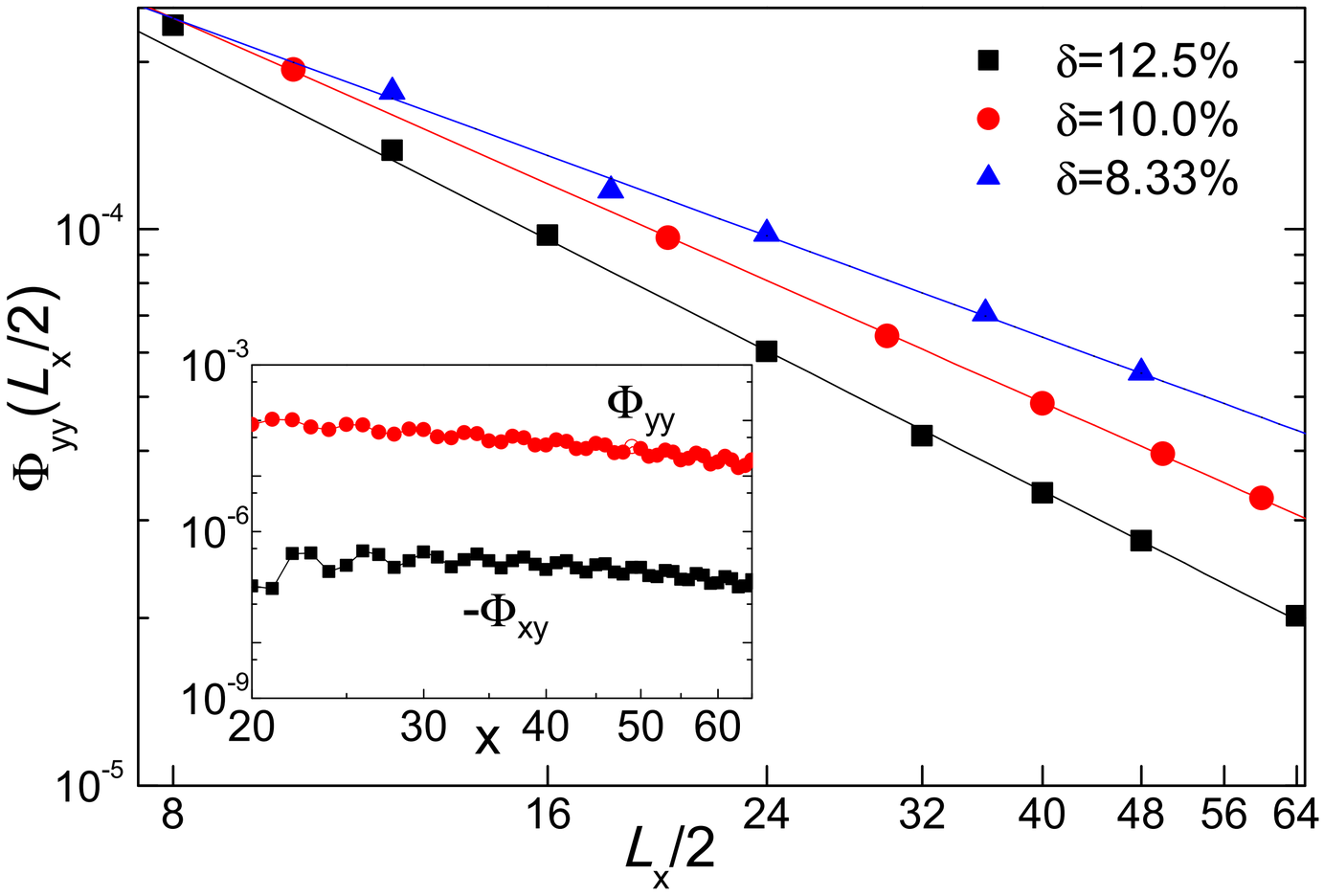
LE phase: CDW

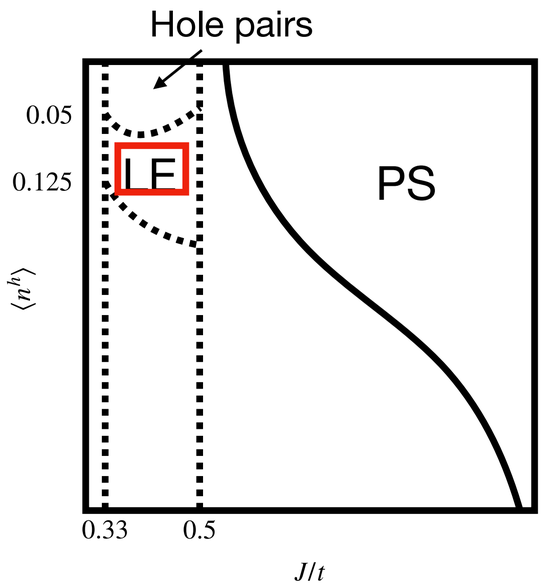




Phase diagram: four-leg ladder

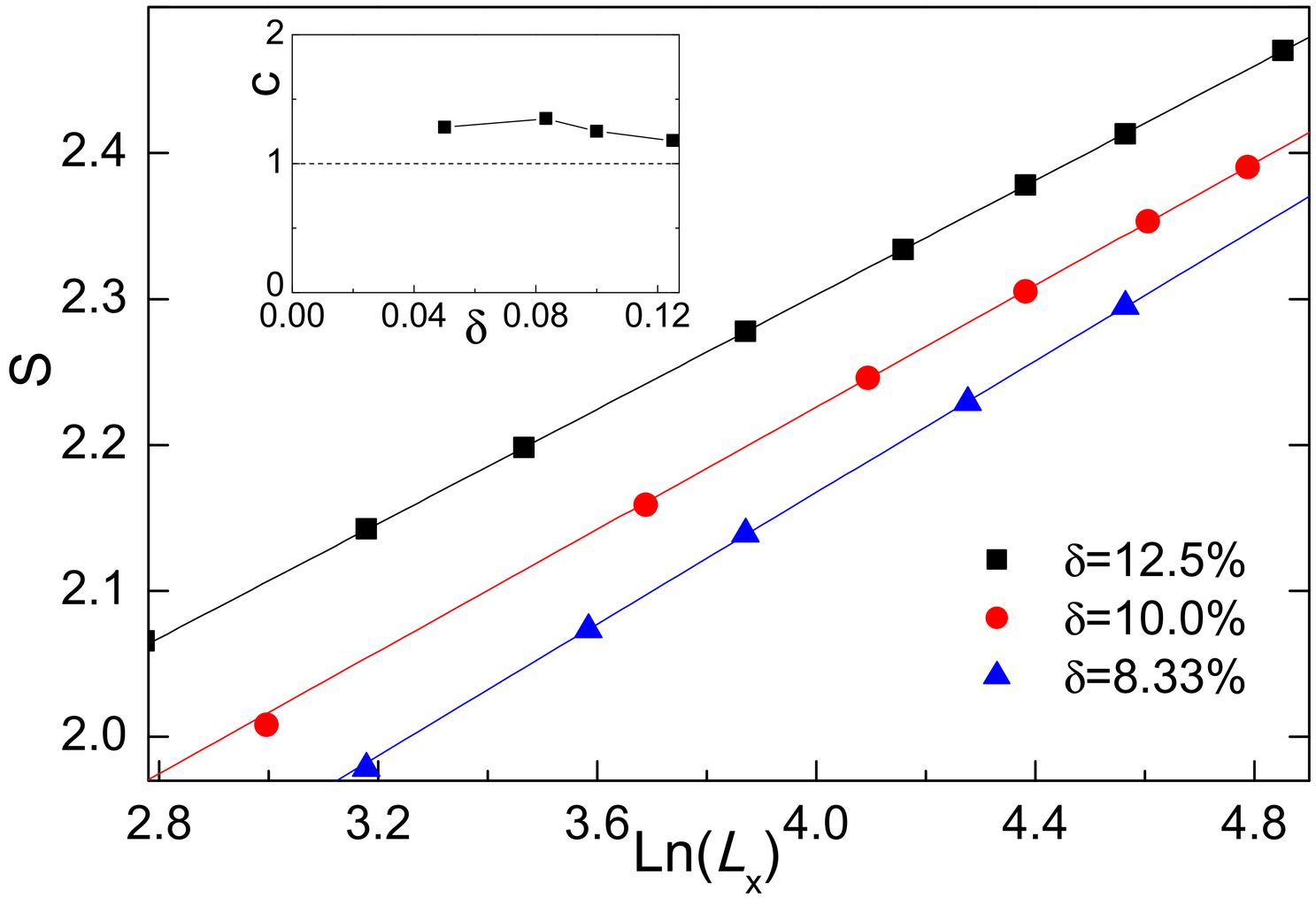
LE phase: SC

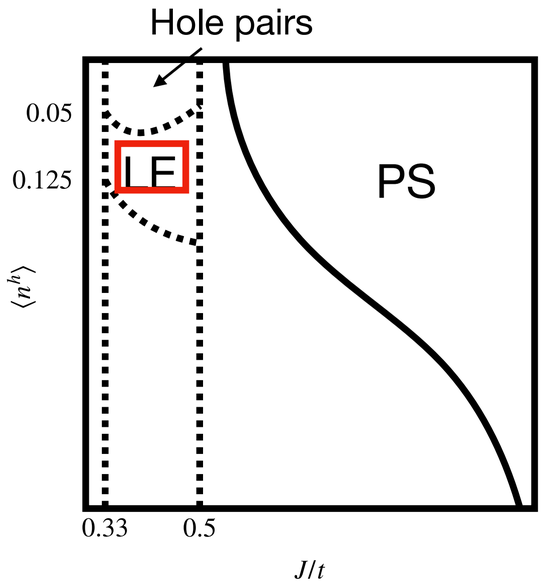




Phase diagram: four-leg ladder

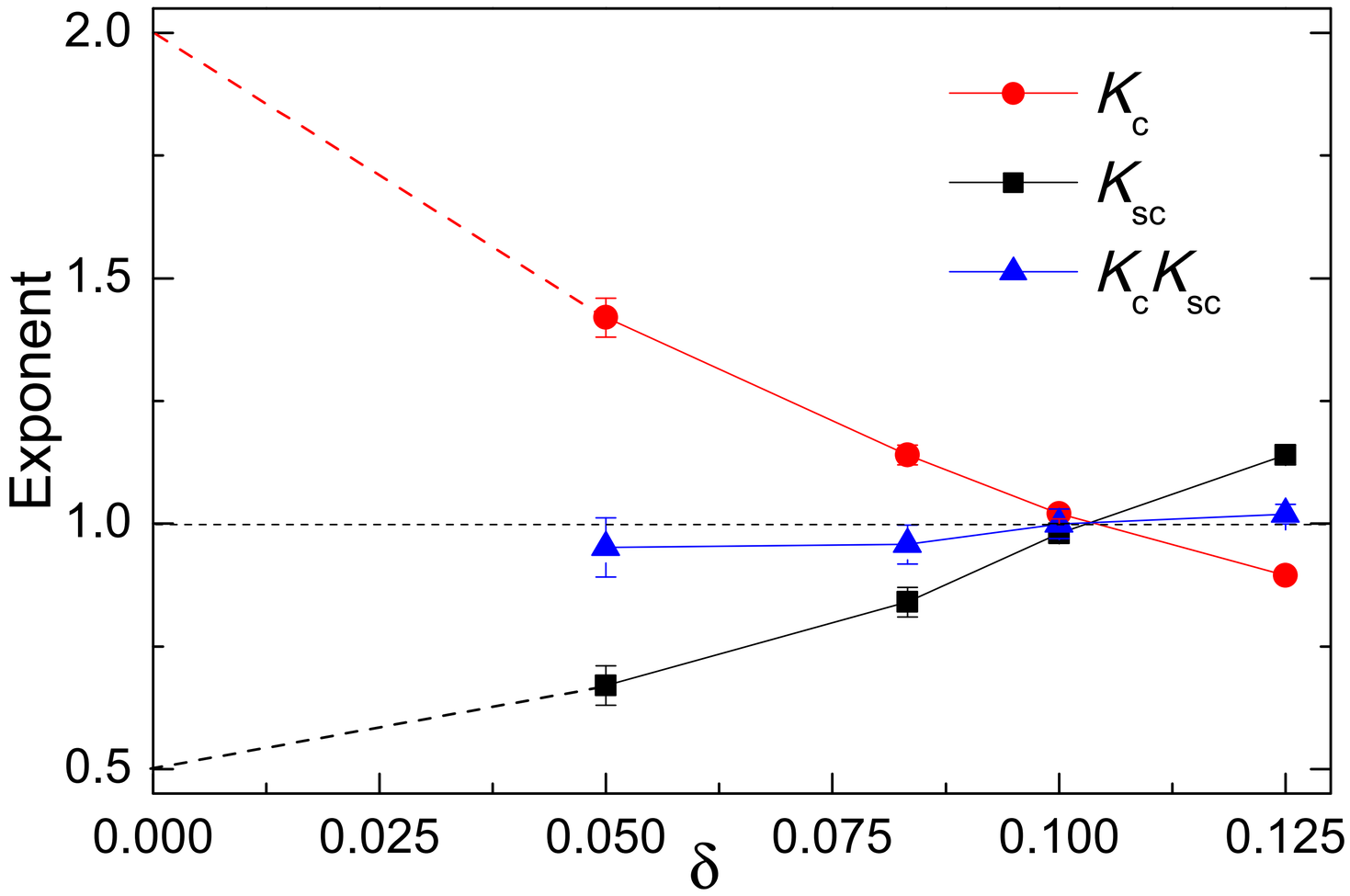
LE phase: Entanglement entropy

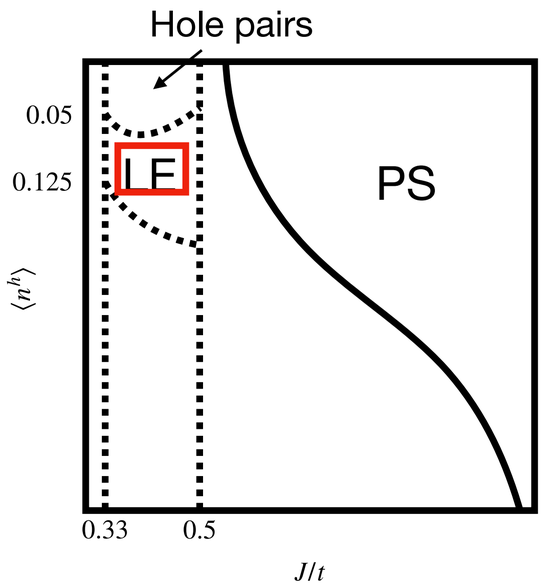




Phase diagram: four-leg ladder

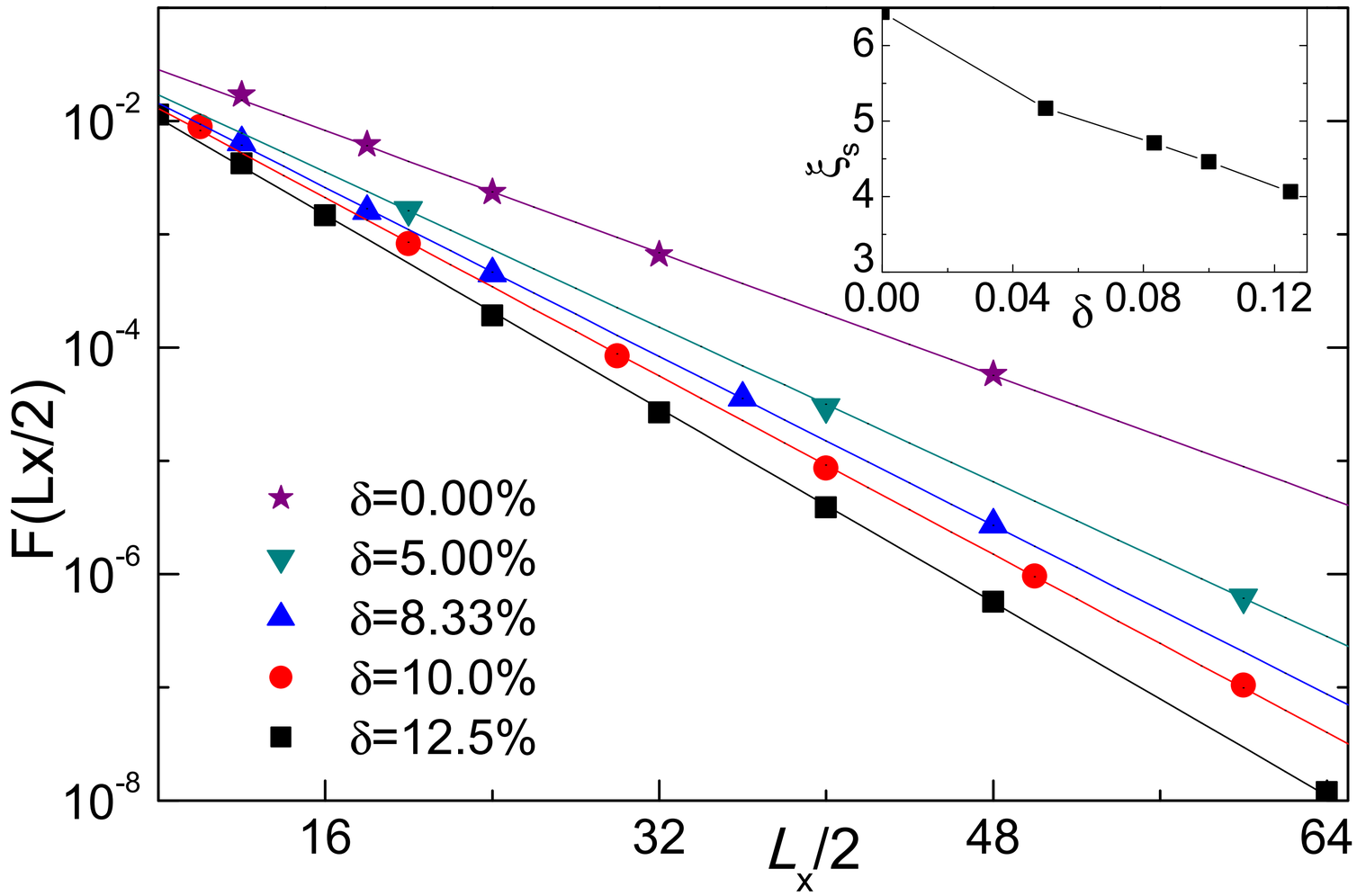
LE phase: Exponents

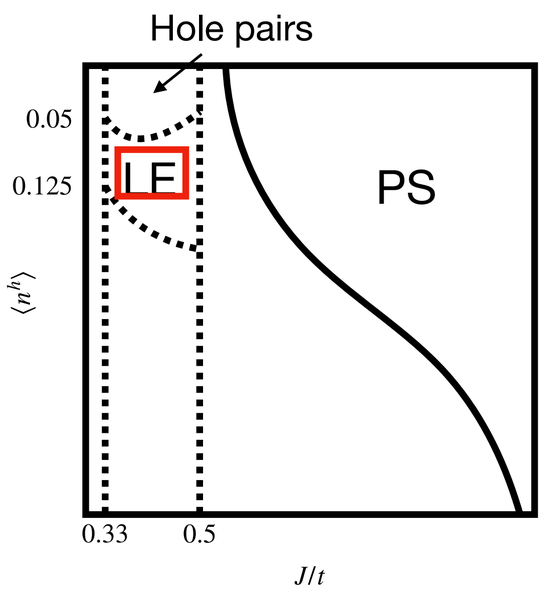




Phase diagram: four-leg ladder

LE phase: Spin-spin correlation





Phase diagram: 2D

Phase diagram: 2D

- White, S. R., & Scalapino, D. J. (1998). Density Matrix Renormalization Group Study of the Striped Phase in the 2D t-J Model. *Physical Review Letters*, *80*(6), 1272–1275.
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- White, S. R., & Scalapino, D. J. (2009). Pairing on striped t-t'-J lattices. *Physical Review B*, *79*(22), 220504.
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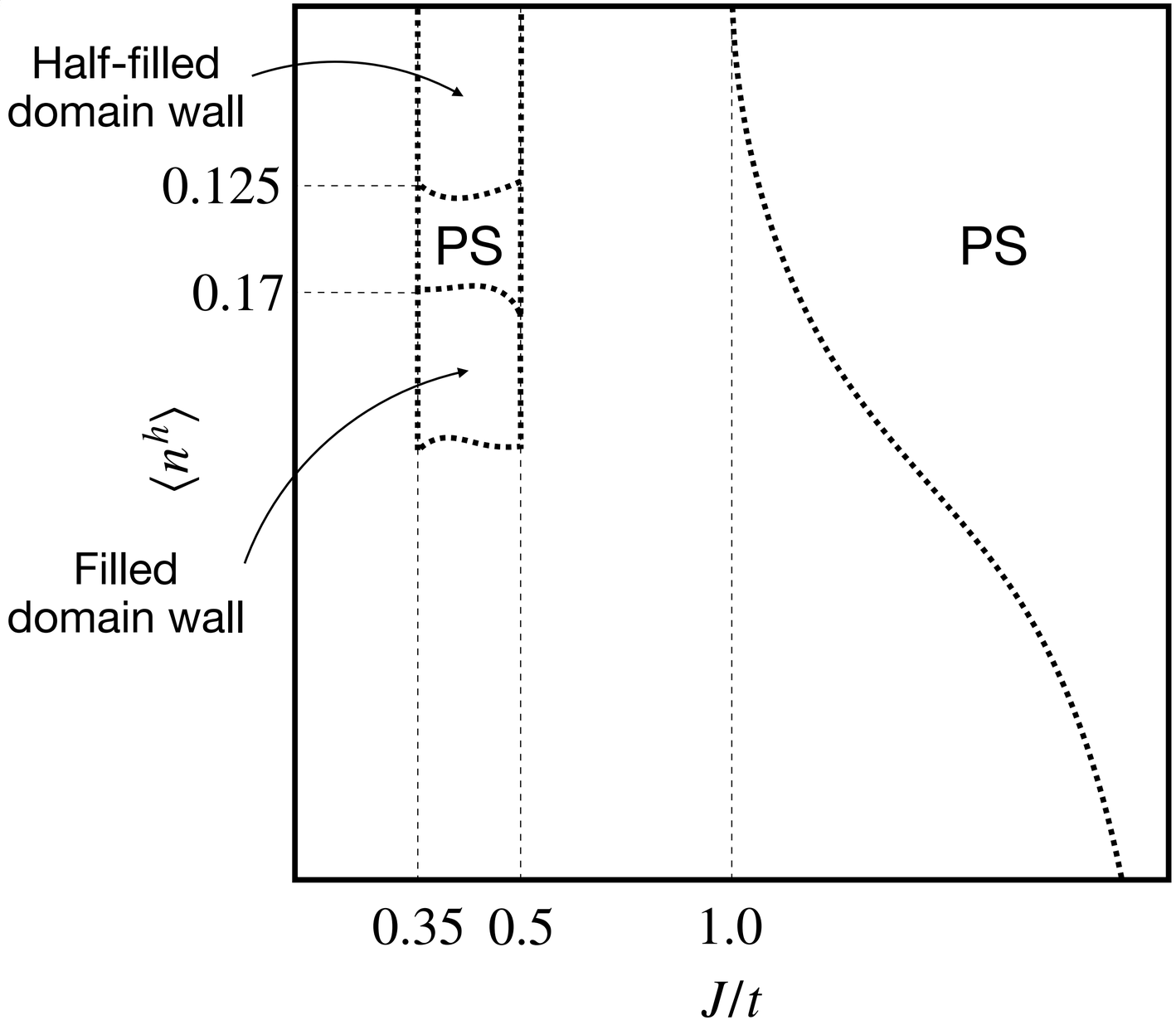
Phase diagram: 2D

Domain wall terminology

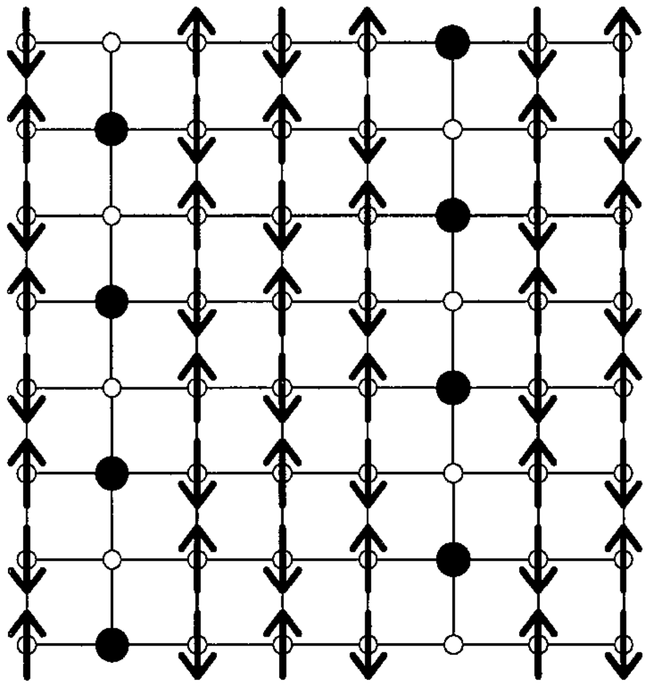
- Domain and domain wall
- Site-centered domain wall and bond-centered domain wall
- Linear filling of the domain wall
- In phase and out of (anti-) phase

Phase diagram: 2D

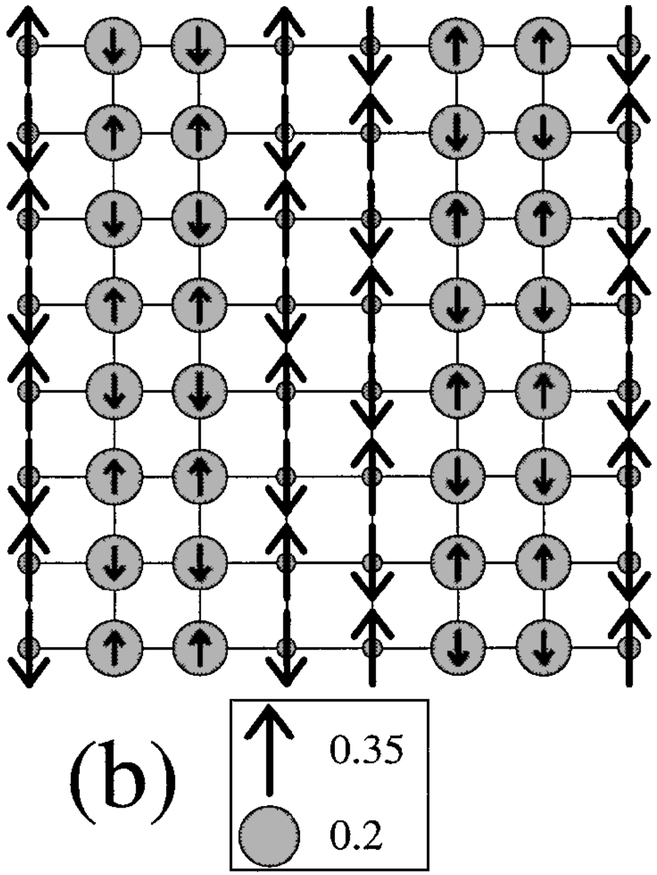


Phase diagram: 2D

Half-filled domain wall: hole and spin distributions



(a)



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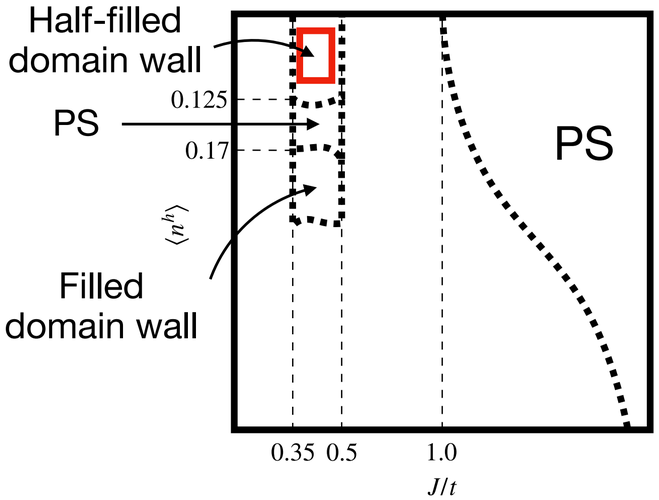
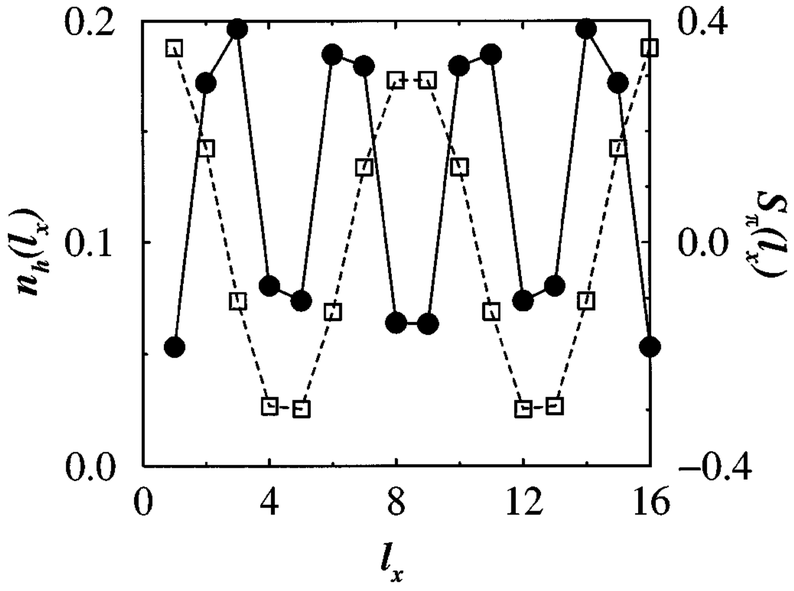
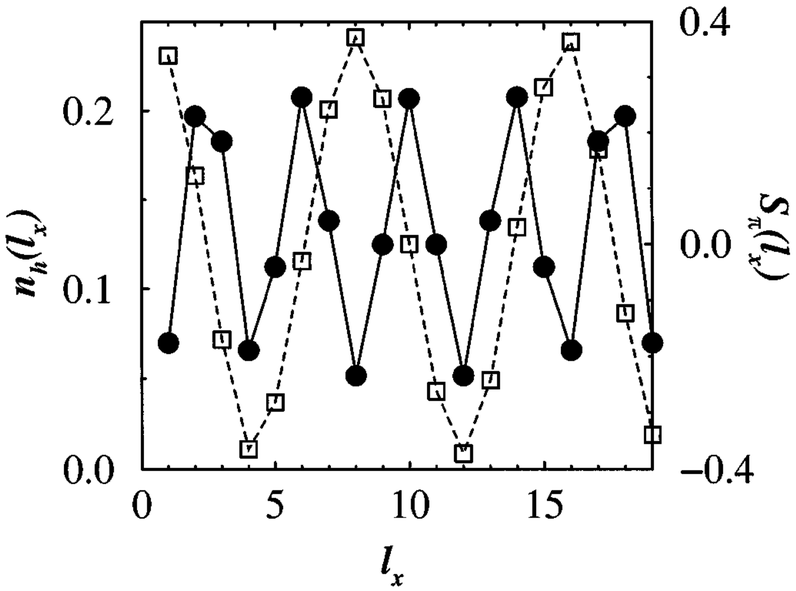
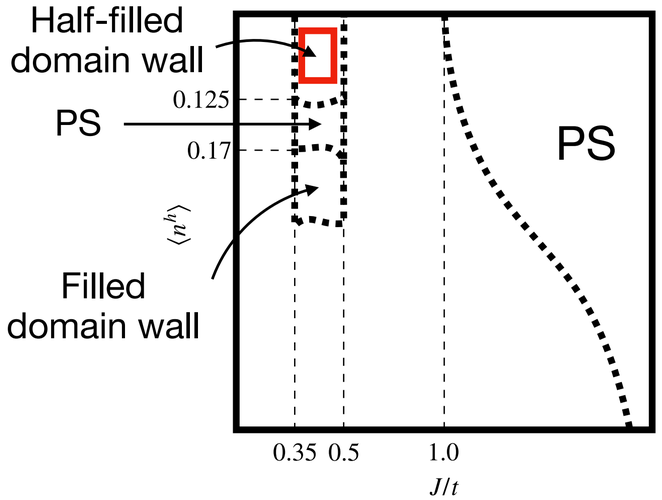


Phase diagram: 2D

Half-filled domain wall: bond-centered vs site-centered

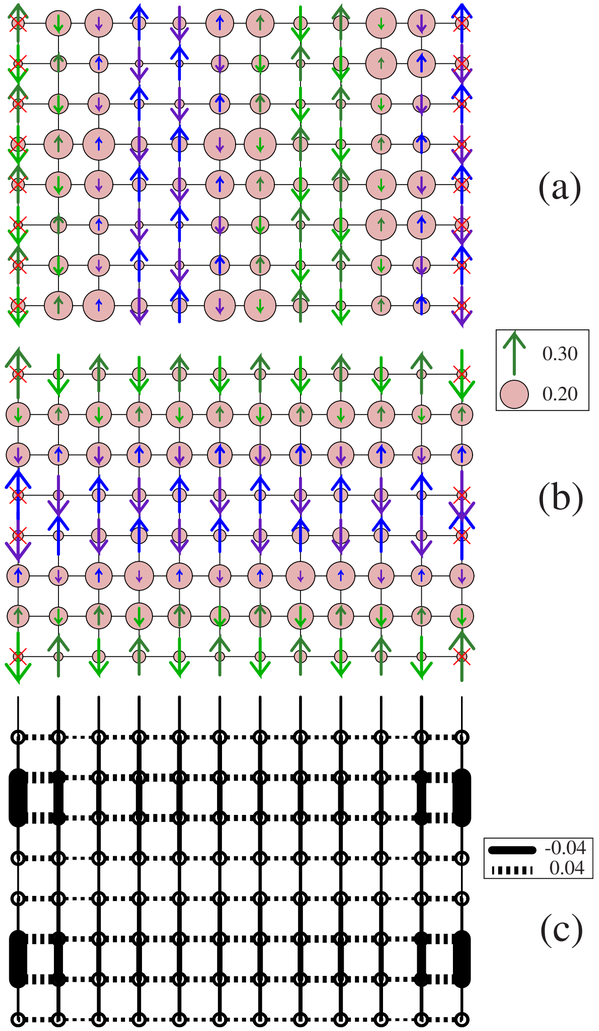


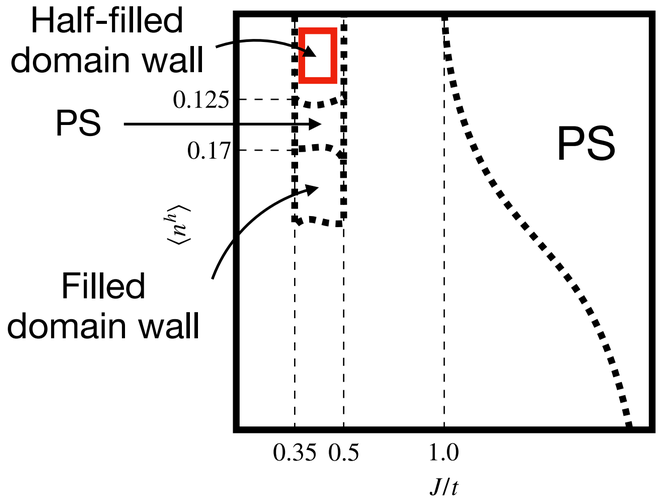




Phase diagram: 2D

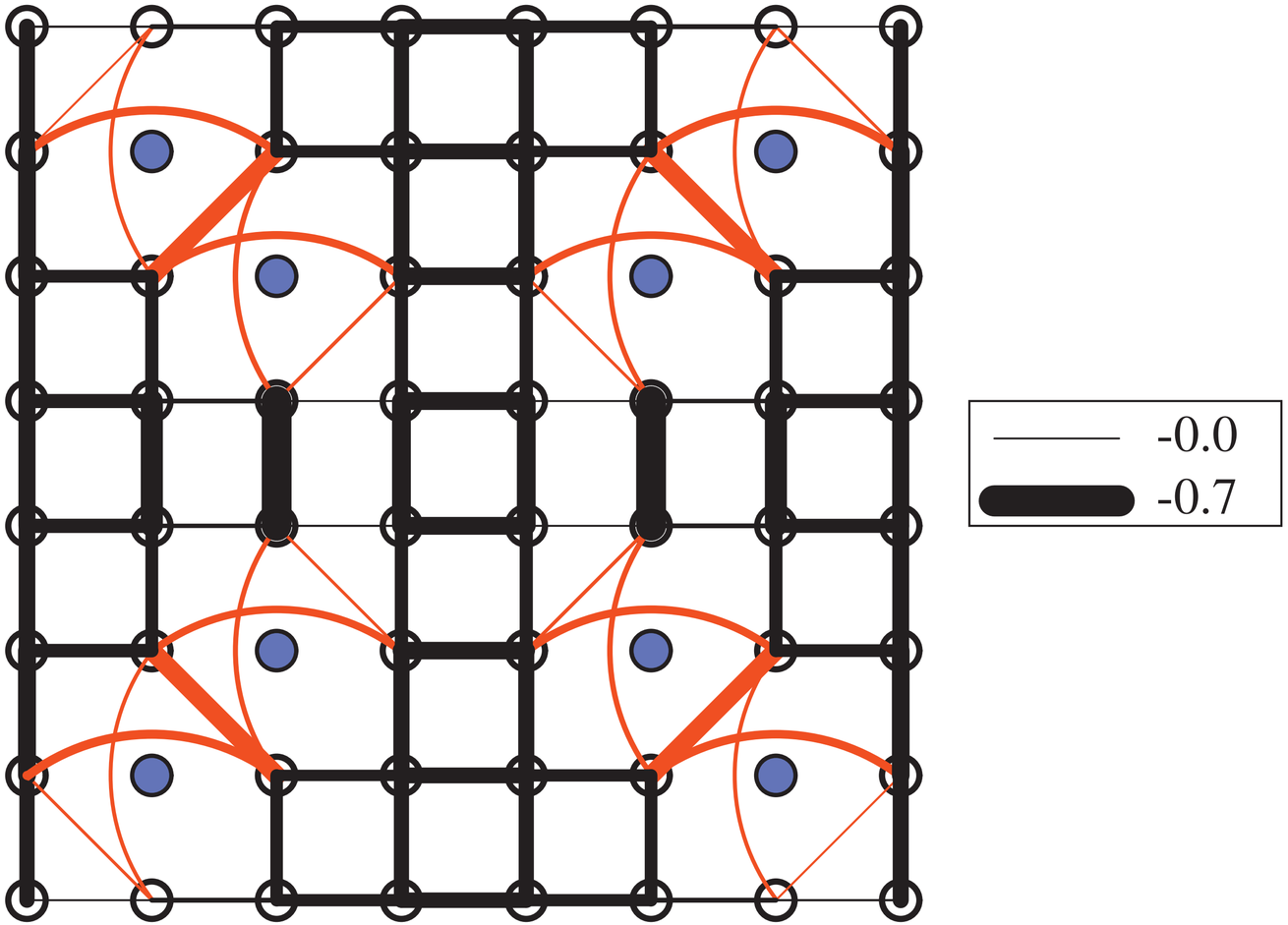
Half-filled domain wall: pairing symmetry

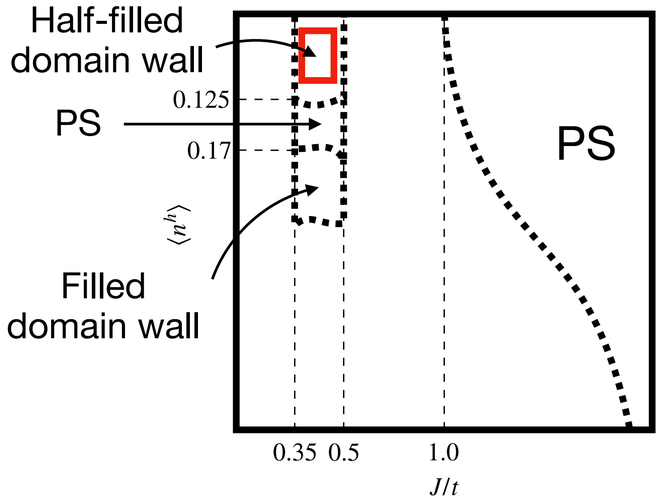




Phase diagram: 2D

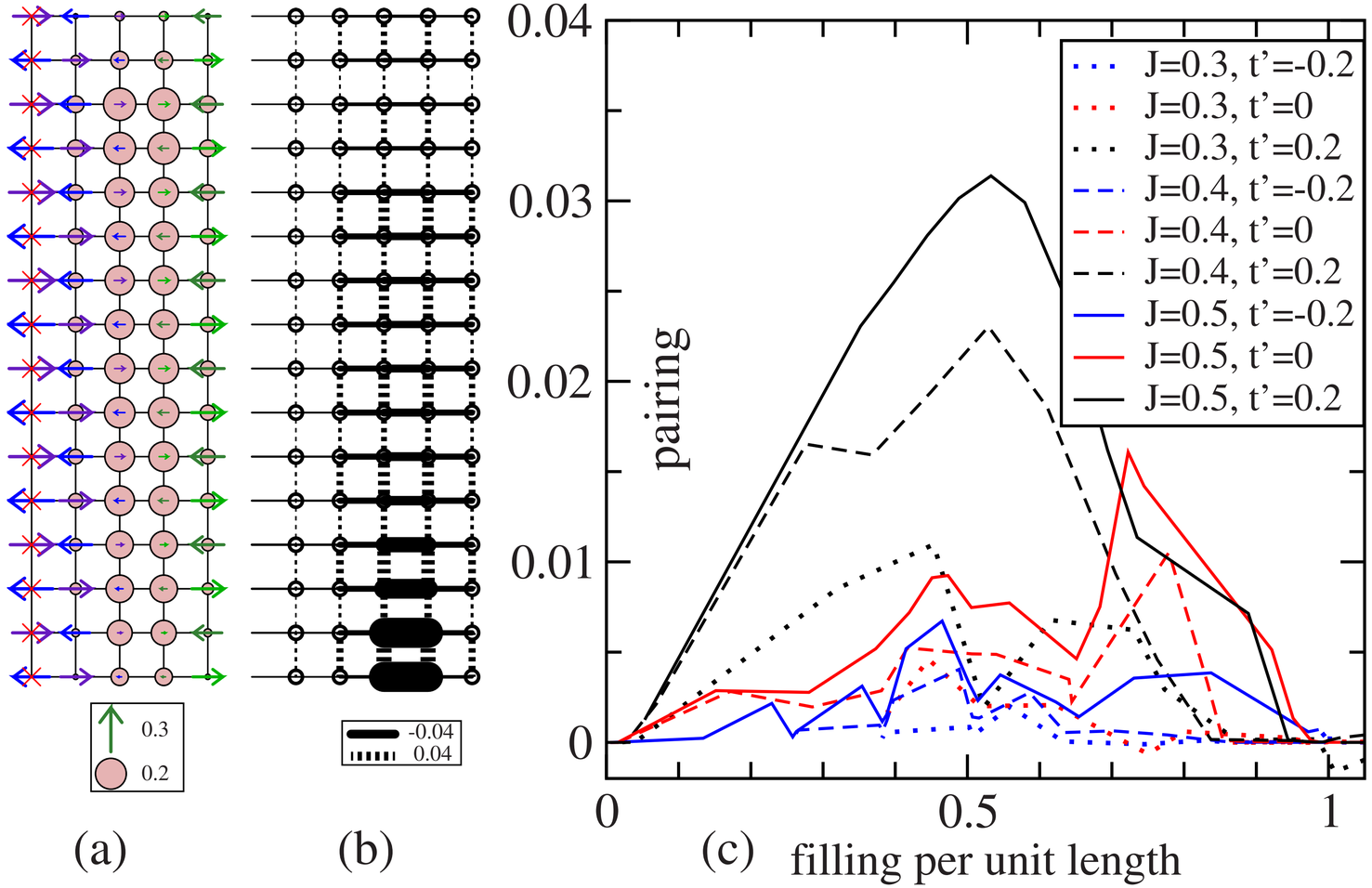
Half-filled domain wall: pairing structure

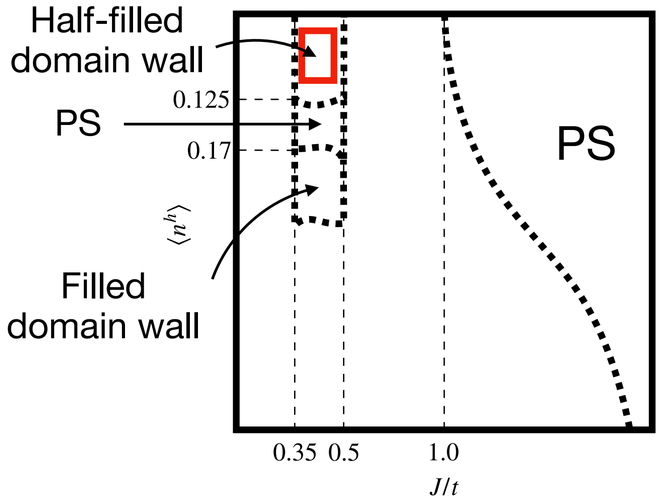




Phase diagram: 2D

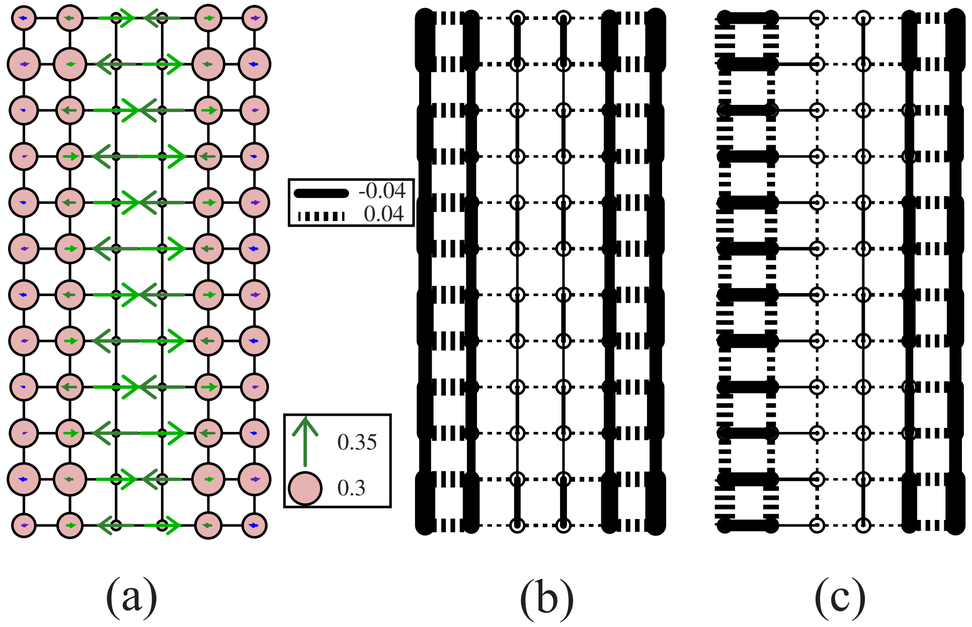
Half-filled domain wall: pairing in a domain wall

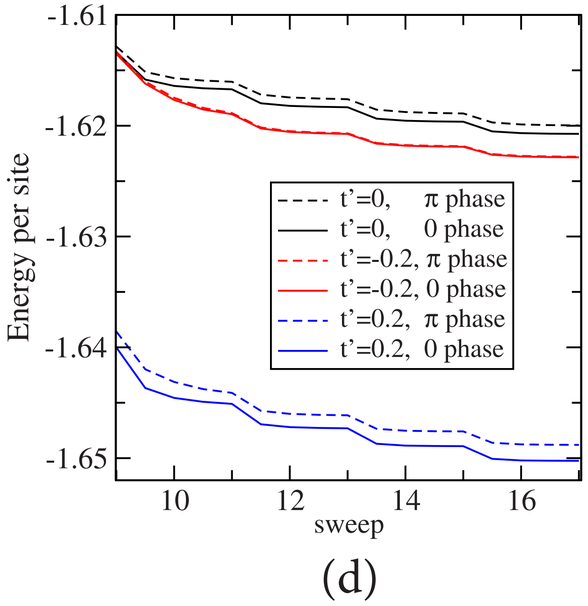


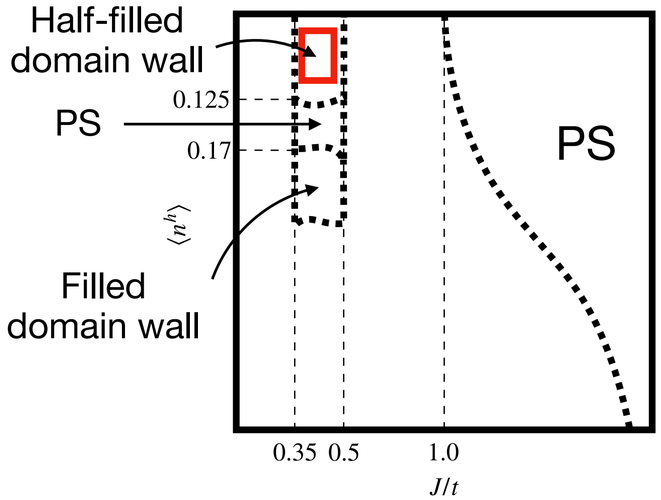


Phase diagram: 2D

Half-filled domain wall: pairing between two domain walls

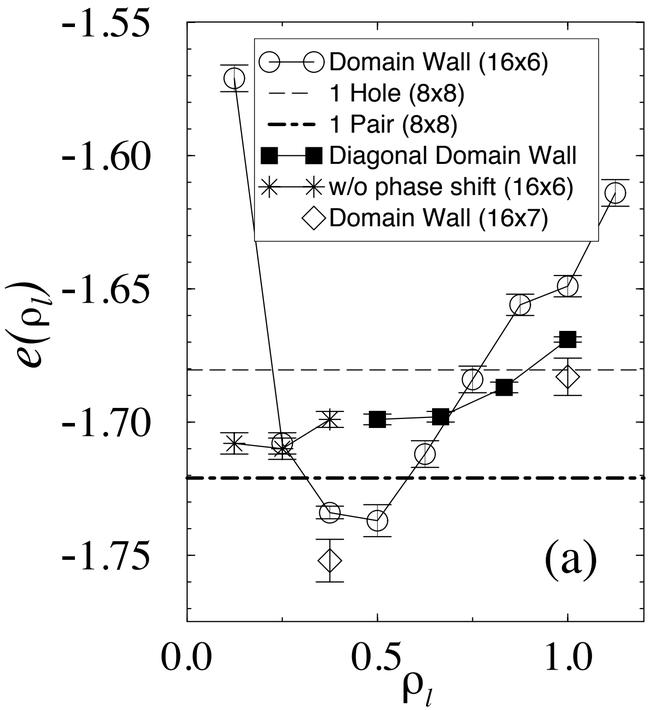


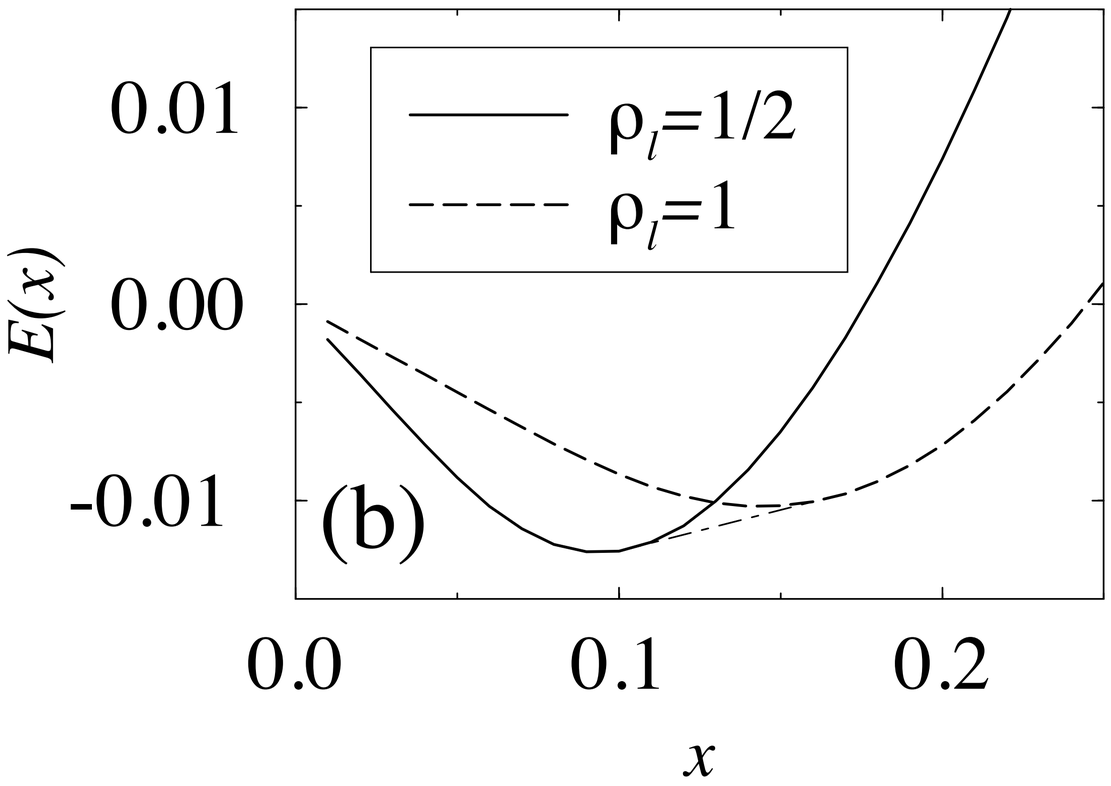


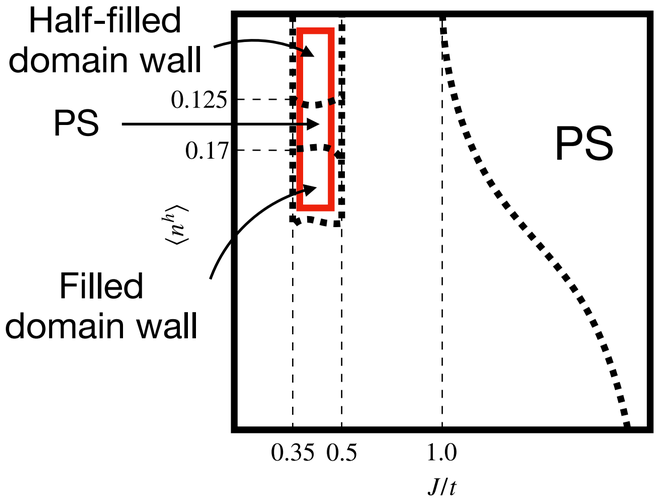


Phase diagram: 2D

Domain wall: linear filling

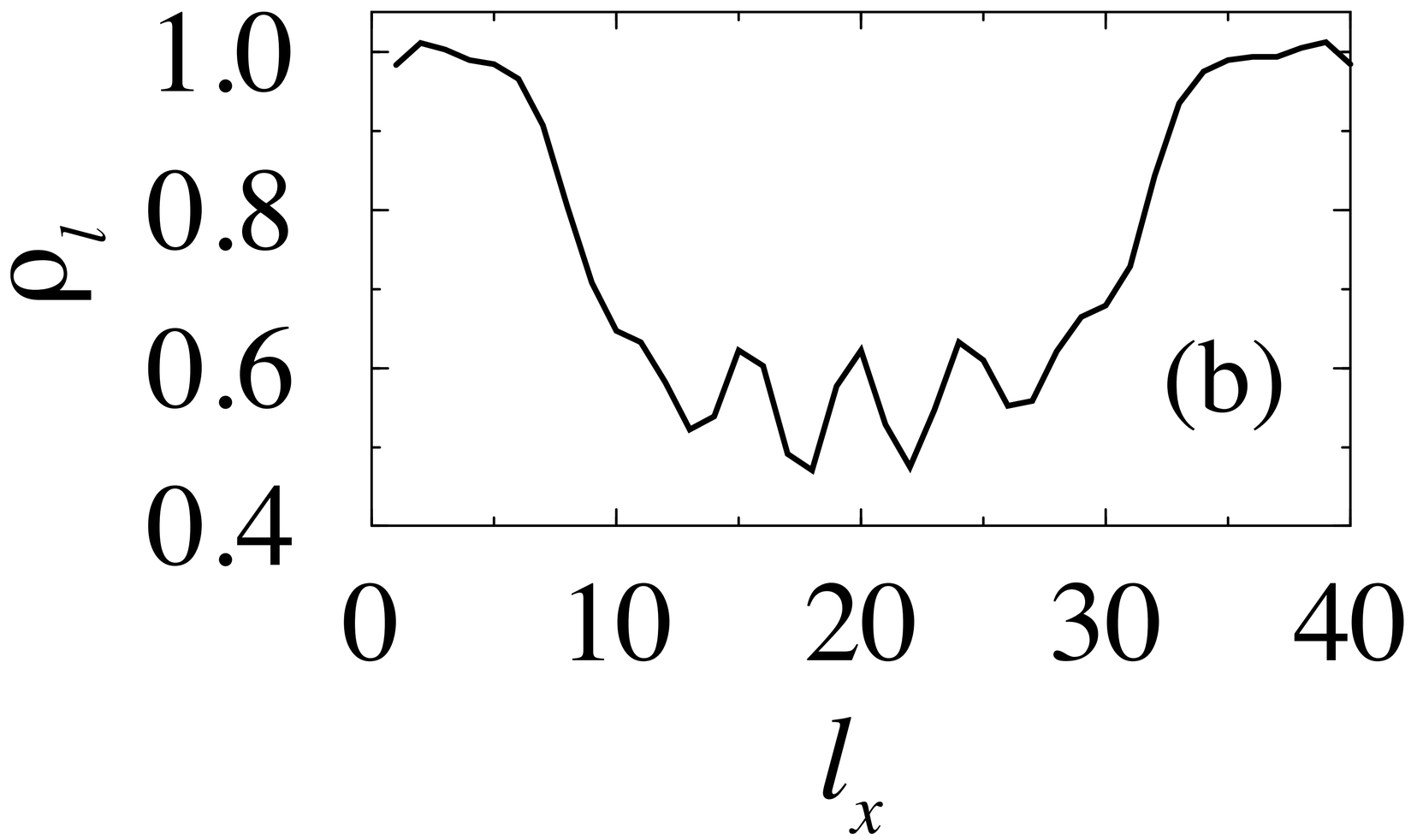


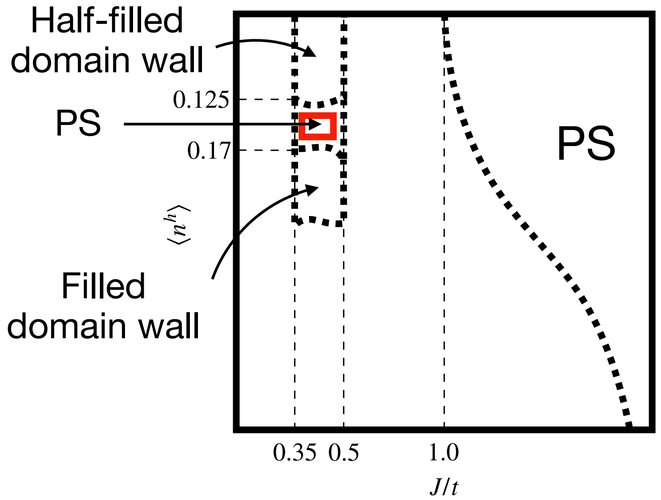




Phase diagram: 2D

Domain wall: phase separation



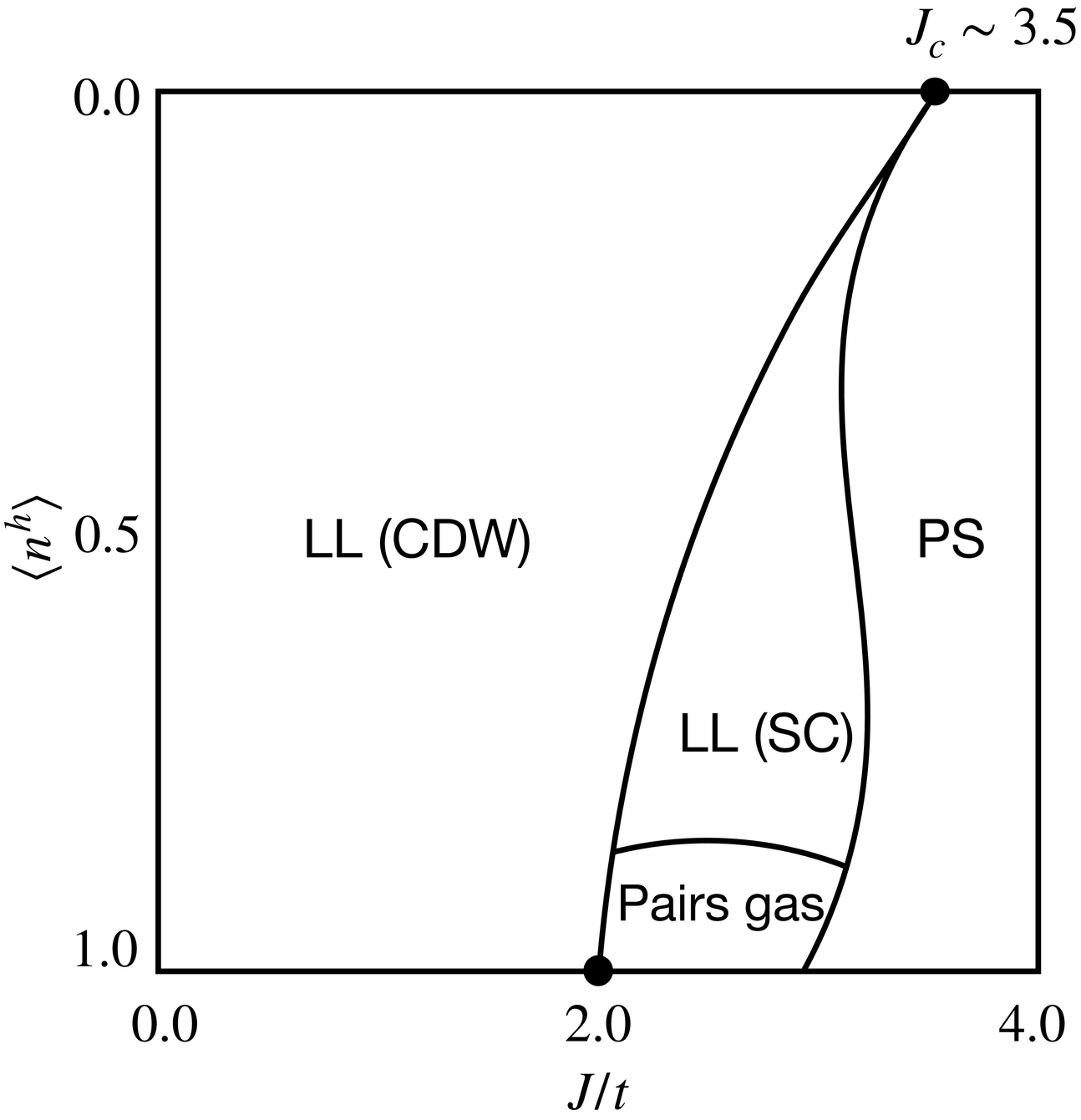


Outlook

Outlook

1D

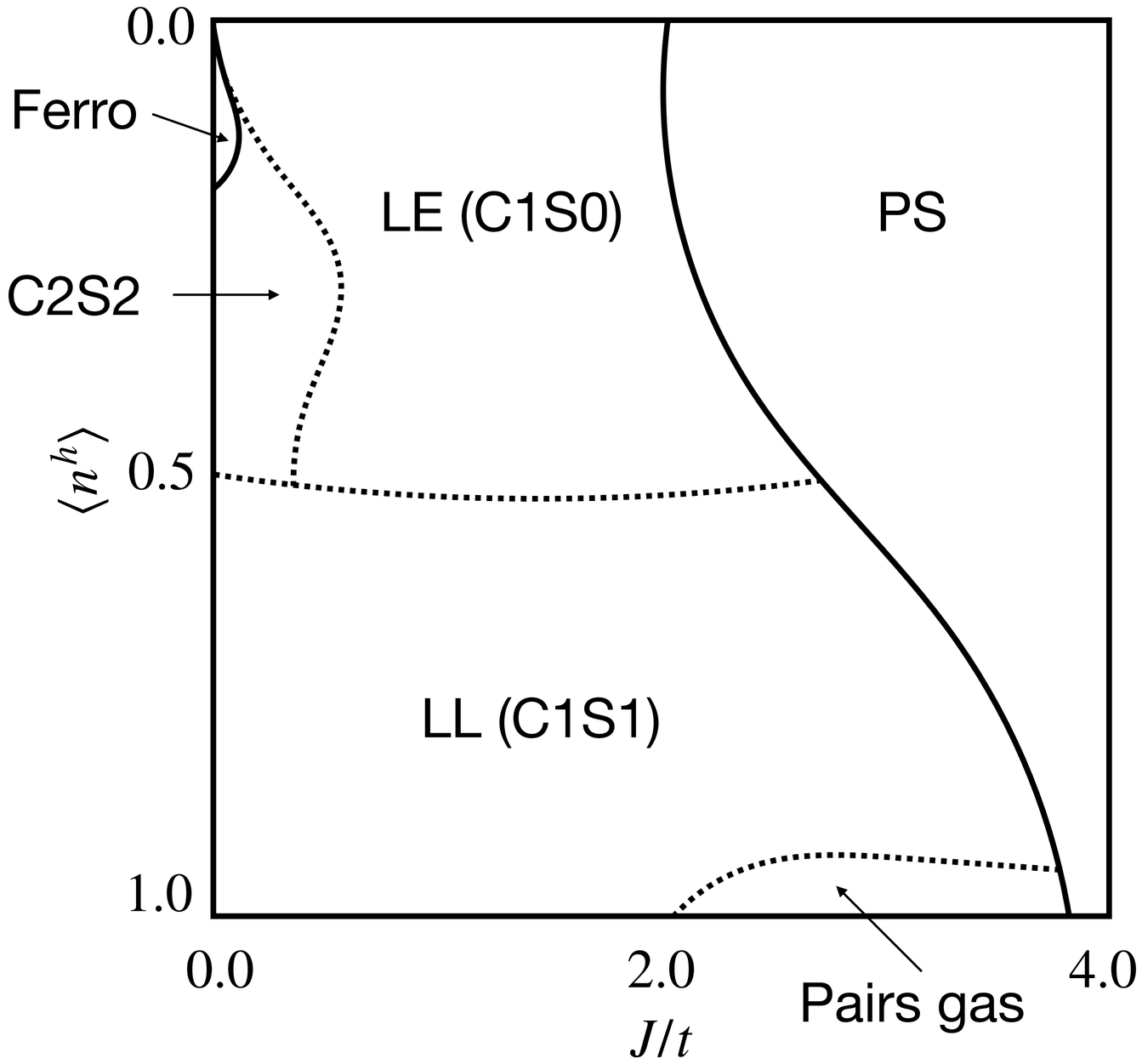
- Pairs gas phase



Outlook

Two-leg ladder

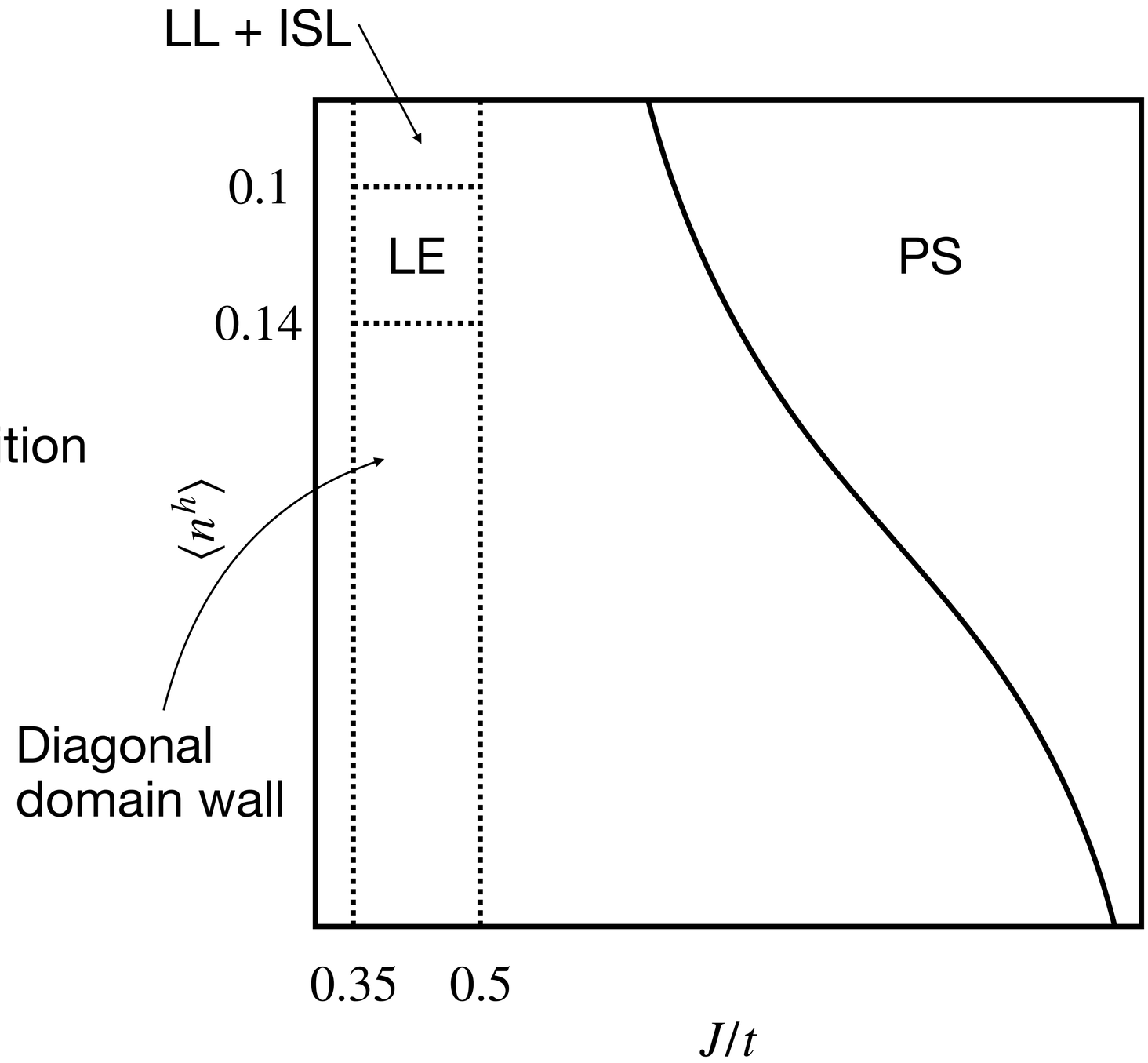
- C2S2 phase
- Phase transition to other phases
- $J/t \sim 0.33$ line



Outlook

Three-leg ladder

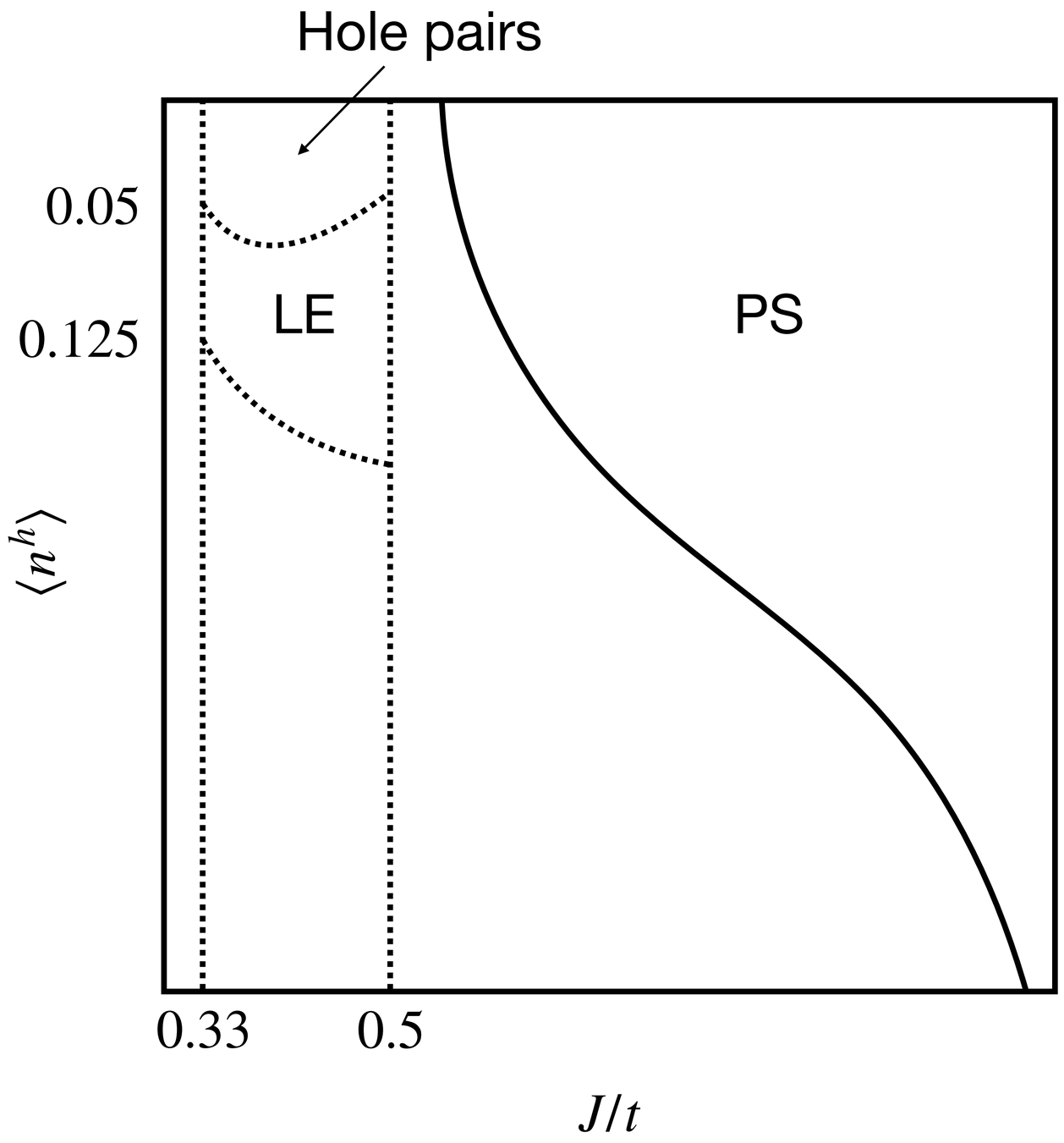
- LE phase
- Doping controled phase transition
- Large doping phase



Outlook

Four-leg ladder

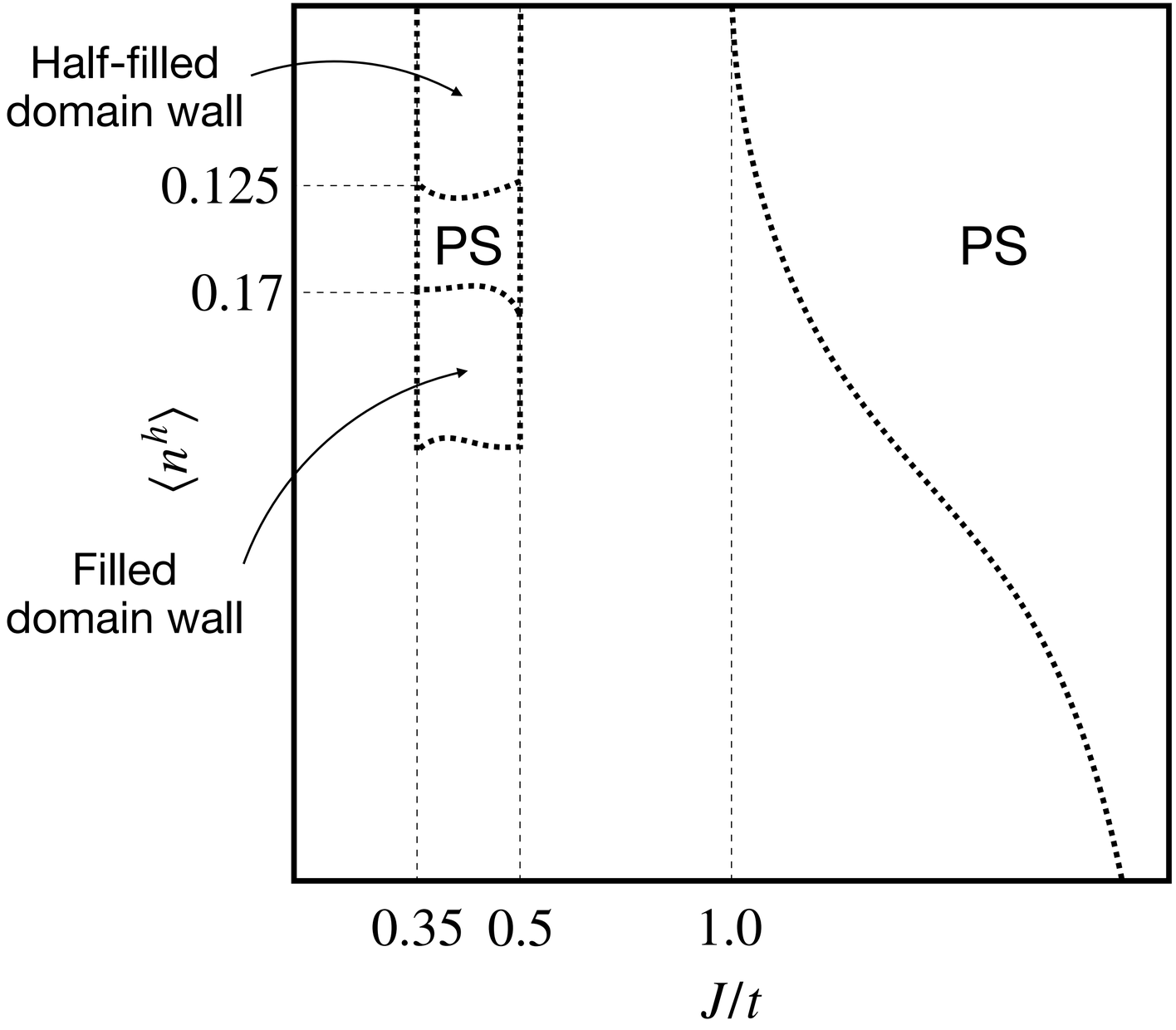
- Small doping ($\delta \leq 0.05$)
- Larger doping
- Phase transitions



Outlook

2D

- Revisit



Thank you!